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Hayashi et al.

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(54) **SEMICONDUCTOR DEVICE**

(56)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(65) **Prior Publication Data**

US 2023/0402432 A1 Dec. 14, 2023

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(63) Continuation of application No. 17/644,452, filed on Dec. 15, 2021, now Pat. No. 11,776,936, which is a (Continued)

(30) **Foreign Application Priority Data**

Apr. 20, 2017 (JP) 2017-083370
Apr. 16, 2018 (JP) 2018-078529

(51) **Int. Cl.**

H01L 25/07 (2006.01)

H01L 23/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H01L 25/072** (2013.01); **H01L 23/049** (2013.01); **H01L 23/296** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC ... H01L 25/072; H01L 23/049; H01L 23/296; H01L 23/3107; H01L 23/3735;

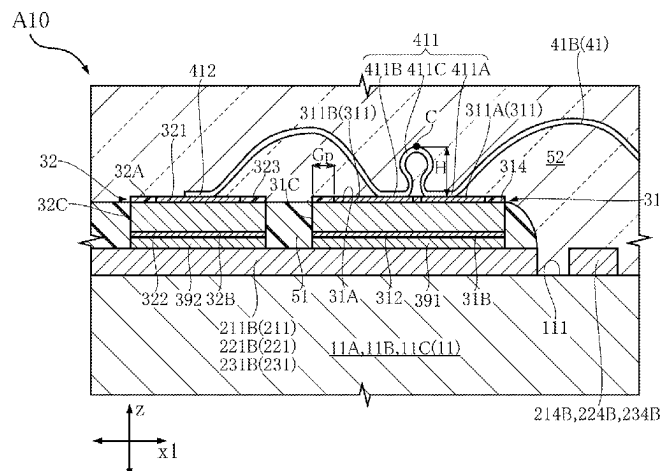
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(57)

ABSTRACT

The present disclosure provides a semiconductor device. The semiconductor device includes a substrate, a mounting layer, switching elements, a moisture-resistant layer and a sealing resin. The substrate has a front surface facing in a thickness direction. The mounting layer is electrically conductive and disposed on the front surface. Each switching element includes an element front surface facing in the same direction in which the front surface faces along the thickness direction, a back surface facing in the opposite direction of the element front surface, and a side surface connected to the element front surface and the back surface. The switching elements are electrically bonded to the mounting layer with their back surfaces facing the front surface. The moisture-resistant layer covers at least one side surface. The sealing resin covers the switching elements and the moisture-resistant layer. The moisture-resistant layer is held in contact with the mounting layer and the side surface so as to be spanned between the mounting layer and the side surface in the thickness direction.

30 Claims, 39 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/492,307, filed as application No. PCT/JP2018/015987 on Apr. 18, 2018, now Pat. No. 11,233,037.

(51) **Int. Cl.**

H01L 23/049 (2006.01)

H01L 23/29 (2006.01)

H01L 23/31 (2006.01)

H01L 23/373 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 23/3107** (2013.01); **H01L 23/3735** (2013.01); **H01L 23/564** (2013.01); **H01L 24/32** (2013.01); **H01L 24/40** (2013.01); **H01L 24/48** (2013.01); **H01L 24/73** (2013.01); **H01L 2224/32225** (2013.01); **H01L 2224/40139** (2013.01); **H01L 2224/40225** (2013.01); **H01L 2224/48096** (2013.01); **H01L 2224/48139** (2013.01); **H01L 2224/48225** (2013.01); **H01L 2224/73263** (2013.01); **H01L 2224/73265** (2013.01); **H01L 2924/13091** (2013.01); **H01L 2924/14252** (2013.01)

(58) **Field of Classification Search**

CPC H01L 23/564; H01L 24/32; H01L 24/40; H01L 24/48; H01L 24/73; H01L 2224/32225; H01L 2224/40139; H01L 2224/40225; H01L 2224/48096; H01L 2224/48139; H01L 2224/48225; H01L 2224/73263; H01L 2224/73265; H01L 2924/13091; H01L 2924/14252

USPC 257/694, 678

See application file for complete search history.

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FIG.1

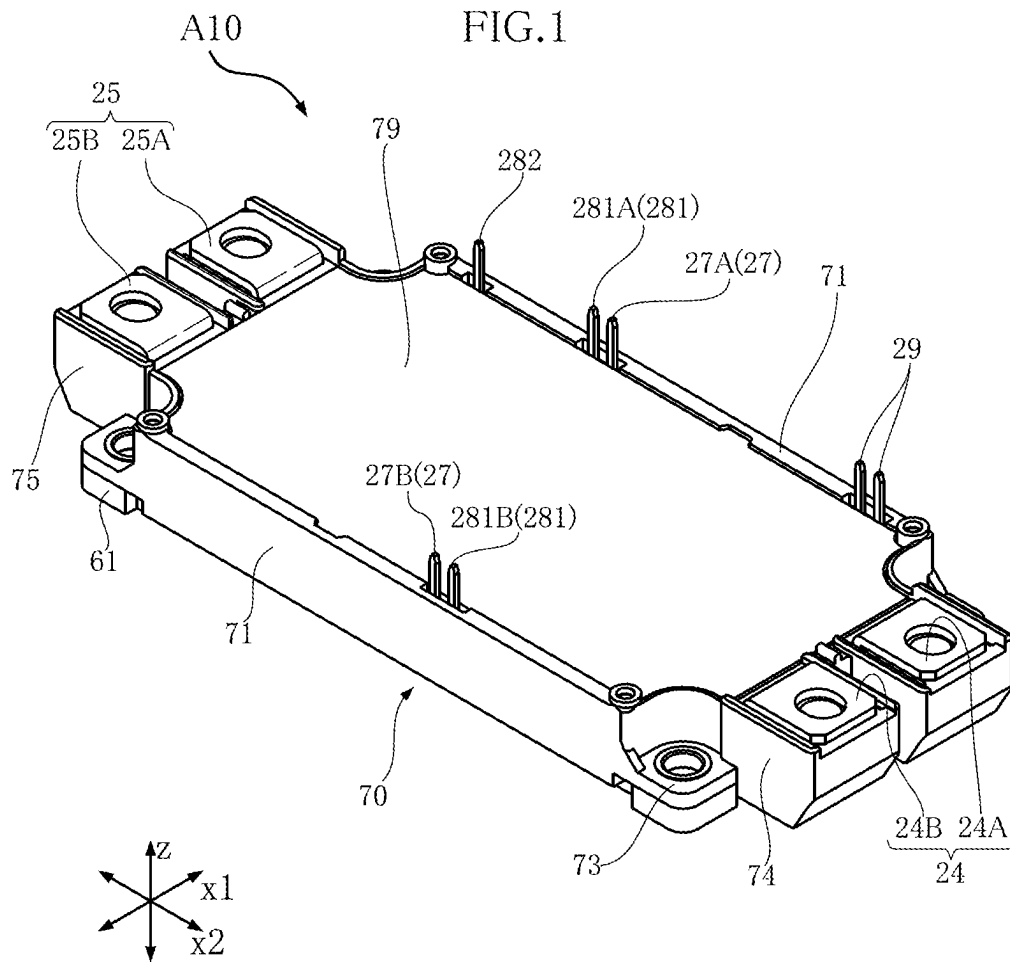


FIG.2

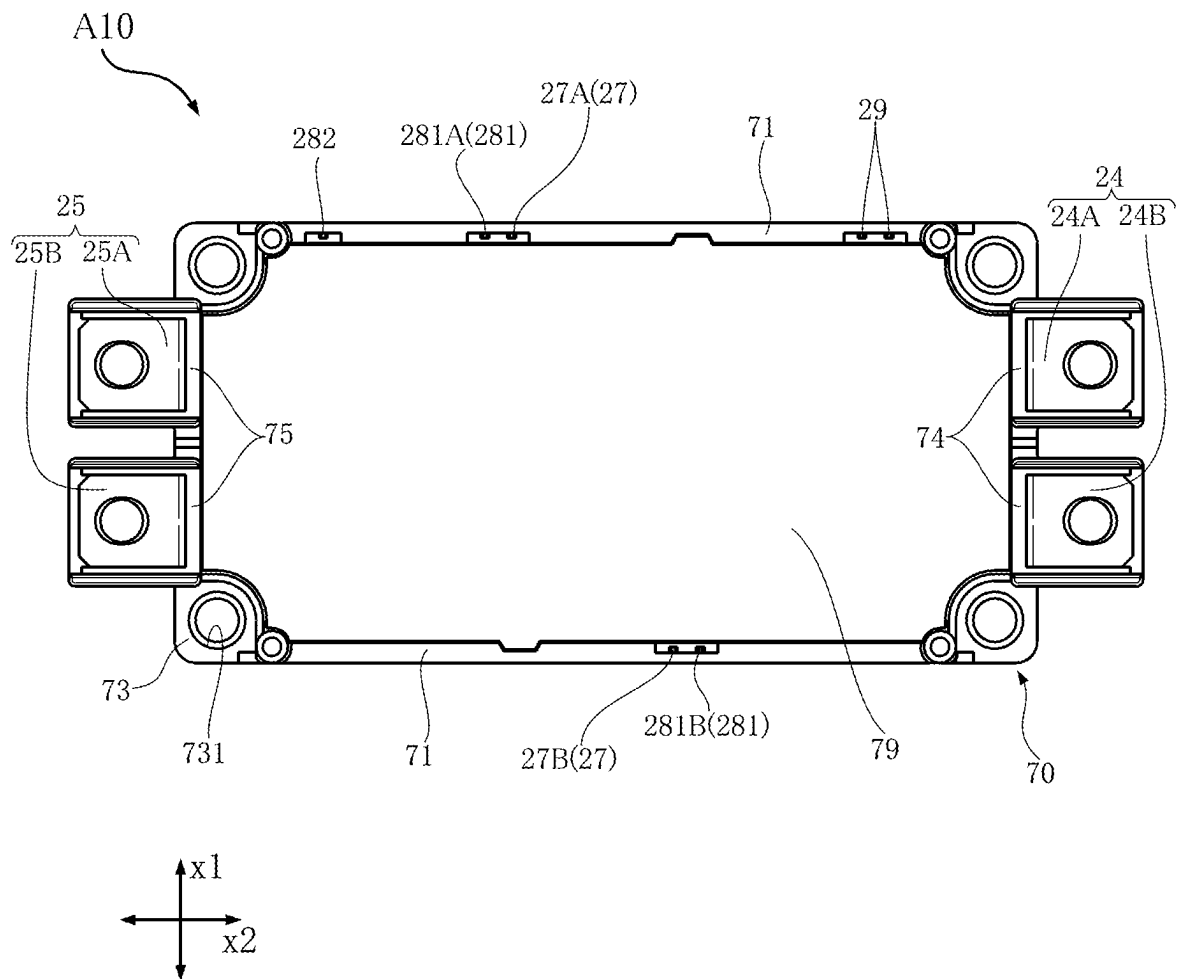


FIG. 3

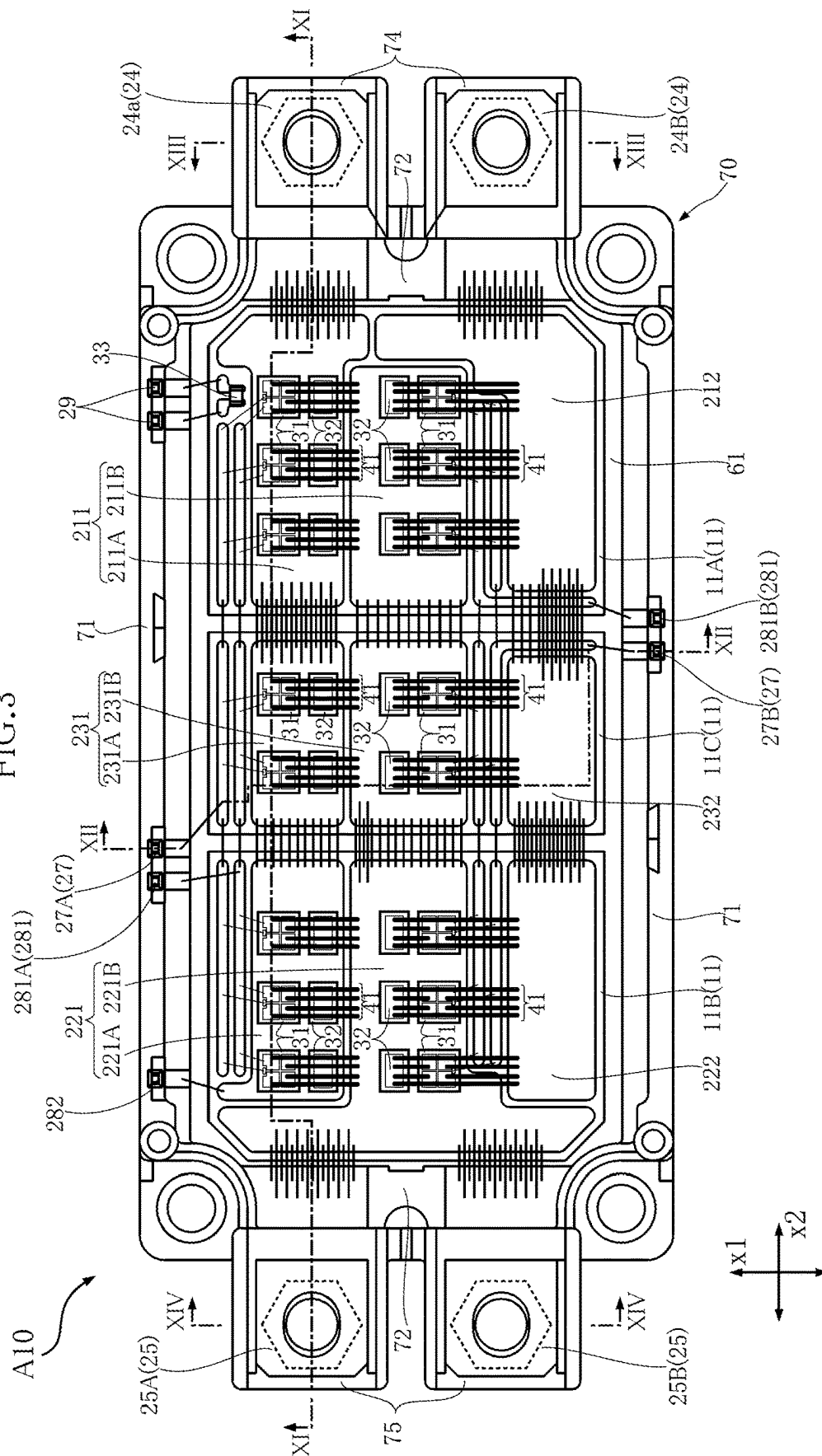
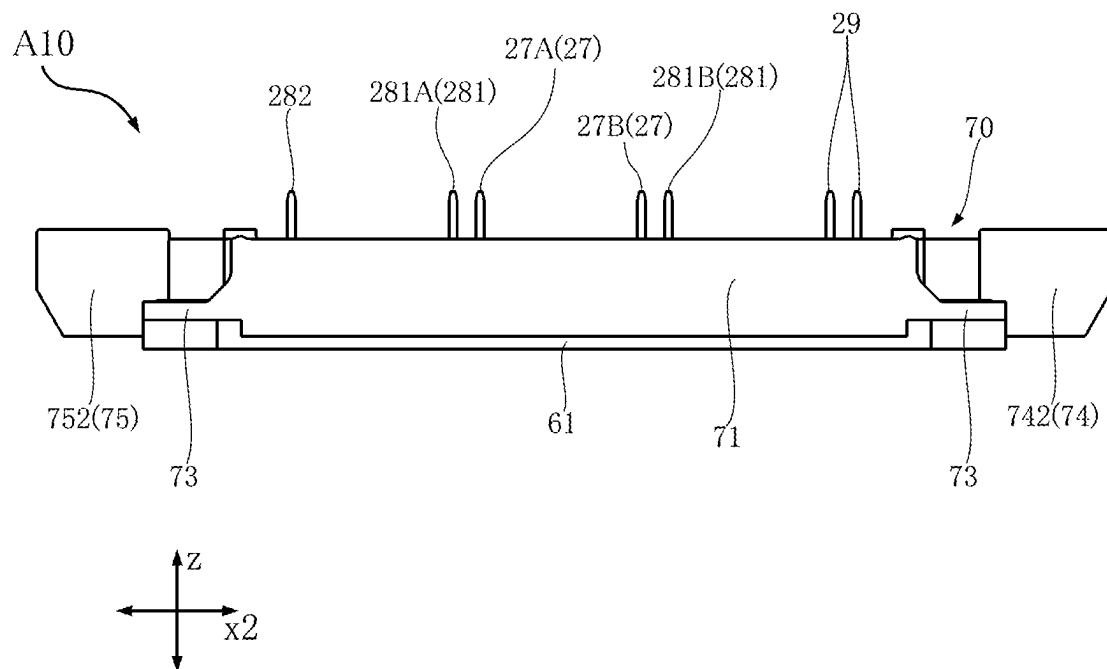


FIG. 4



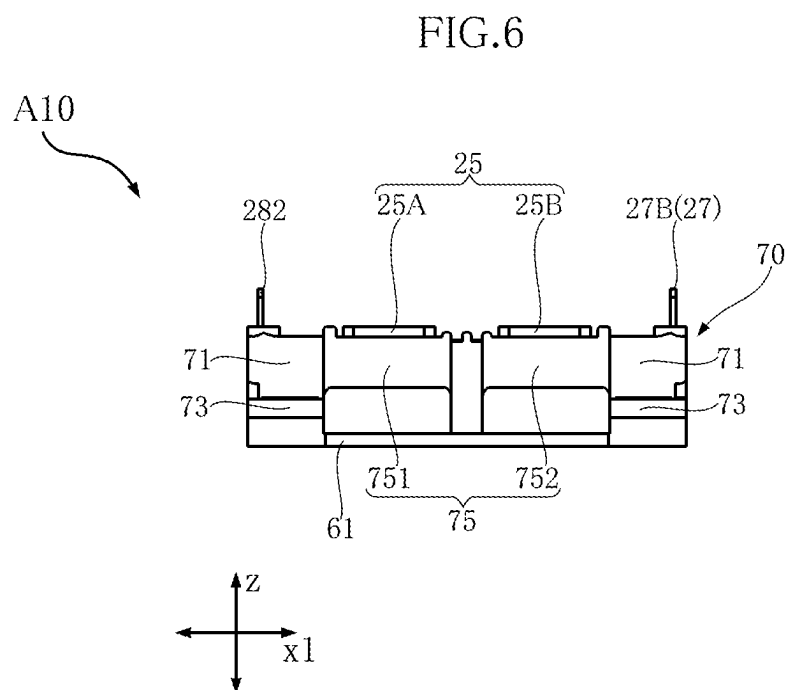
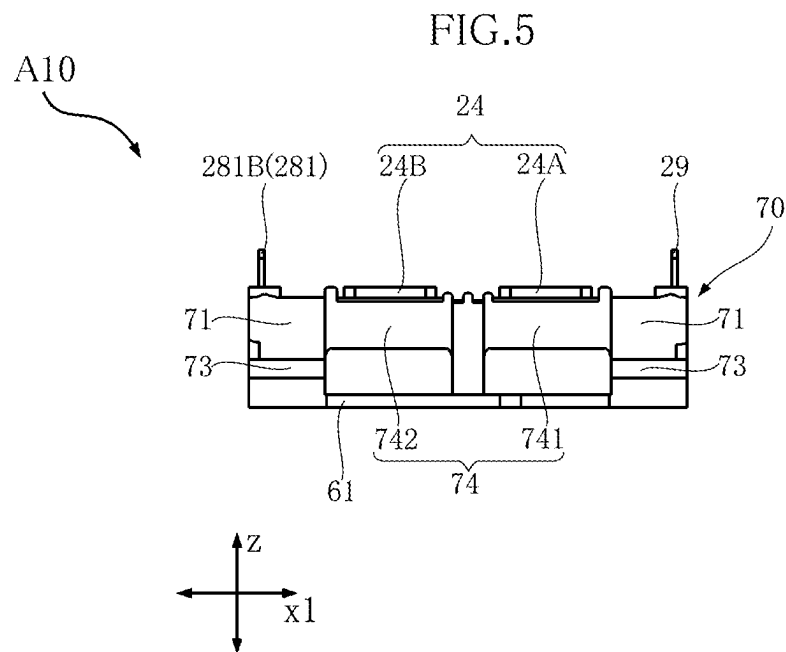


FIG. 7

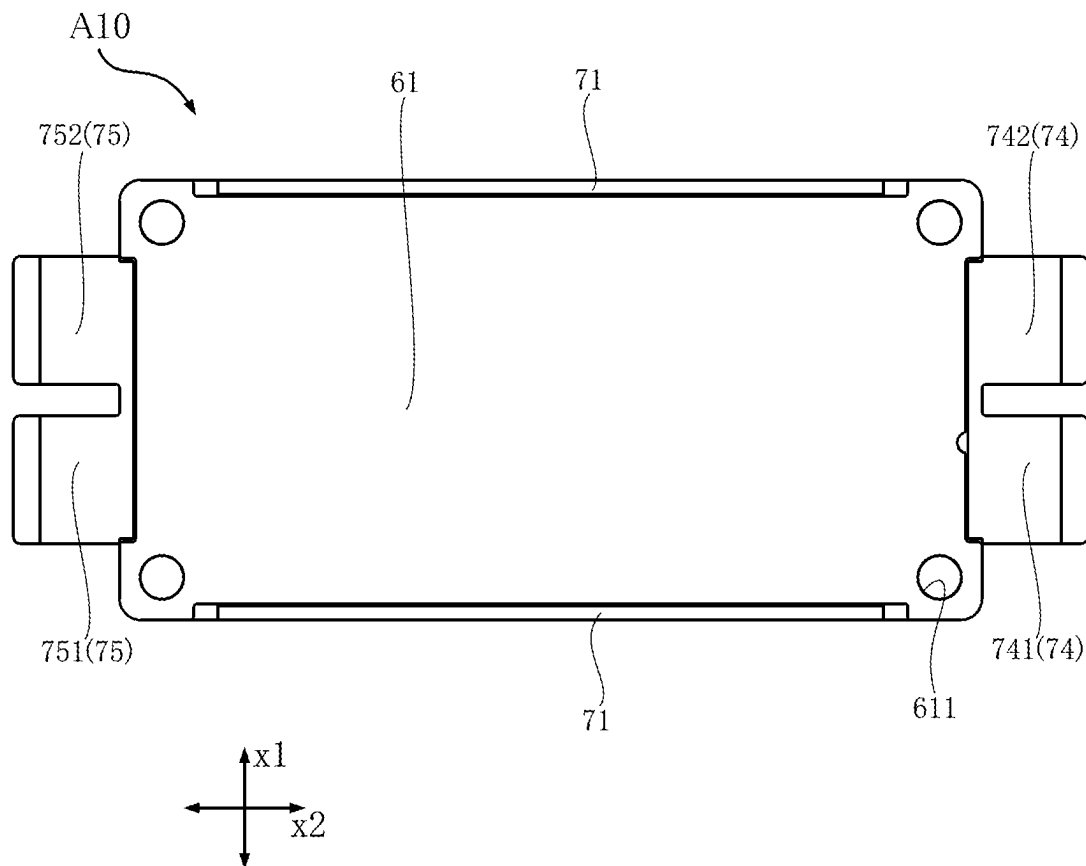


FIG.8

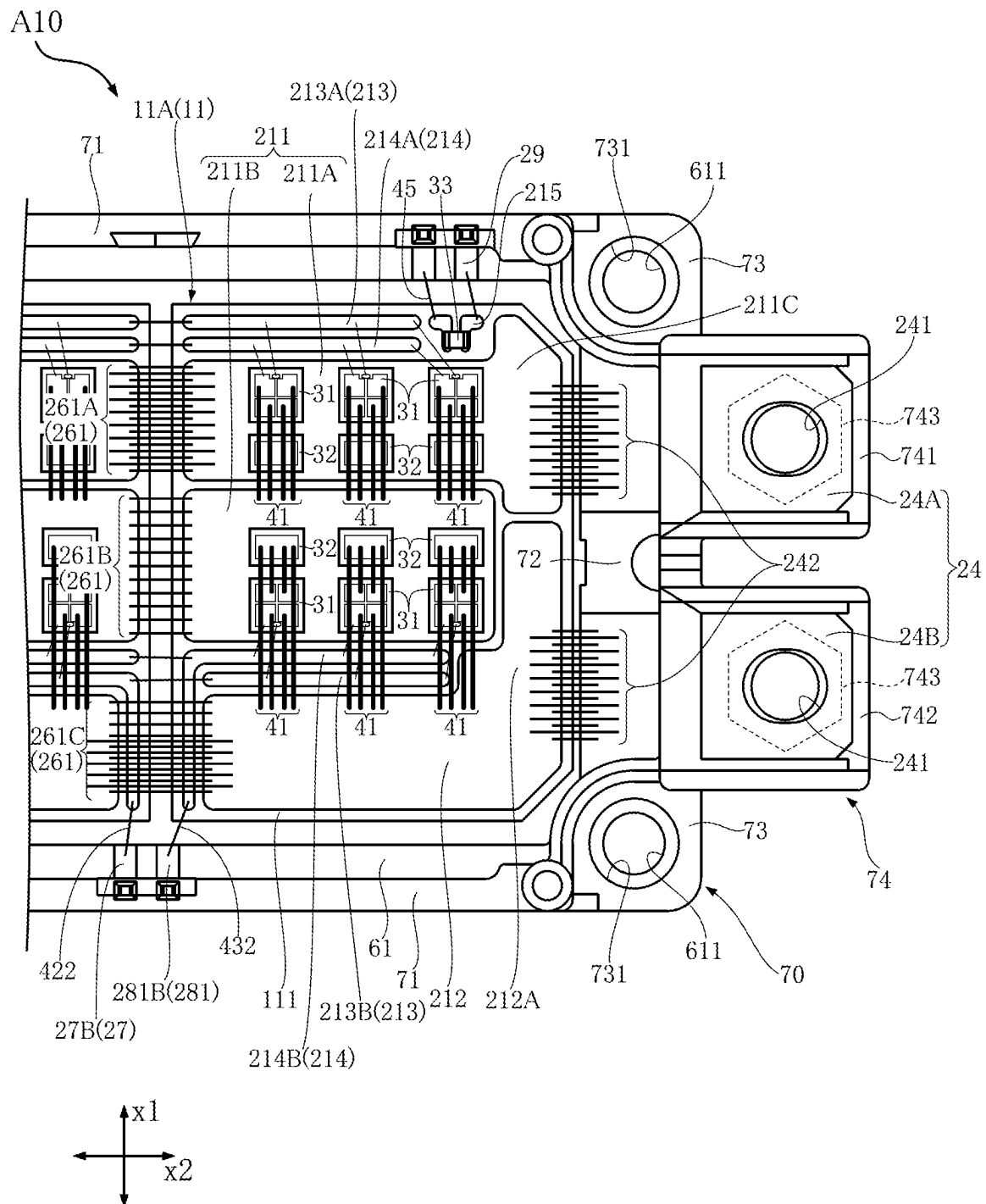


FIG. 9

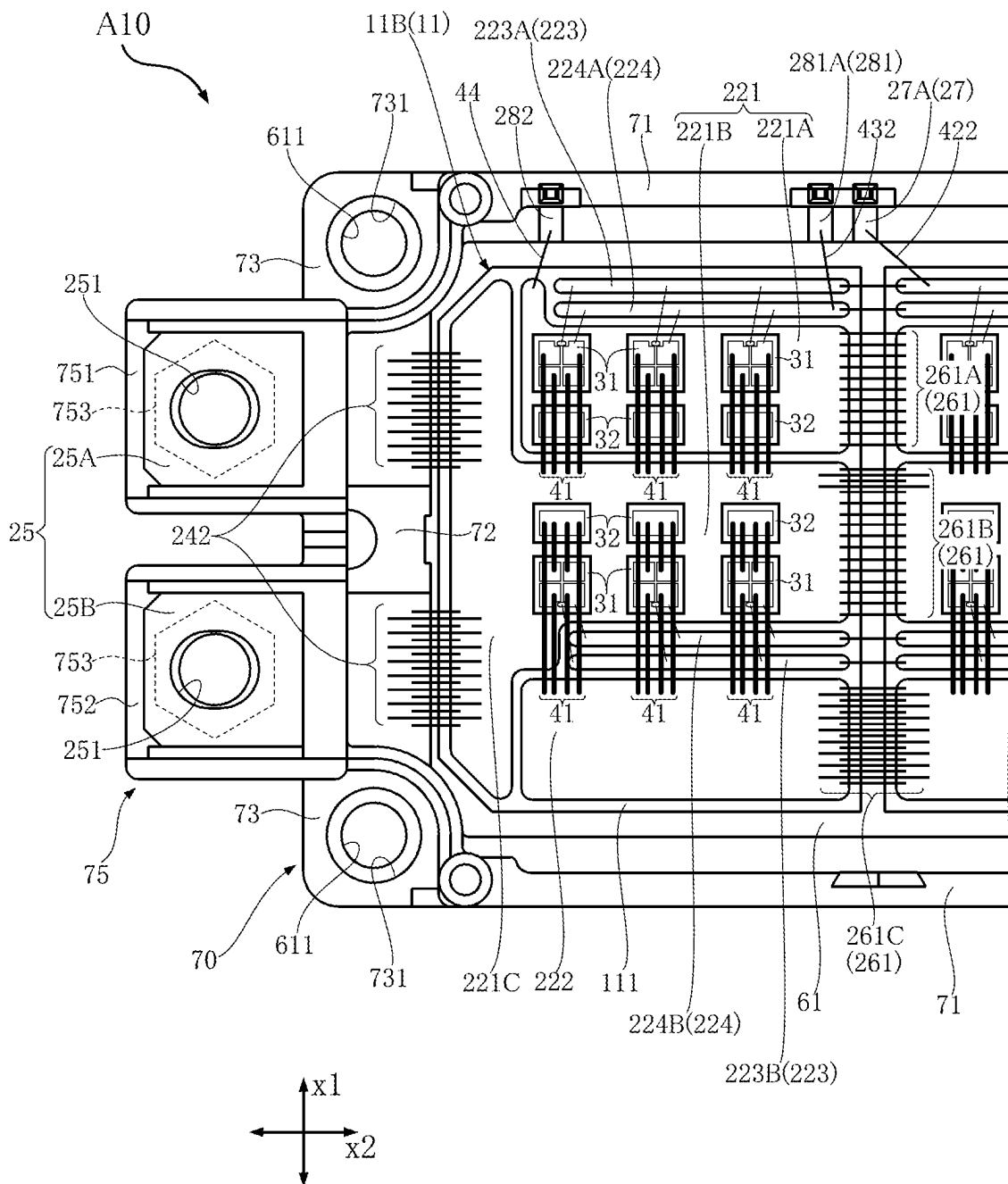


FIG. 10

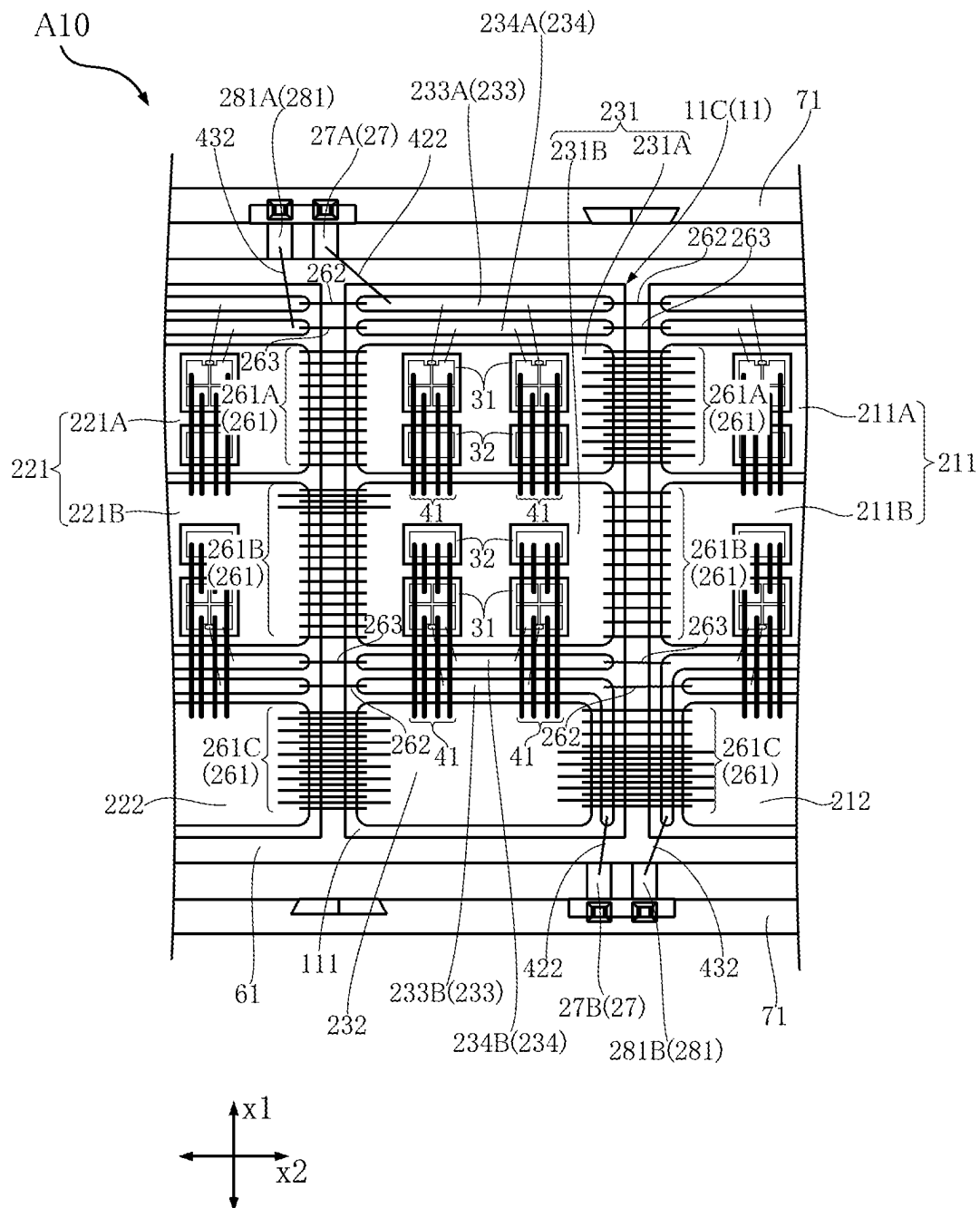


FIG.11

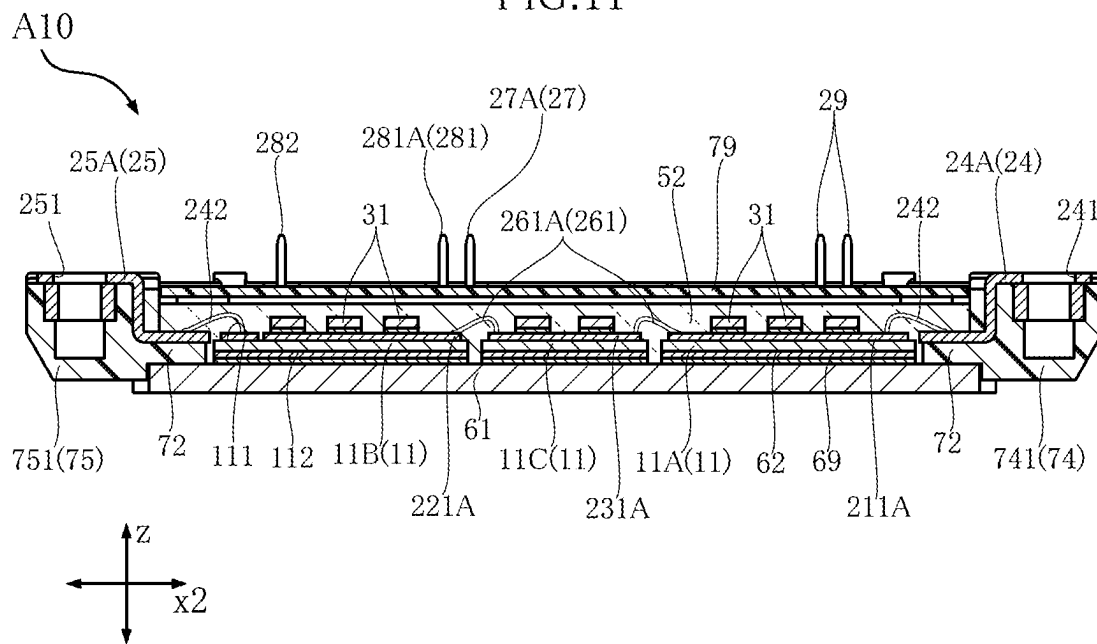


FIG.12

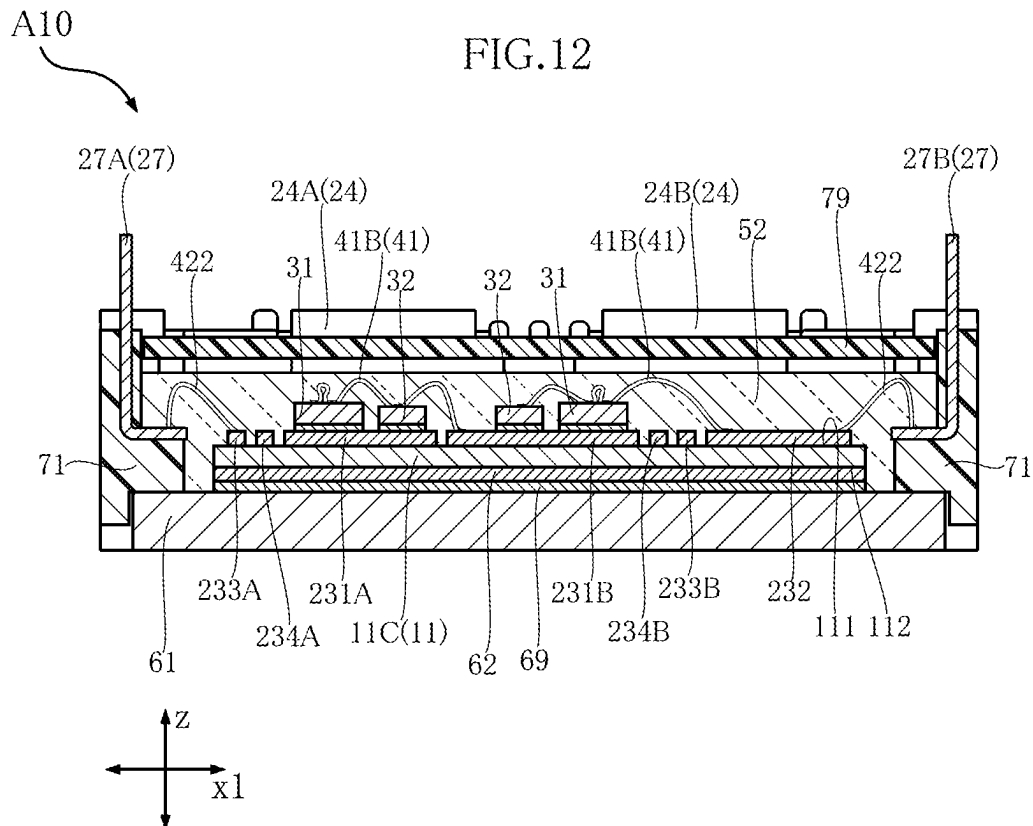


FIG.13

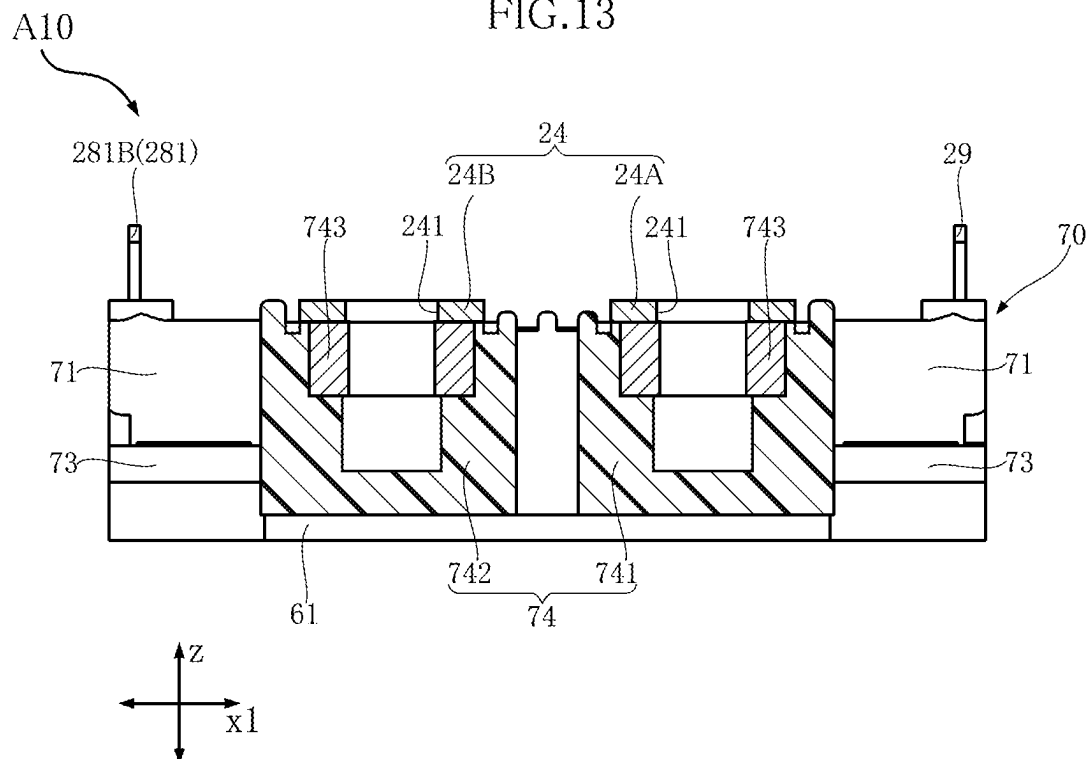


FIG.14

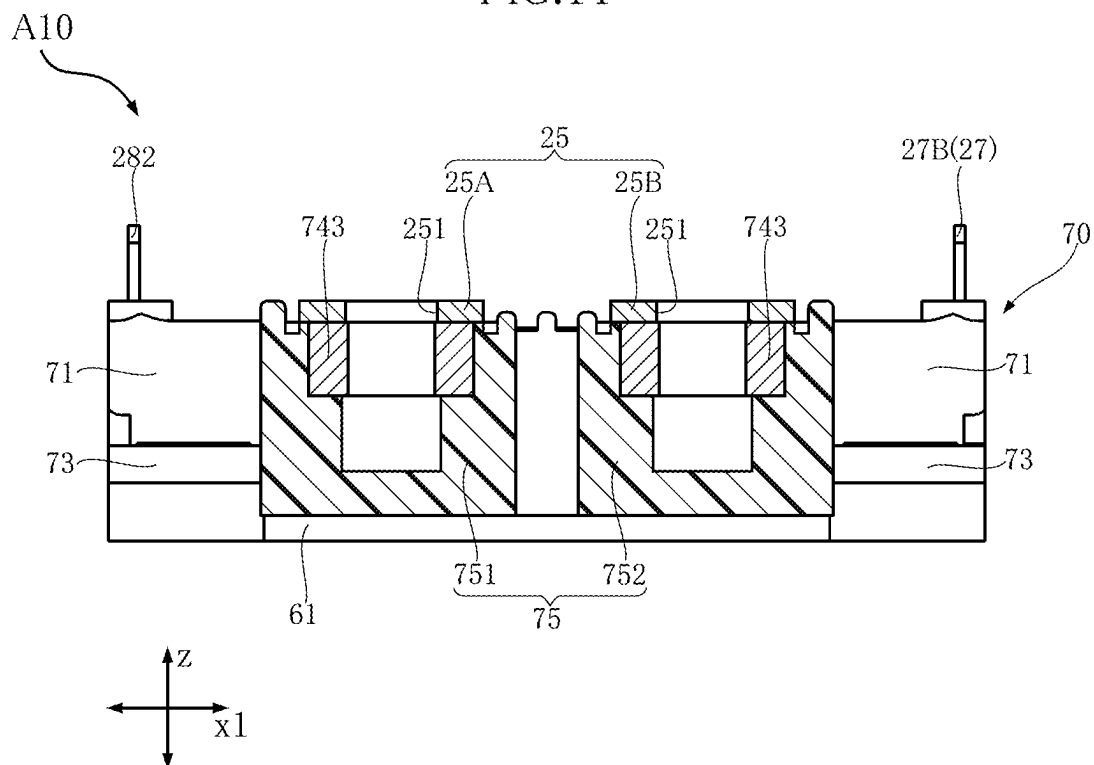


FIG.16

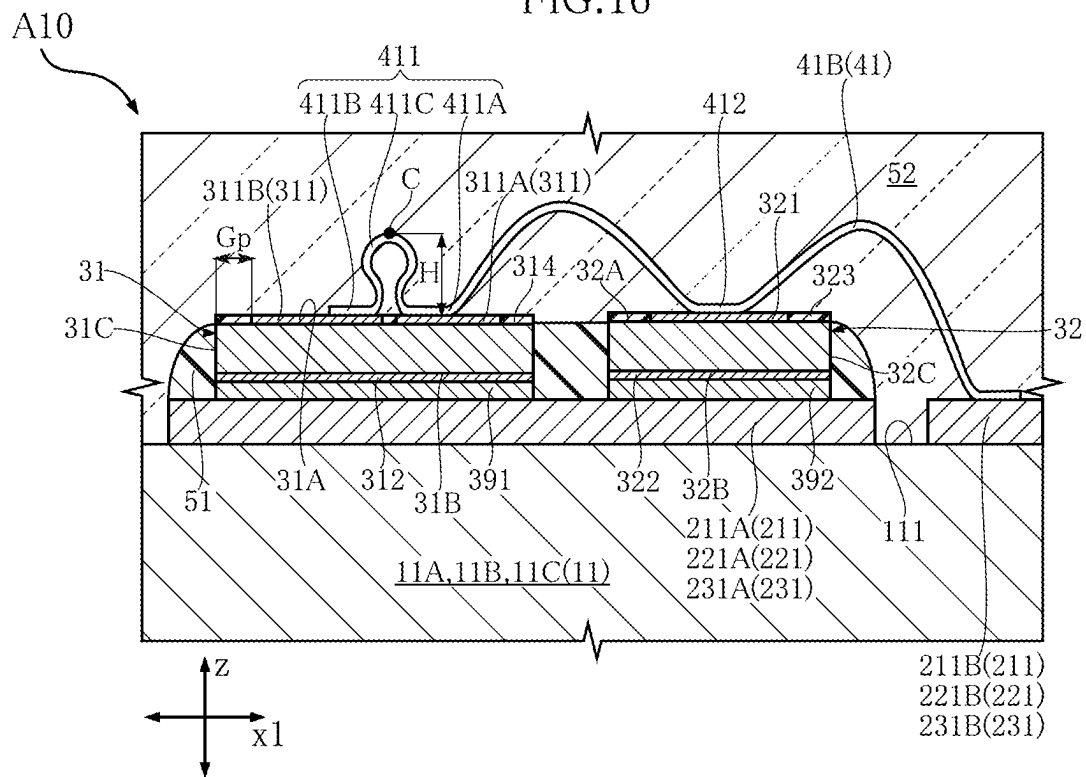


FIG.17

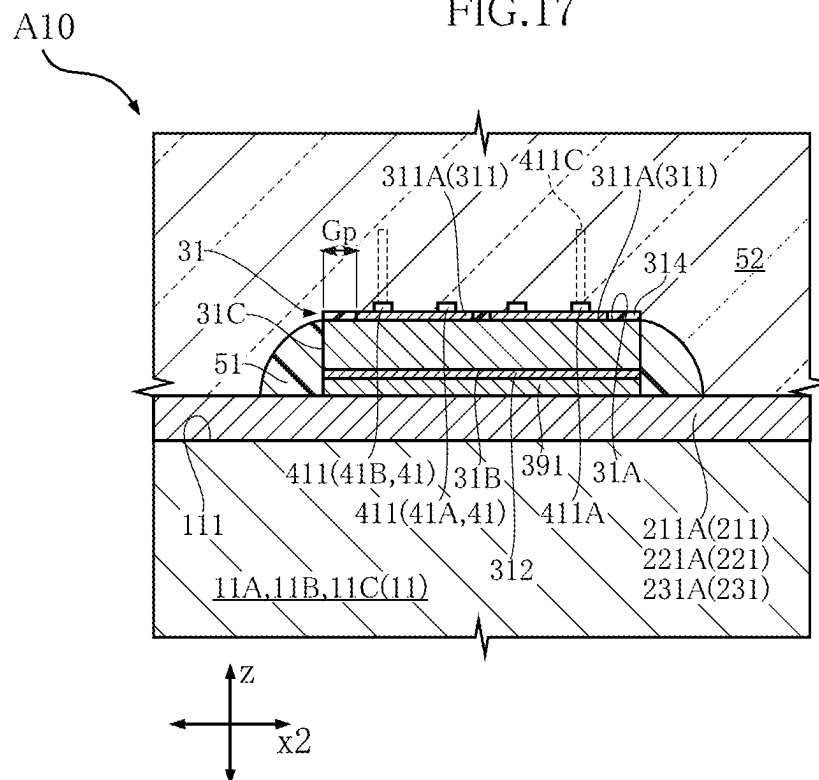


FIG.18

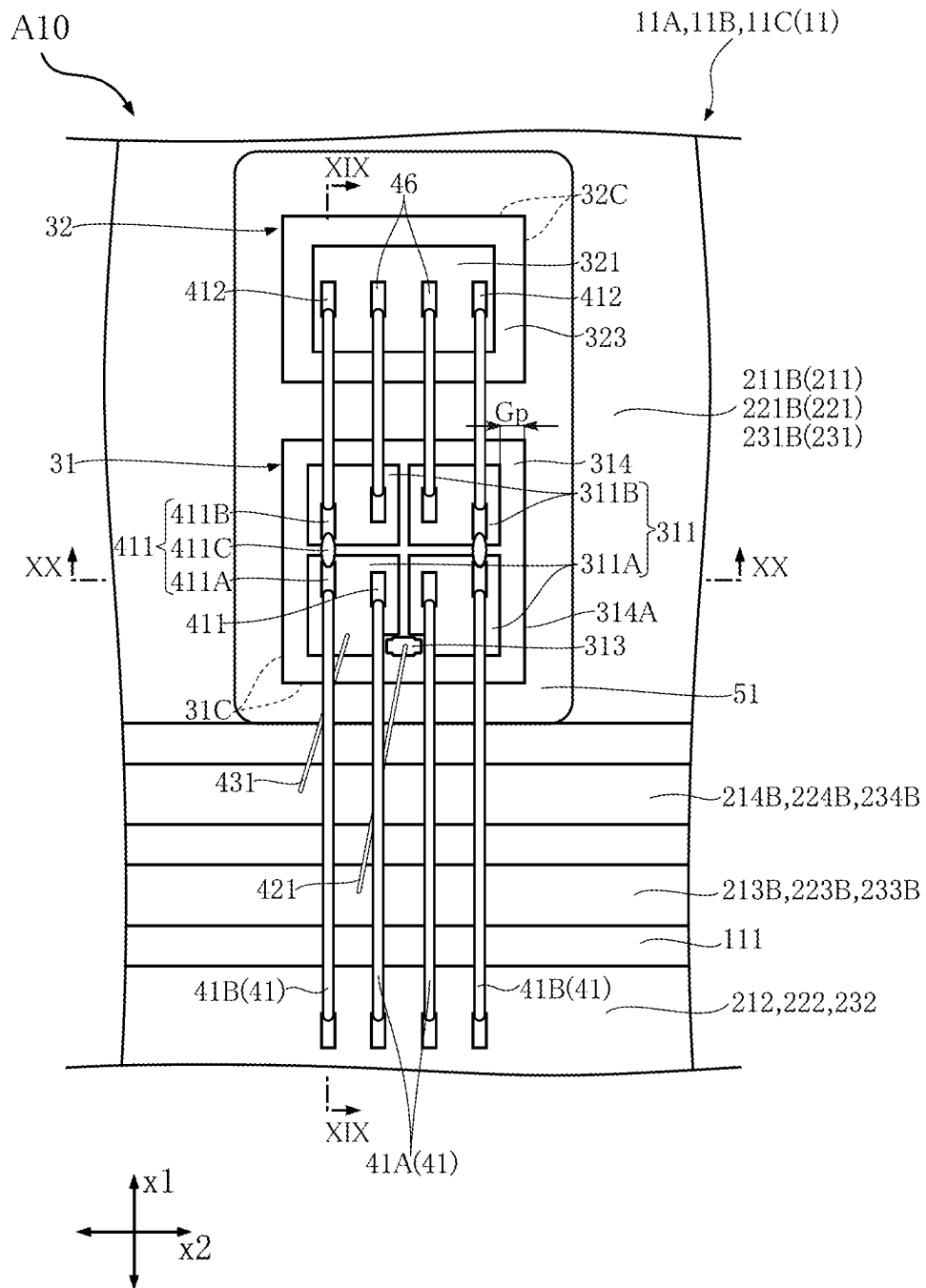


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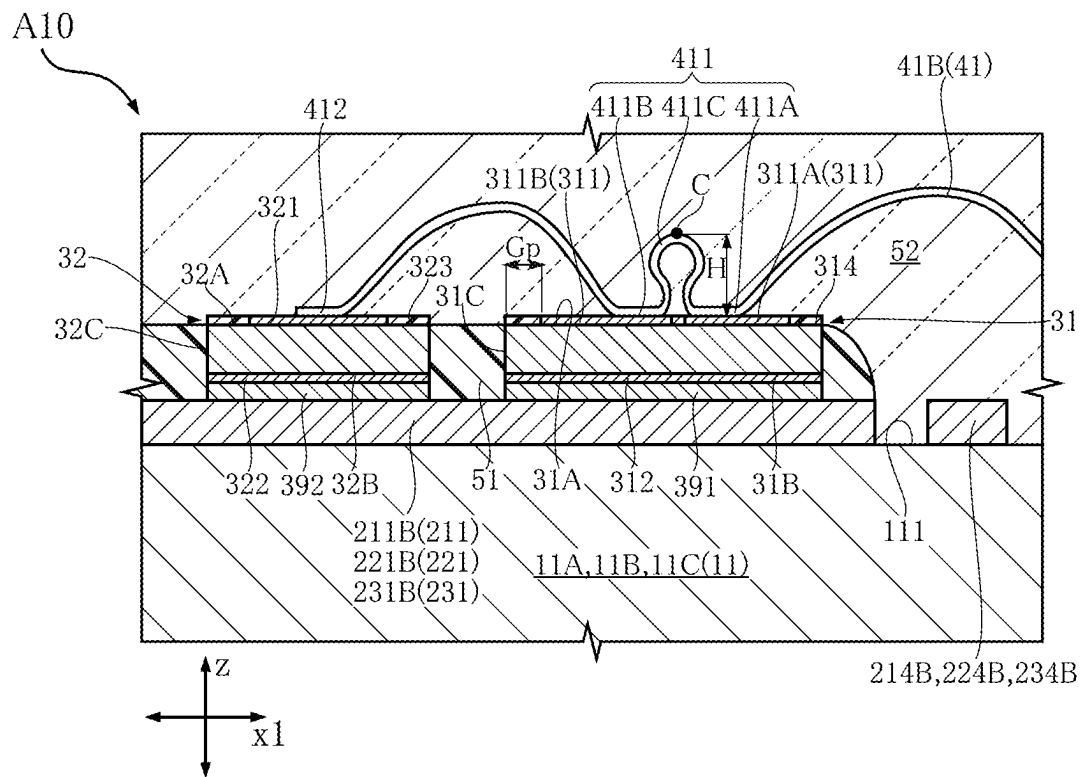


FIG.20

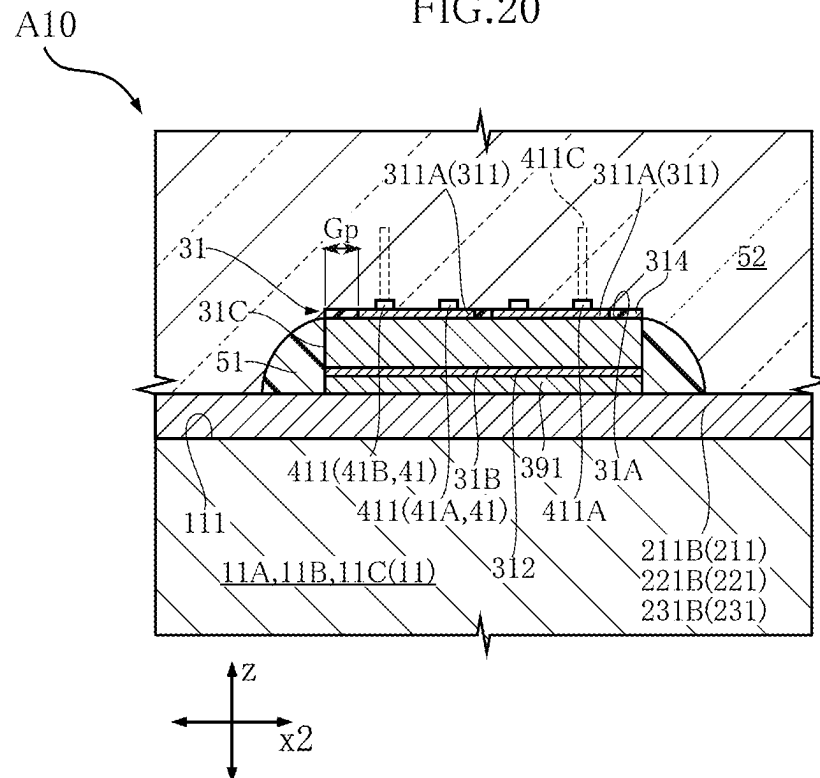


FIG.21

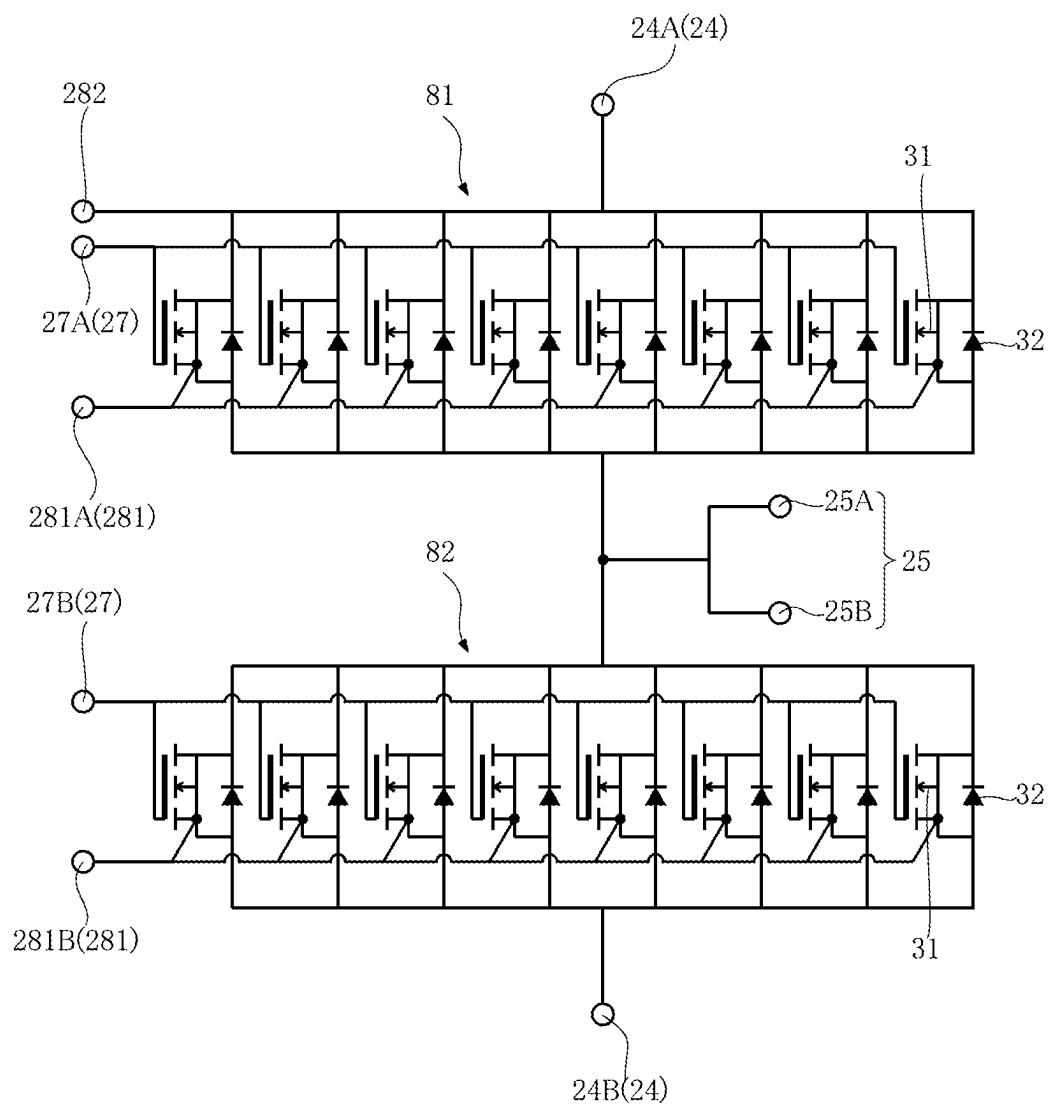


FIG.22

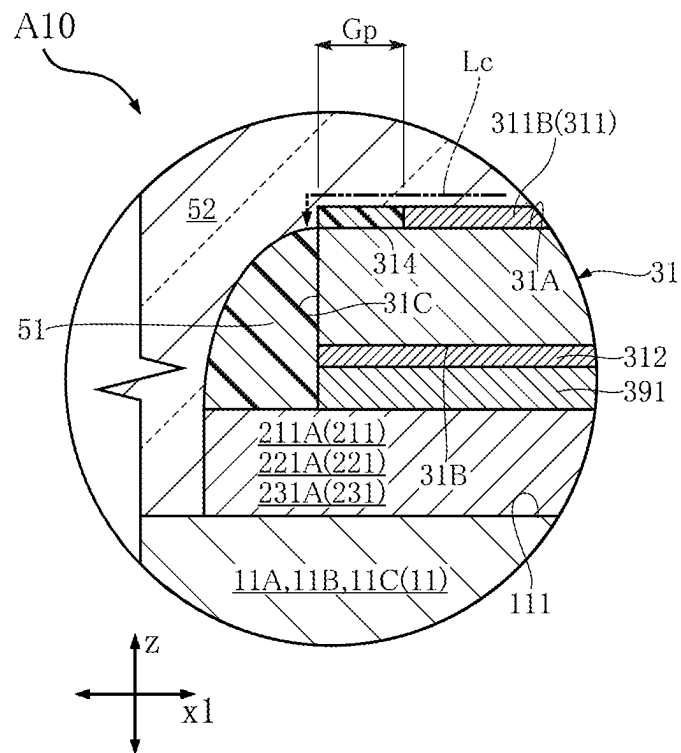


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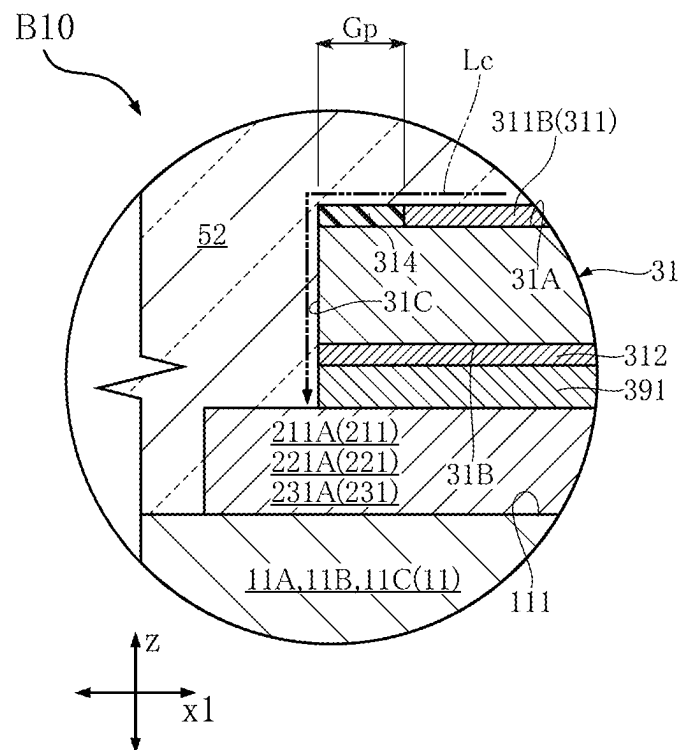


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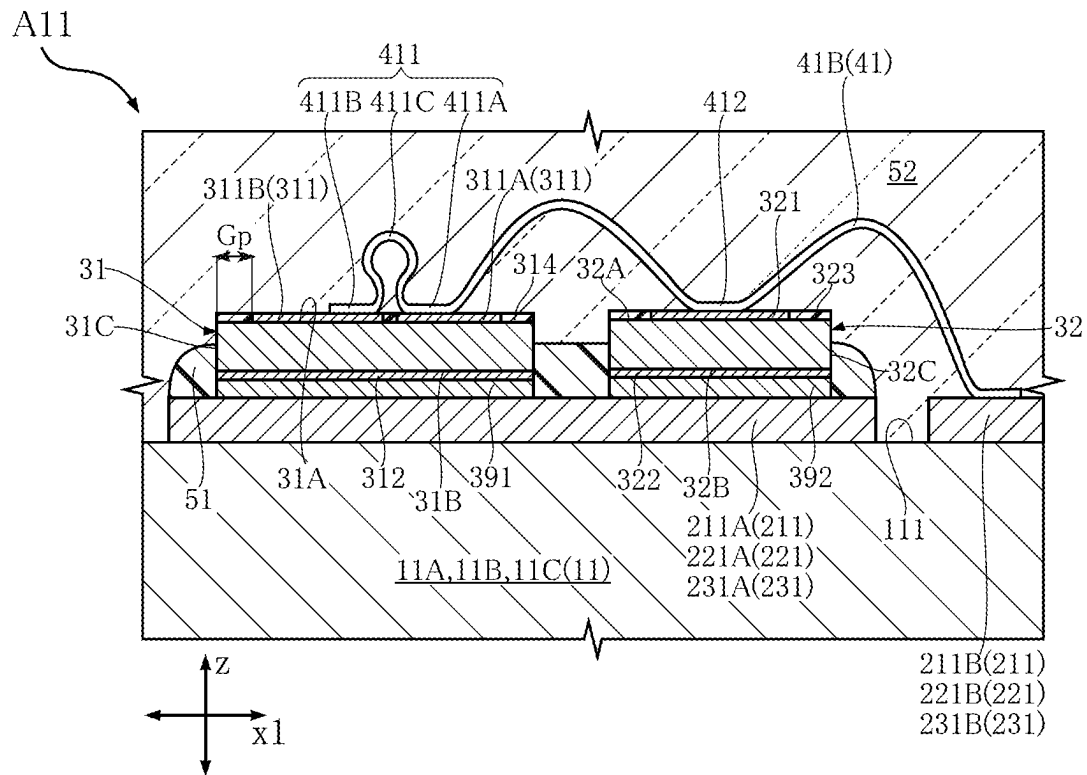
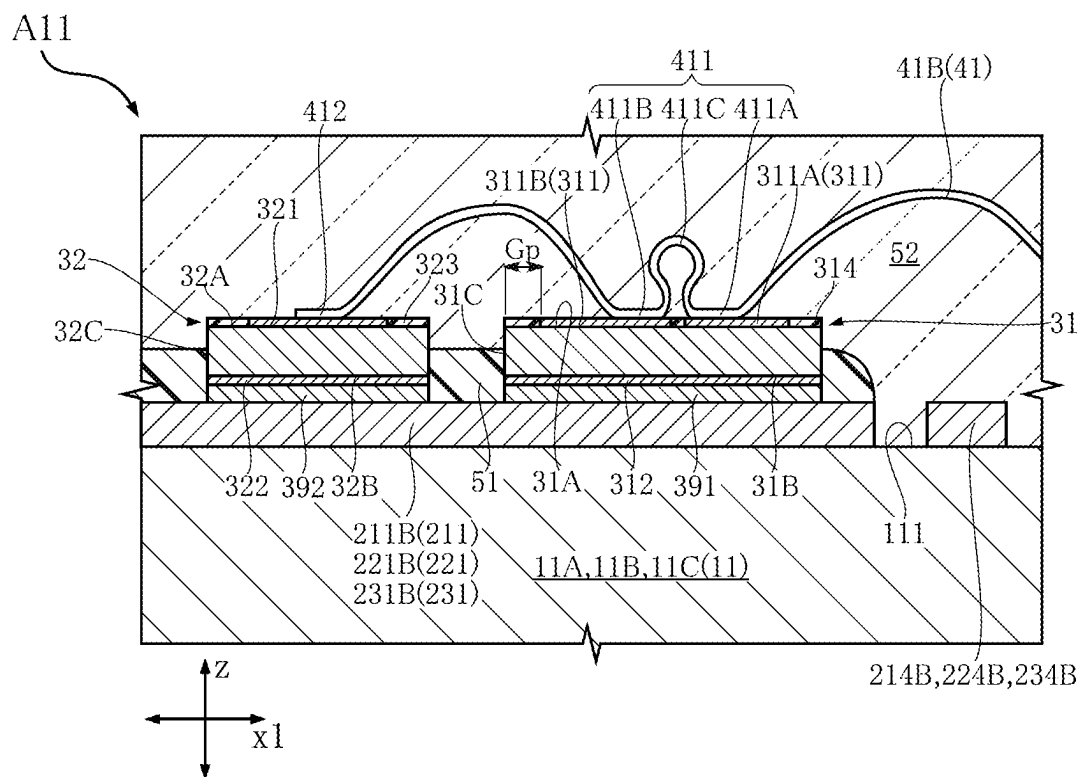


FIG.25



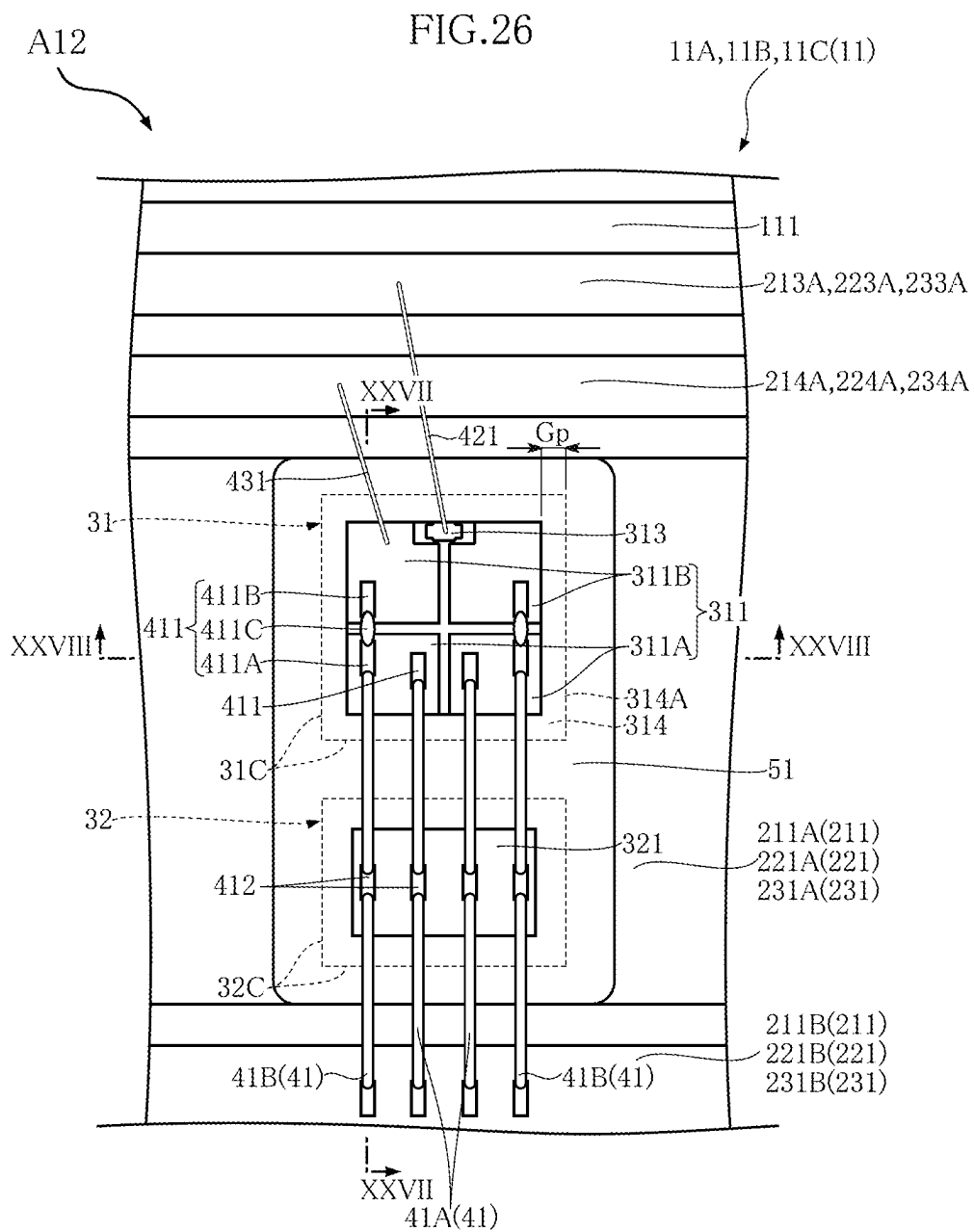


FIG.27

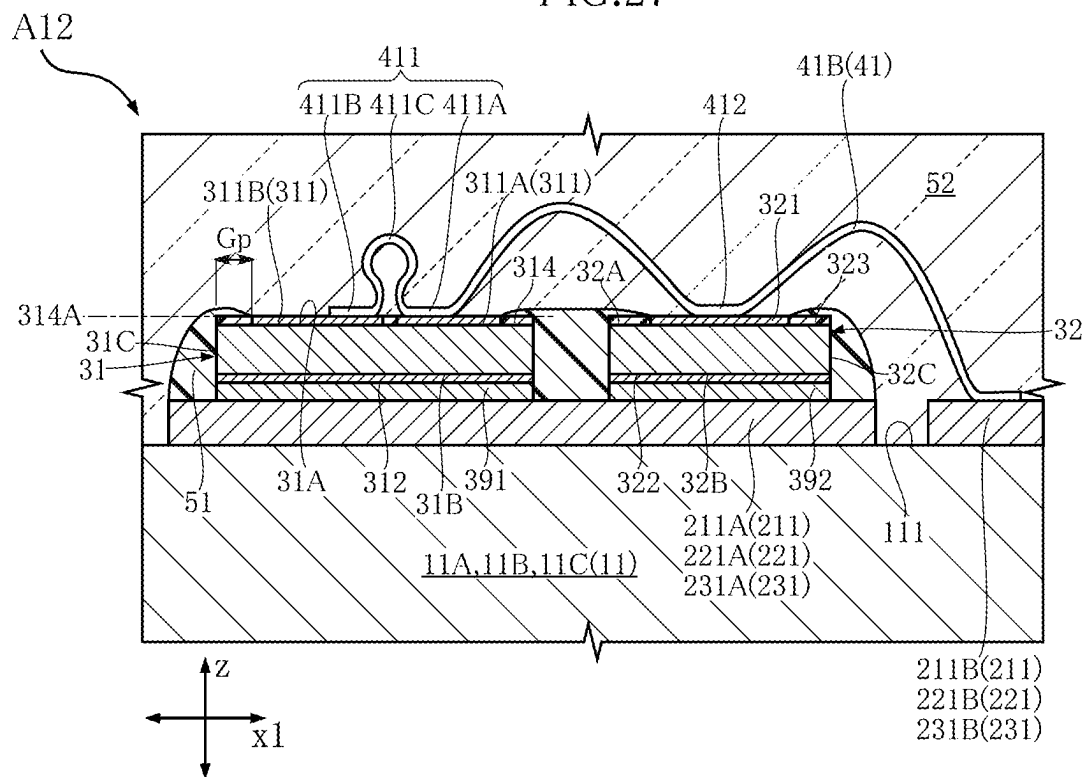


FIG.28

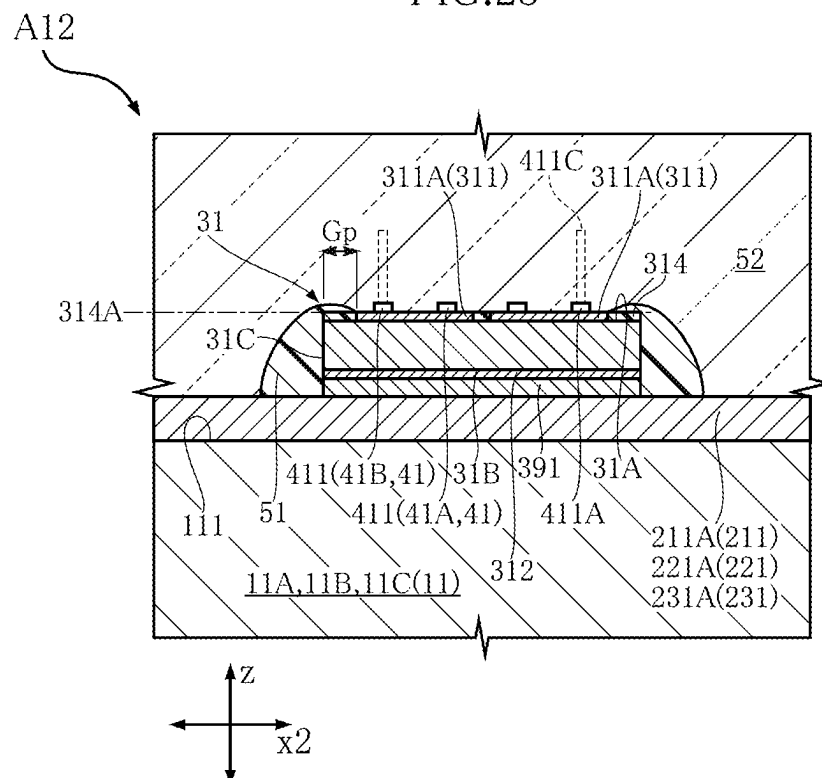


FIG.29

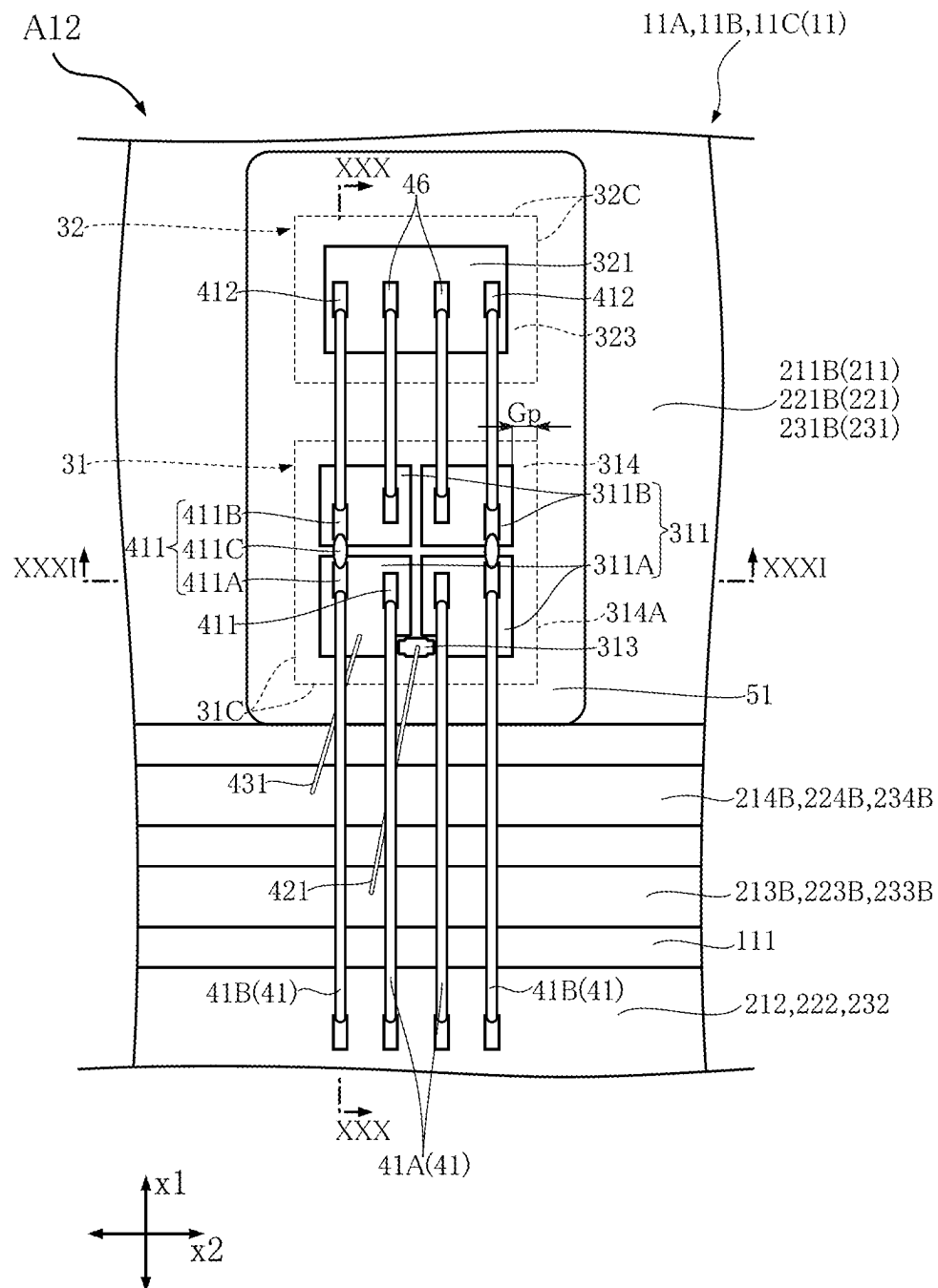


FIG.30

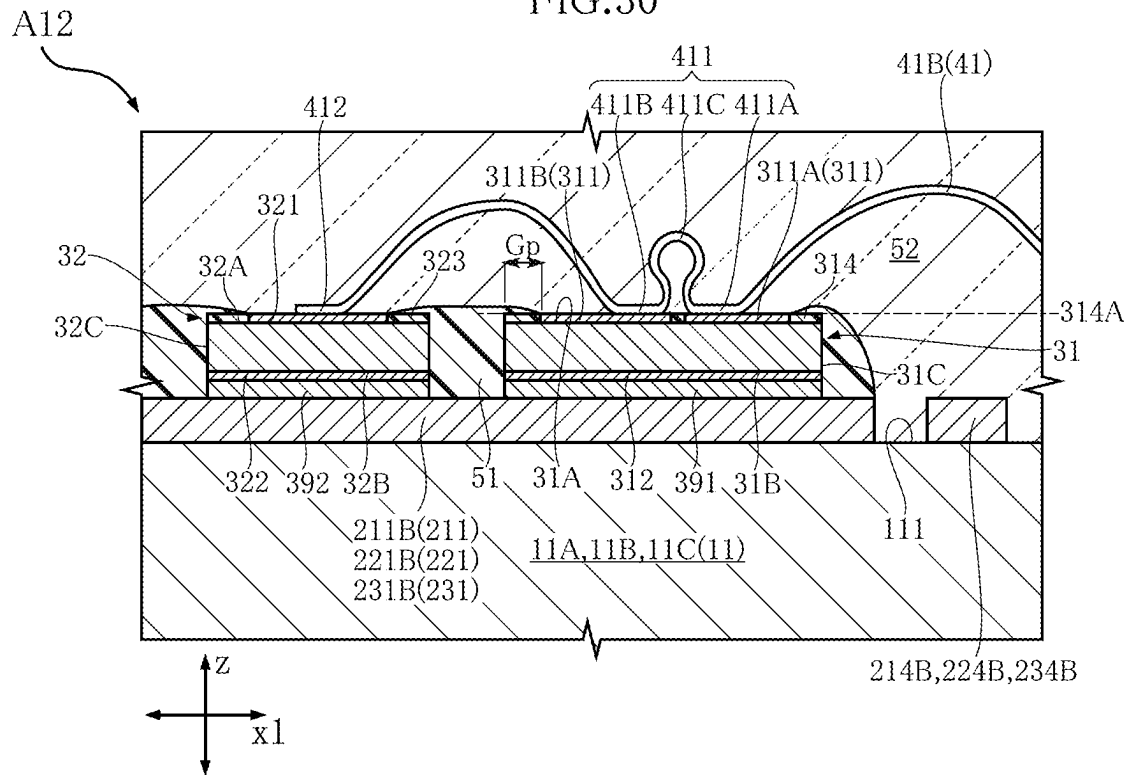


FIG.31

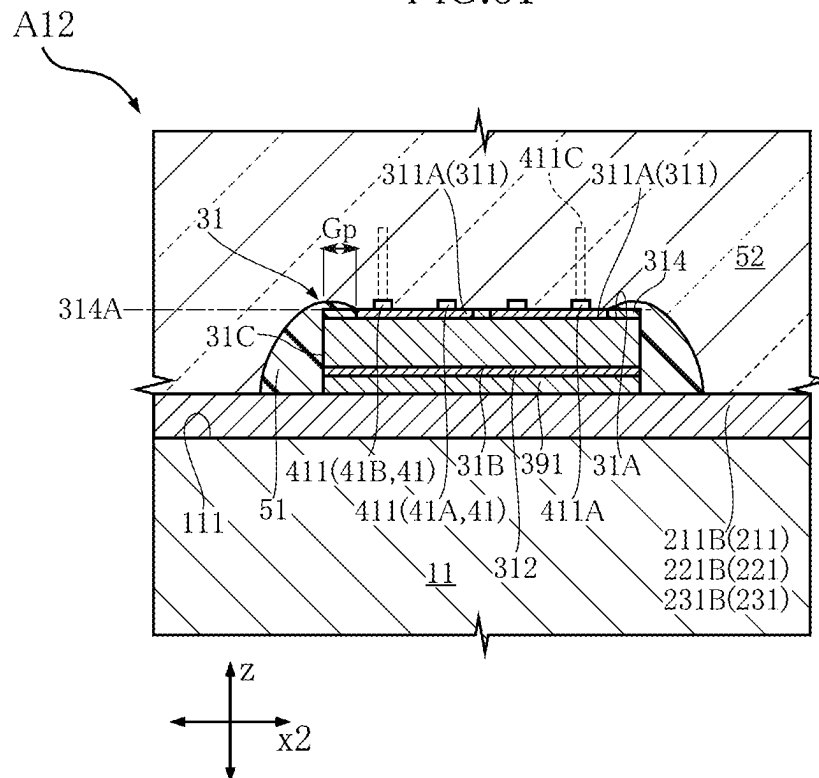


FIG.32

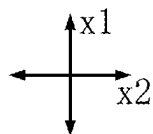
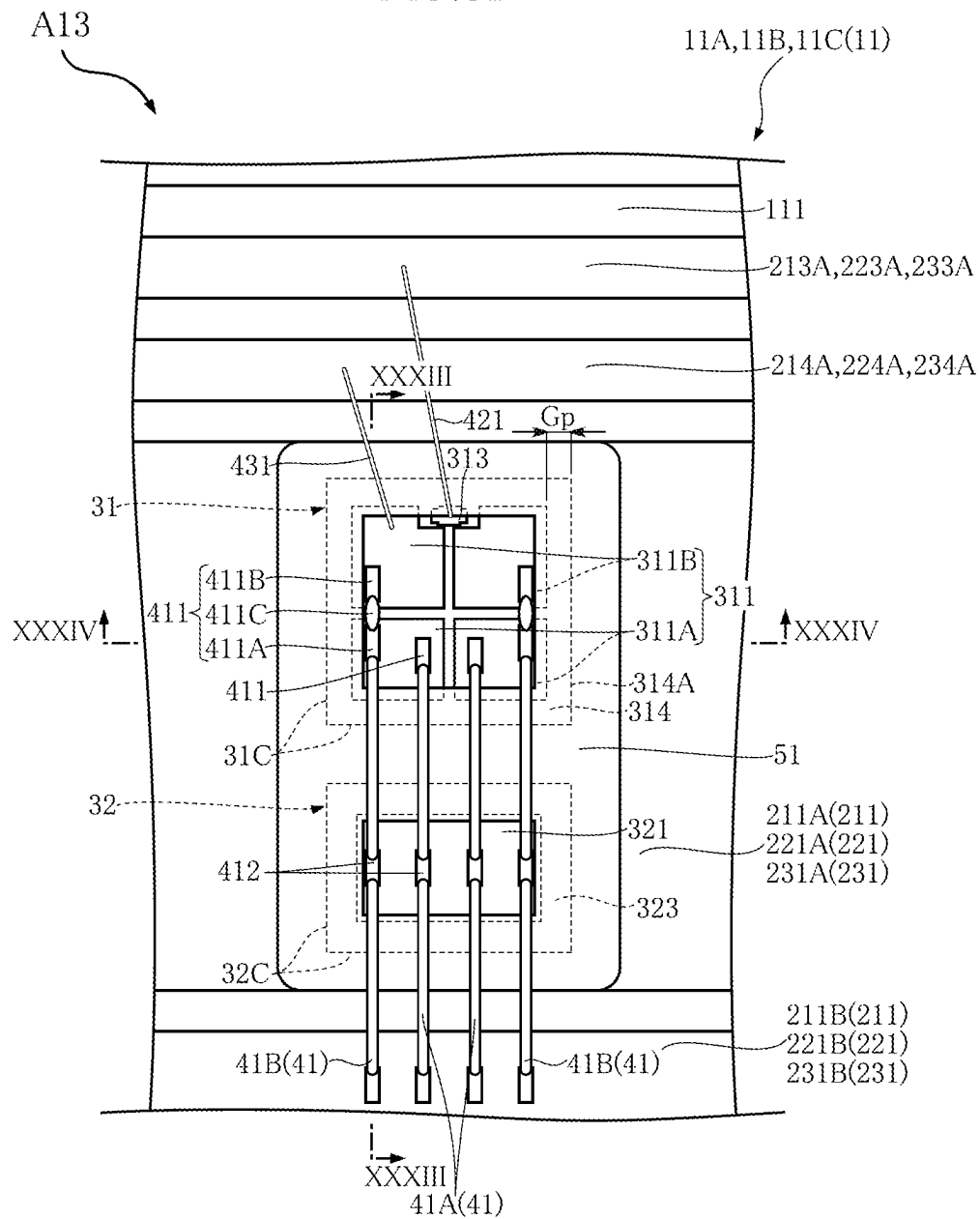
 $\vec{x}_2$

FIG.33

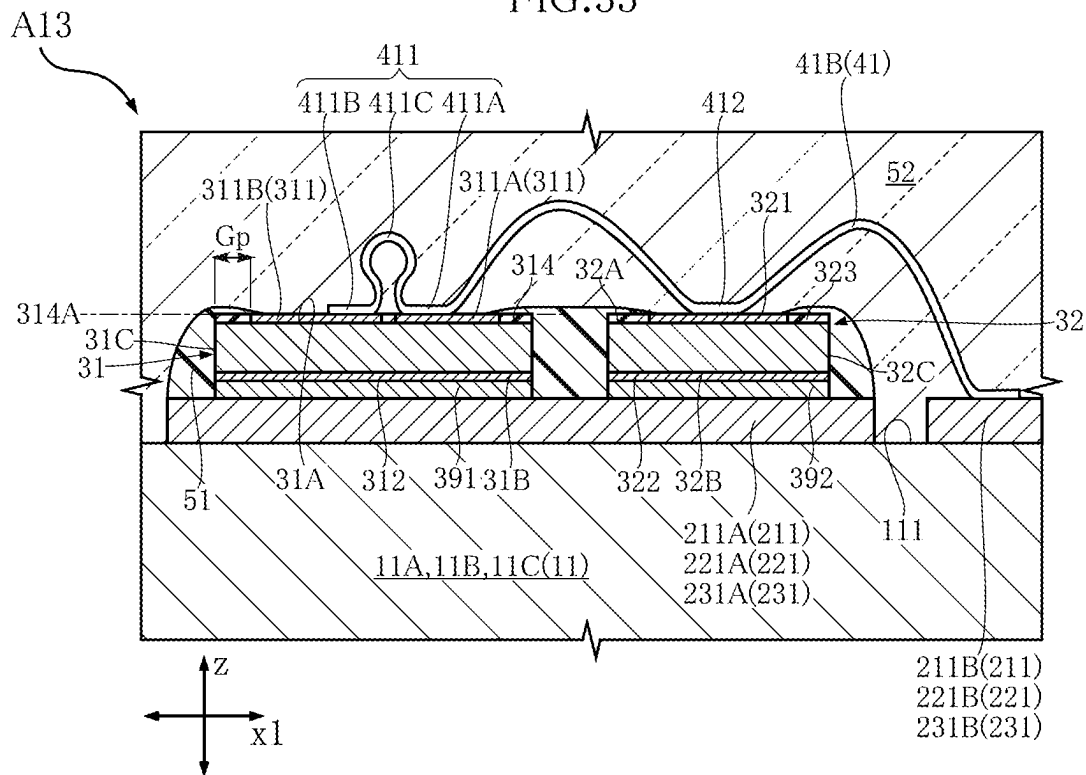


FIG.34

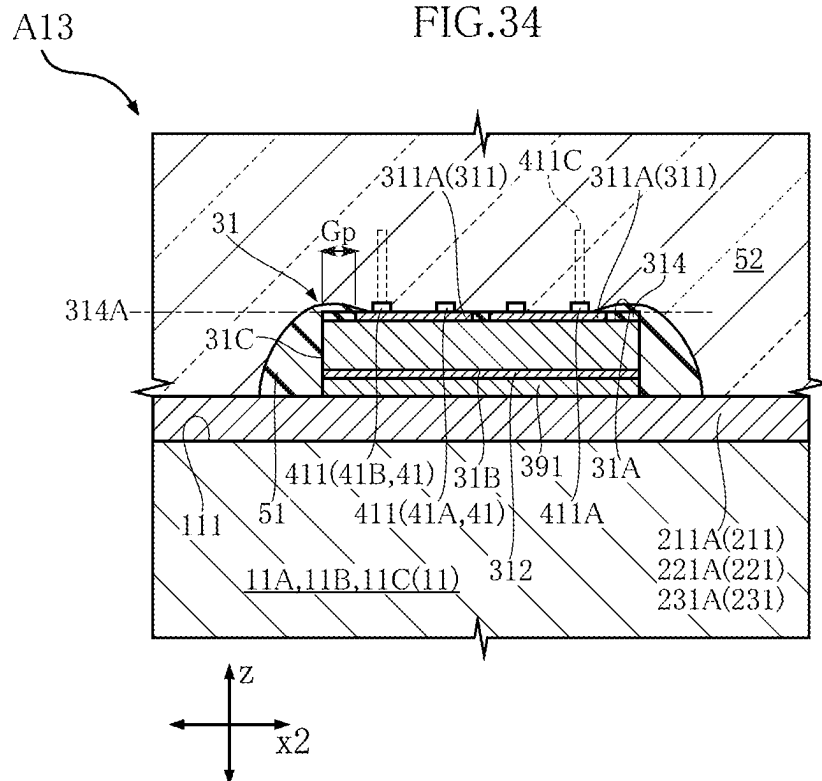


FIG.35

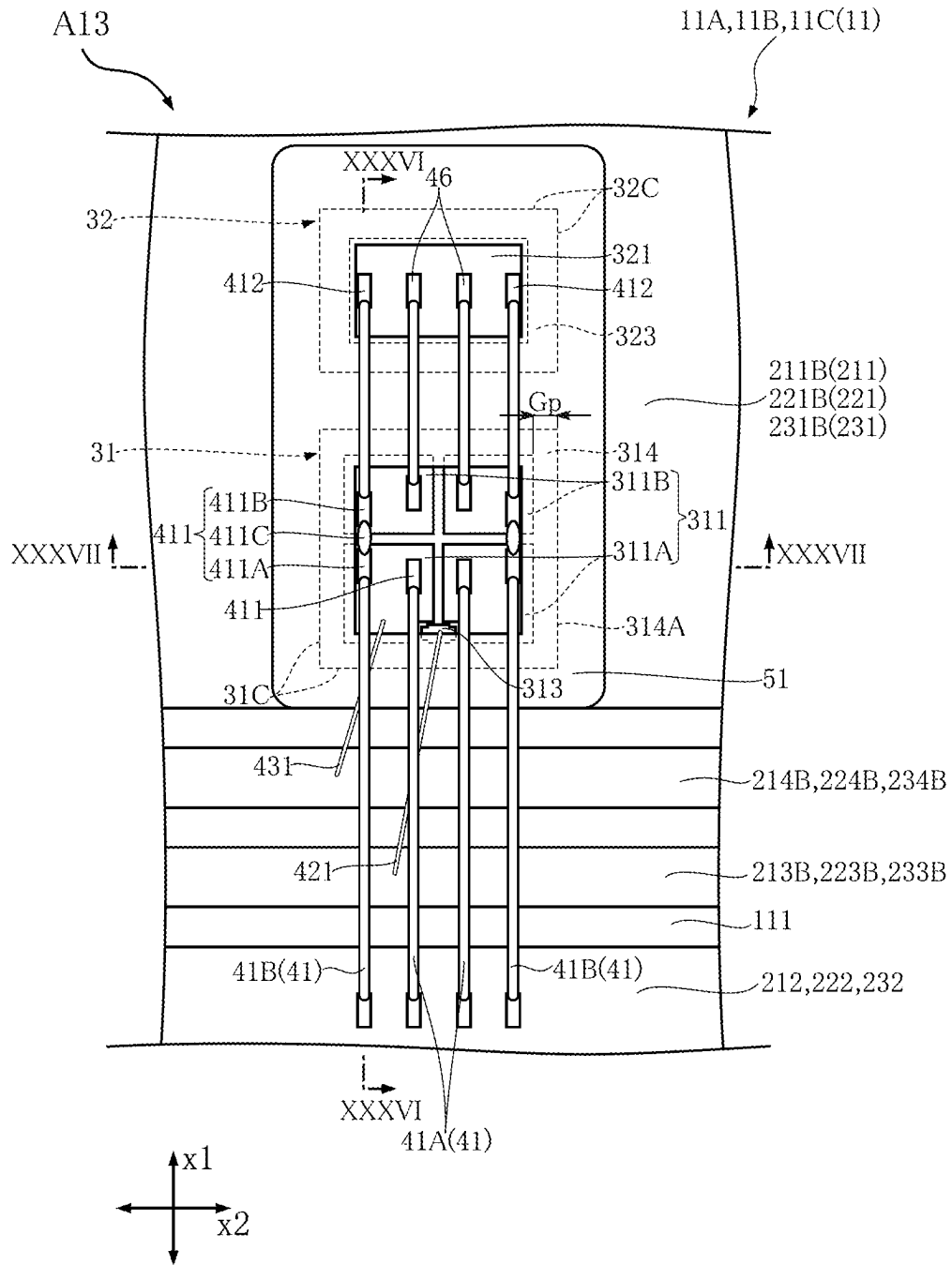


FIG.36

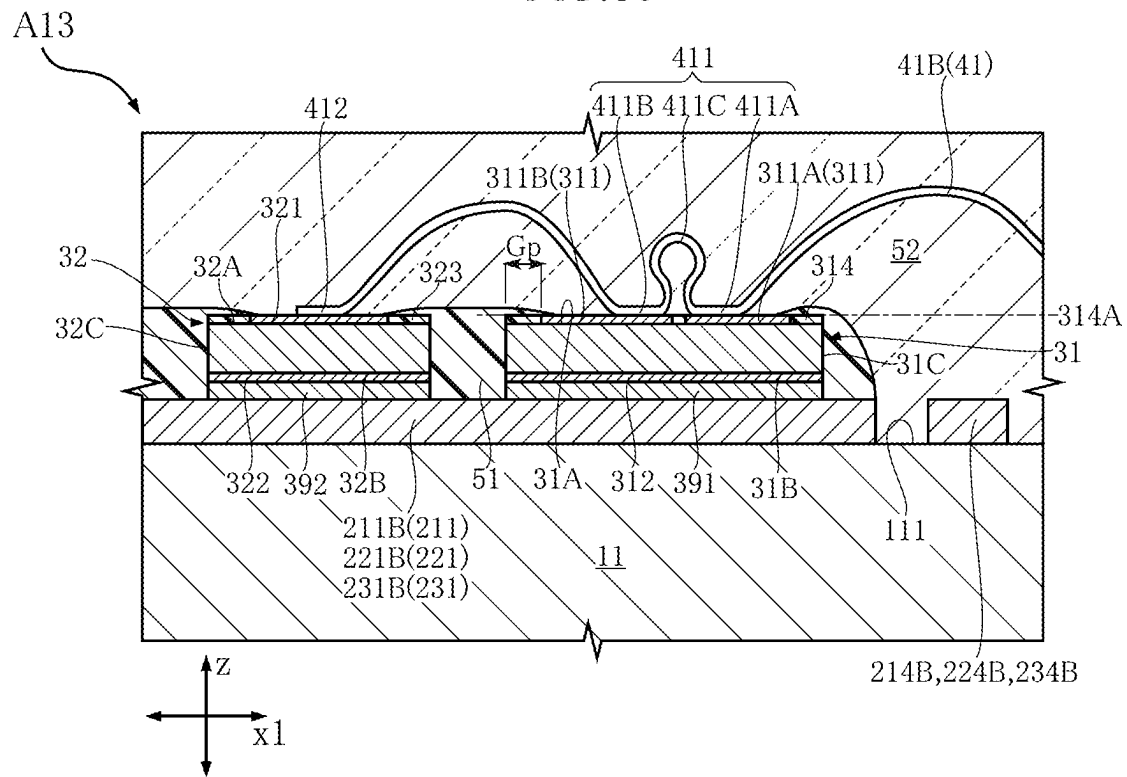


FIG.37

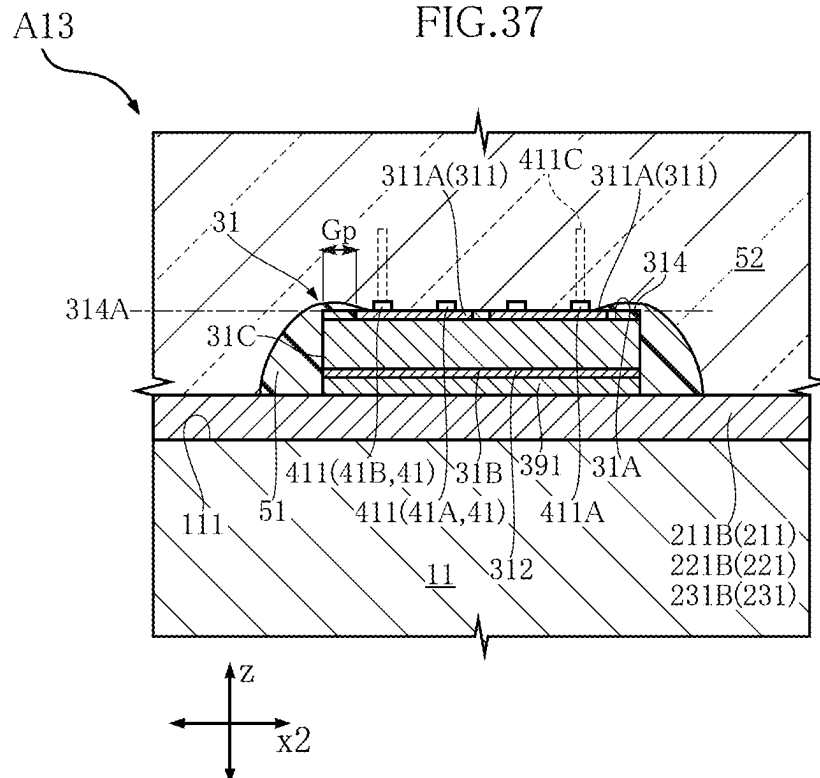


FIG.38

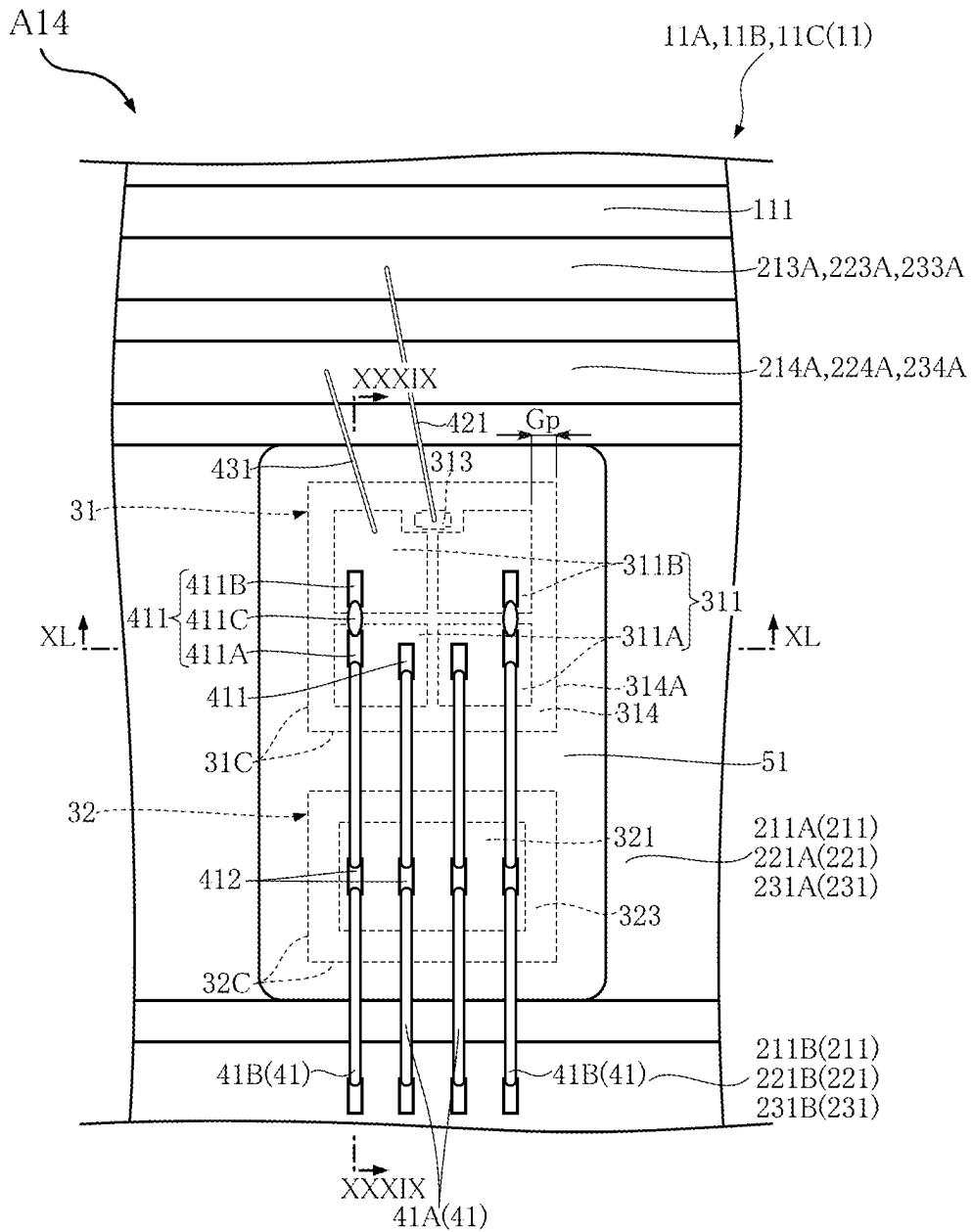


FIG.39

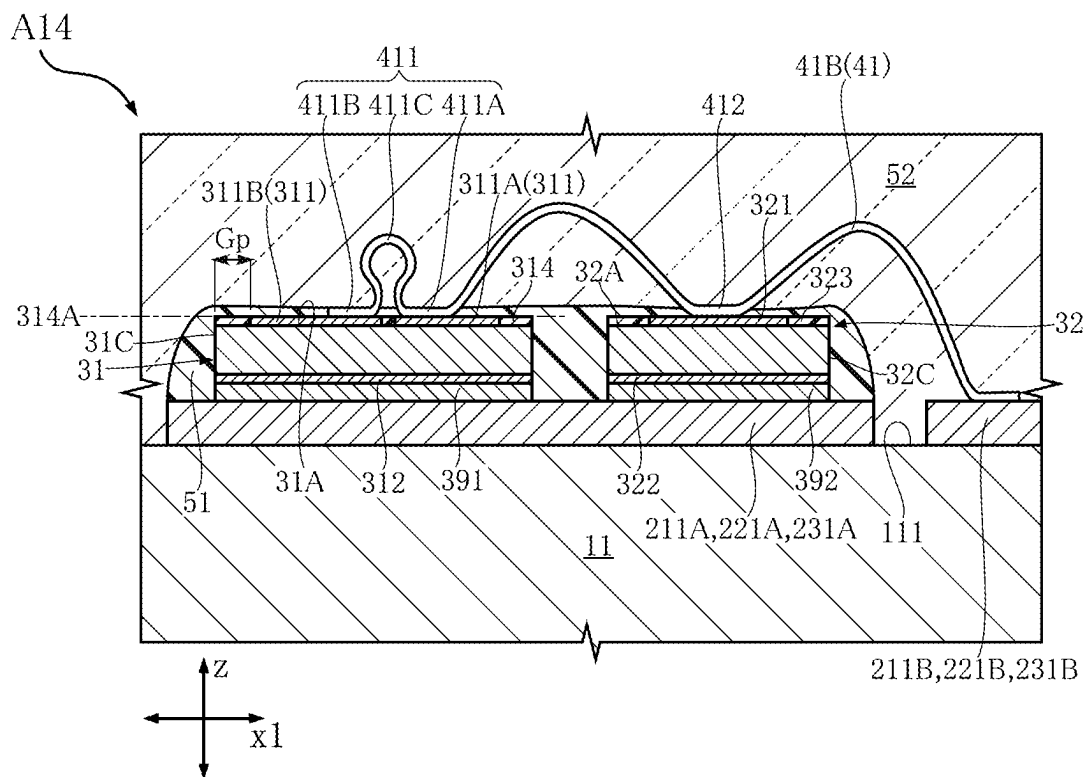


FIG.40

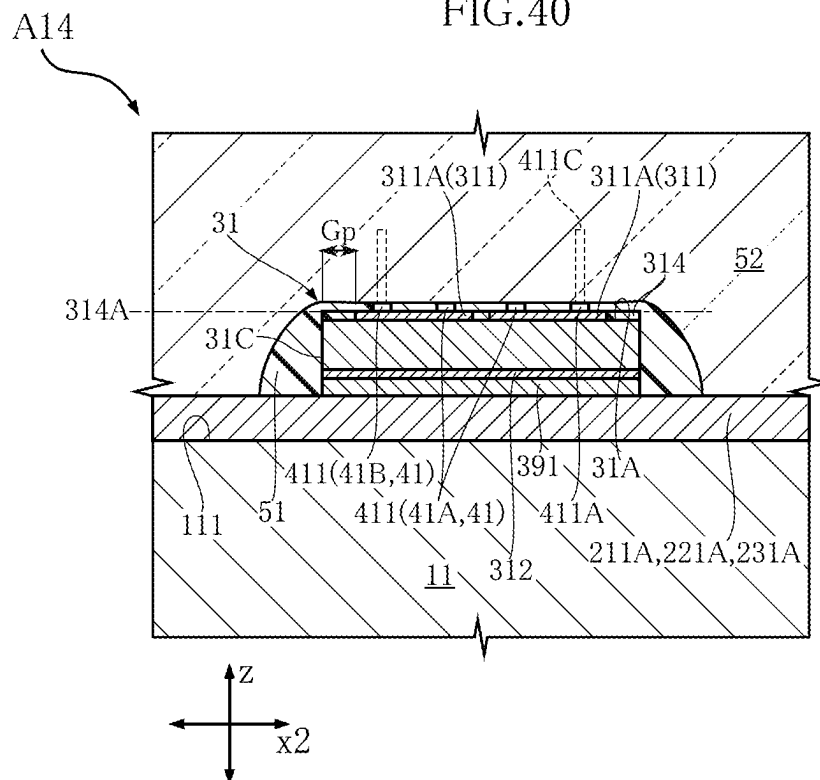


FIG. 41

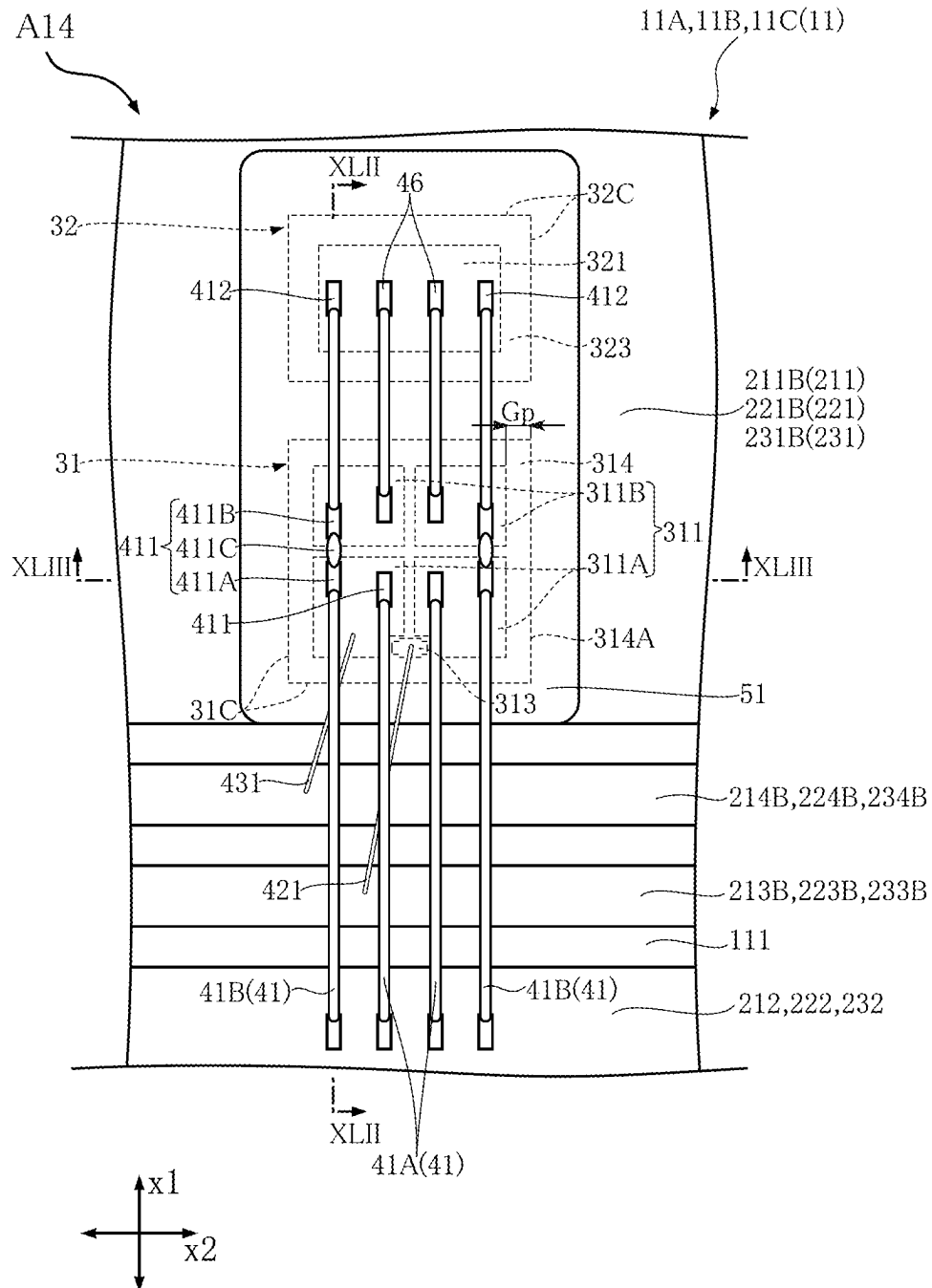


FIG.42

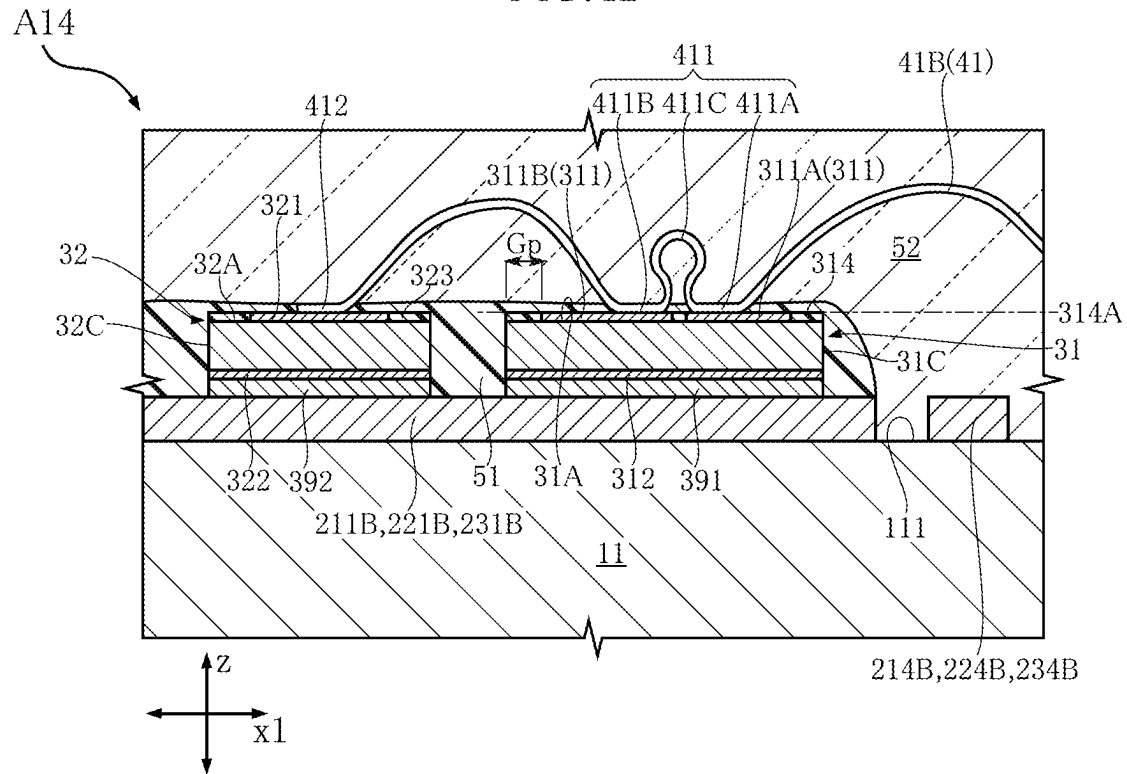
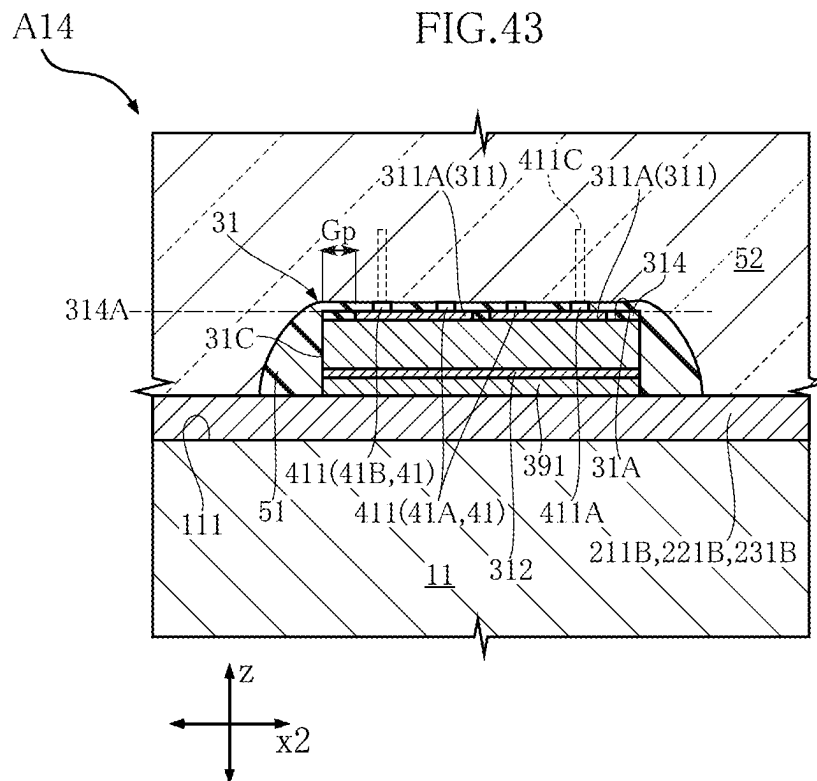


FIG.43



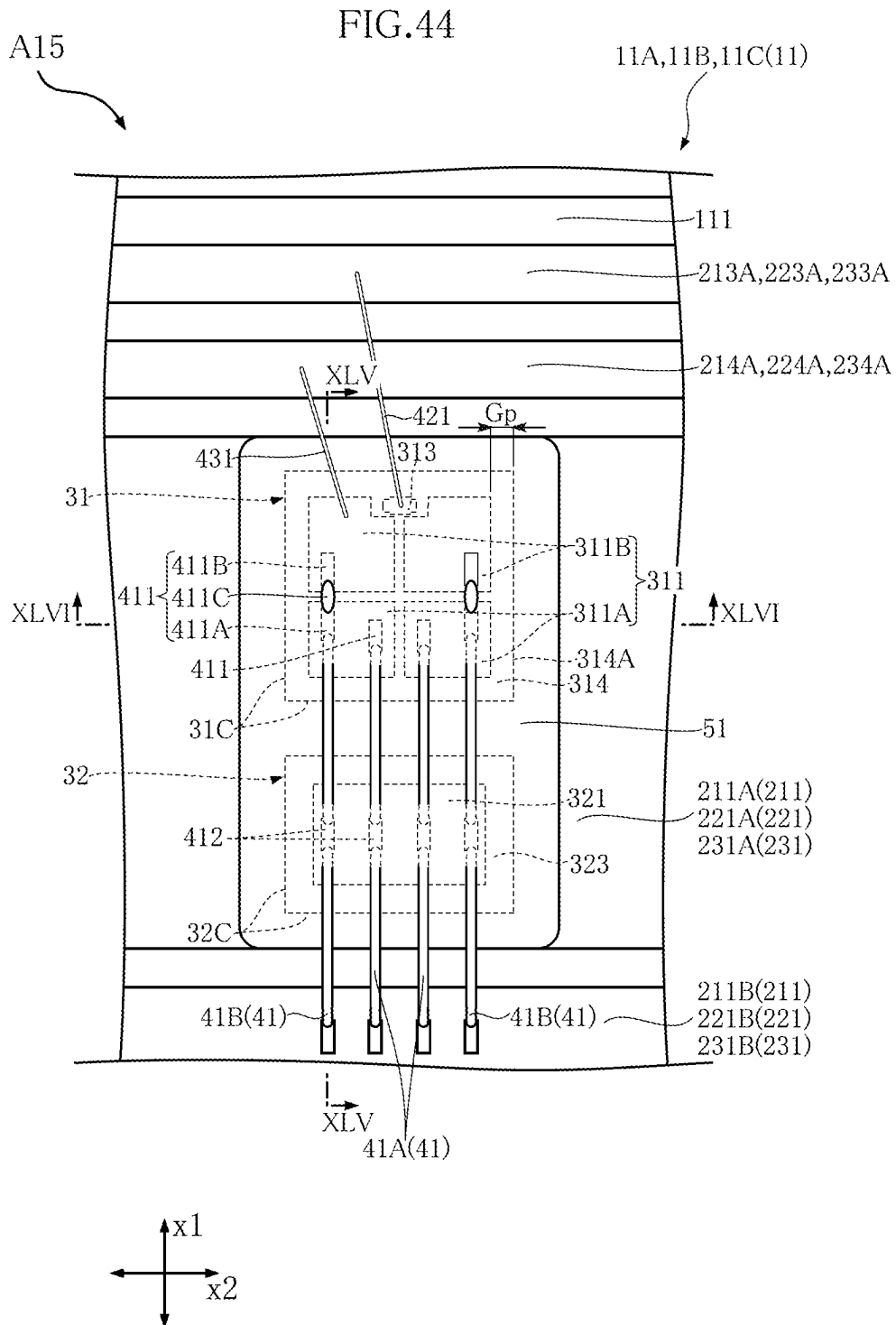


FIG.45

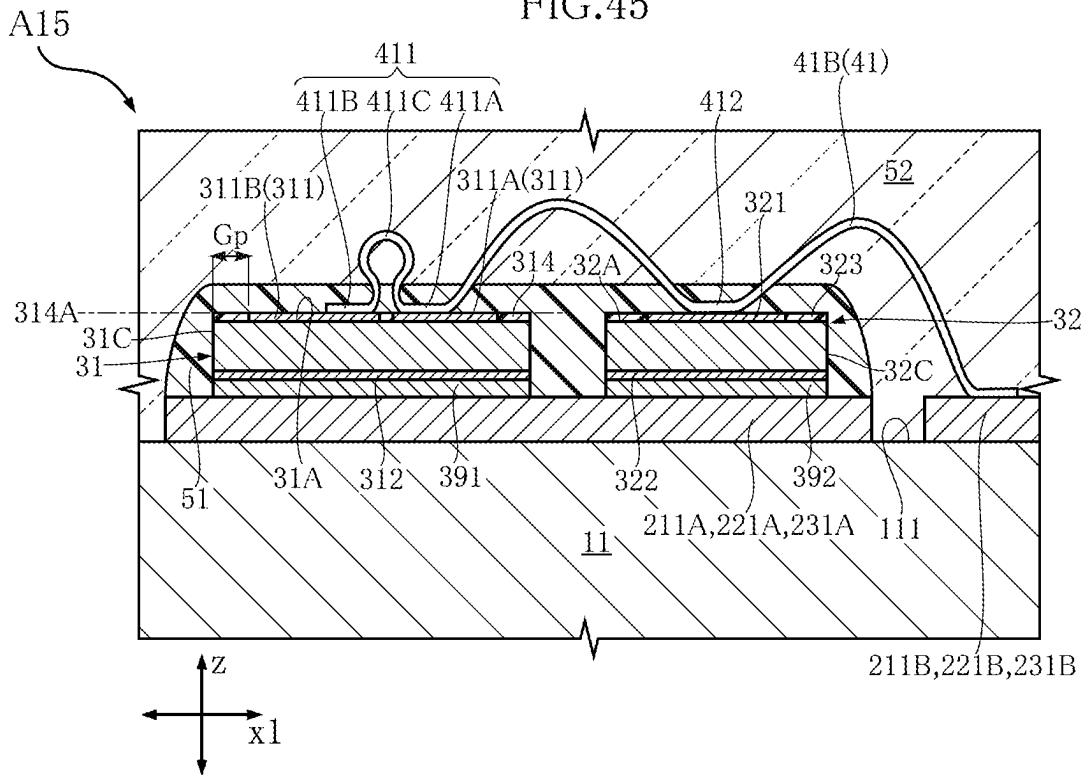


FIG.46

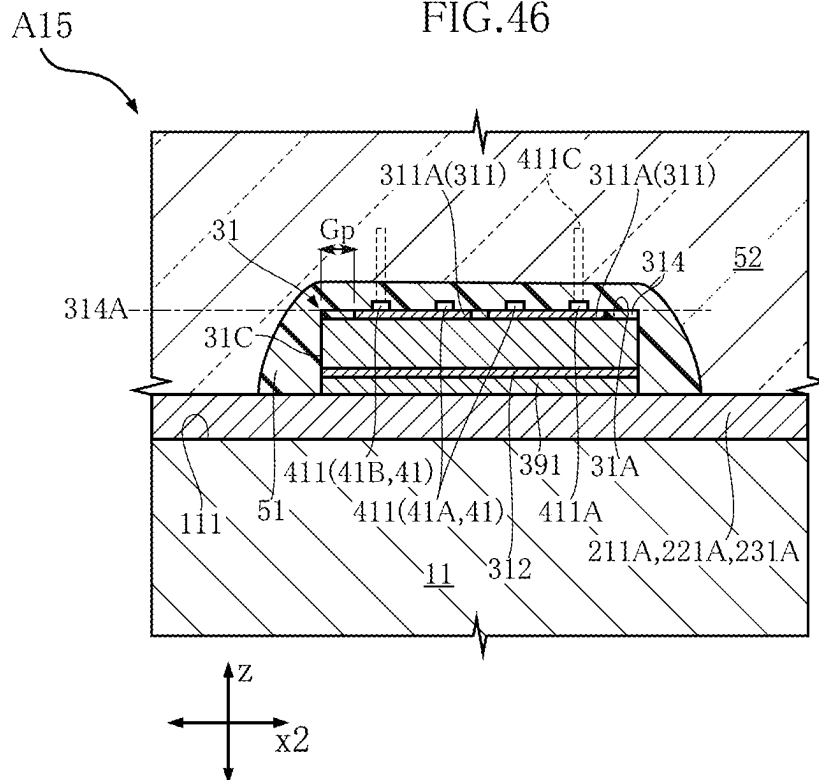


FIG. 47

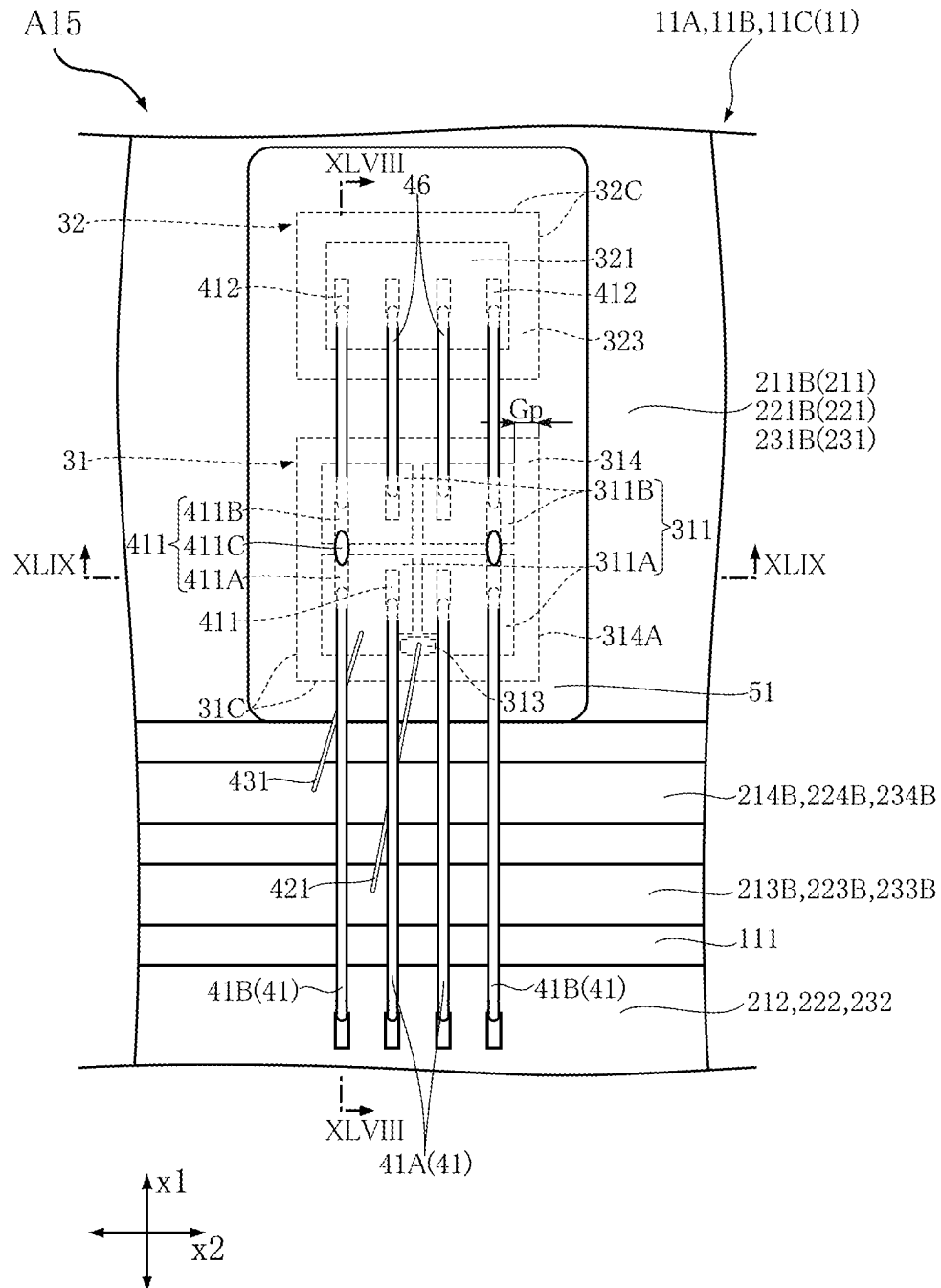


FIG.50

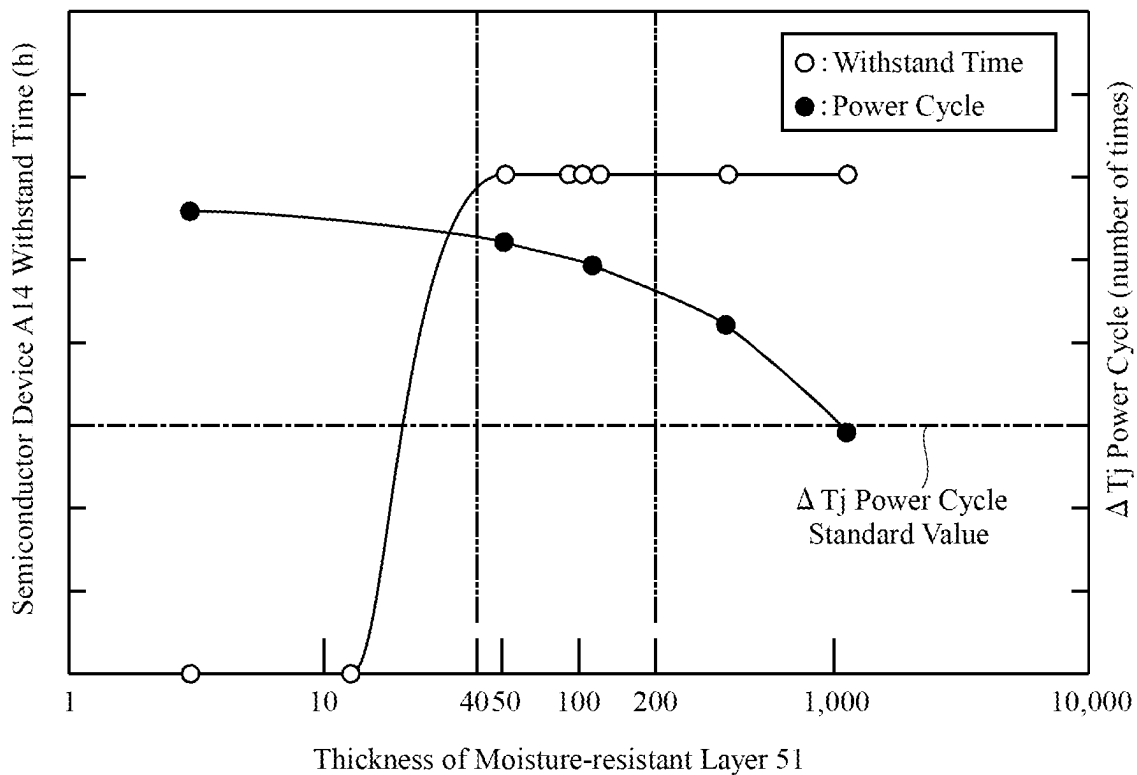


FIG.51

Test Items	Present Invention (A14)	Comparative Example (B10)
H3TRB Test Device Withstand Time (h)	1000 or more (Acceptable)	10~500 (Non-acceptable)
Insulation Resistance Reduction Rate of Sealing Resin 52 During H3TRB Test (unit: %)	20	84

FIG.52

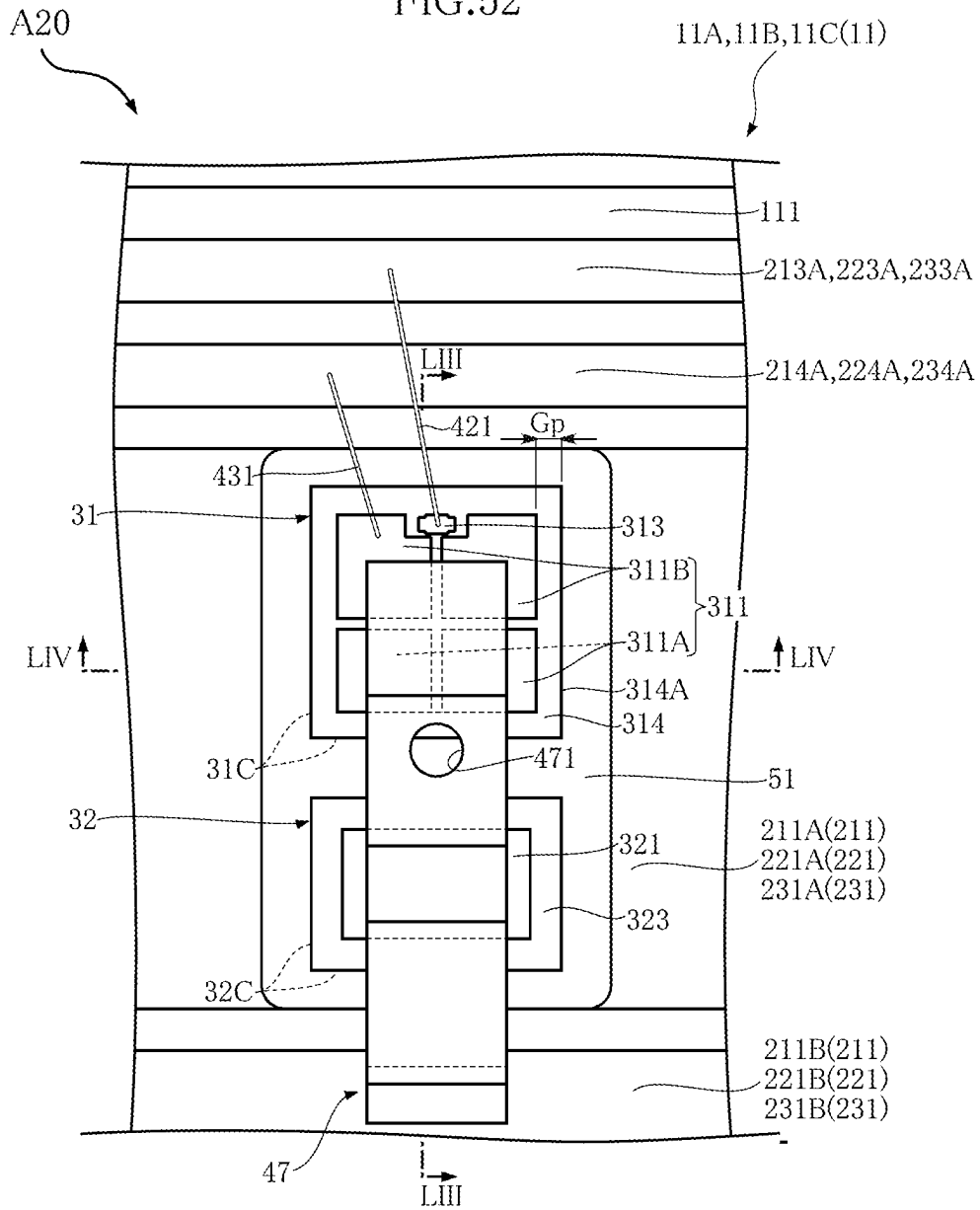


FIG.56

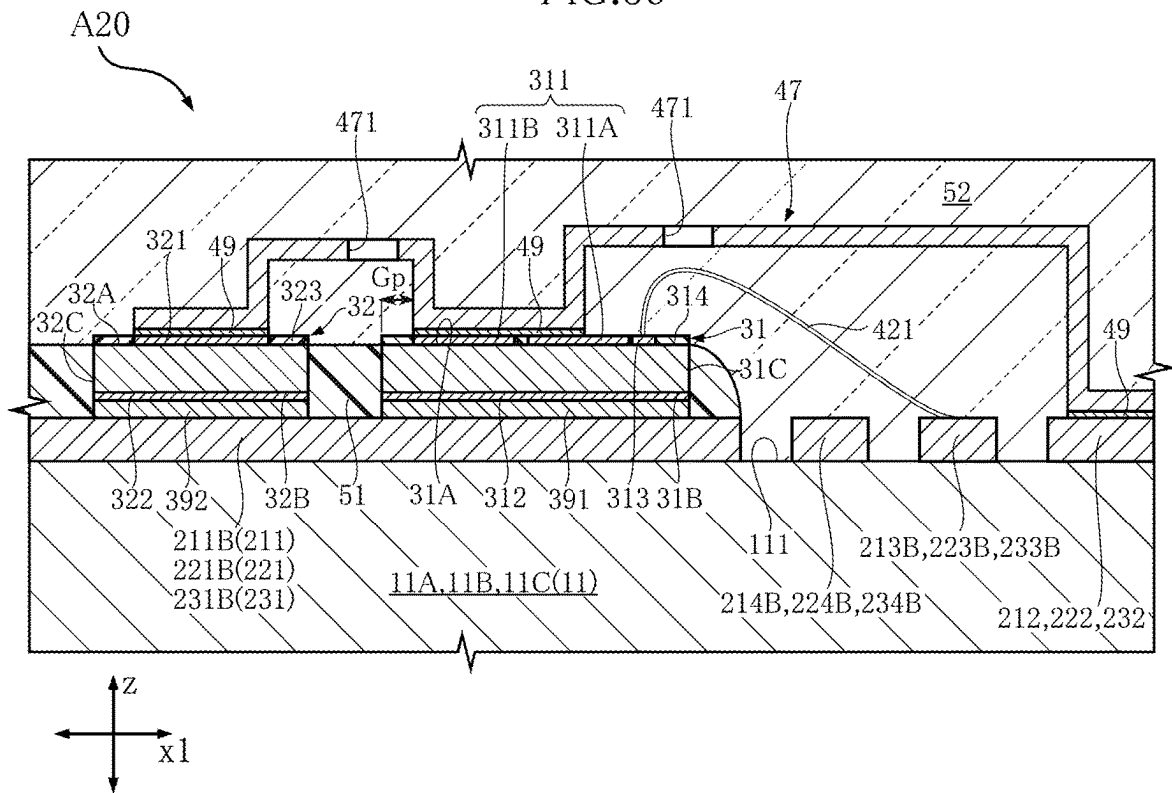
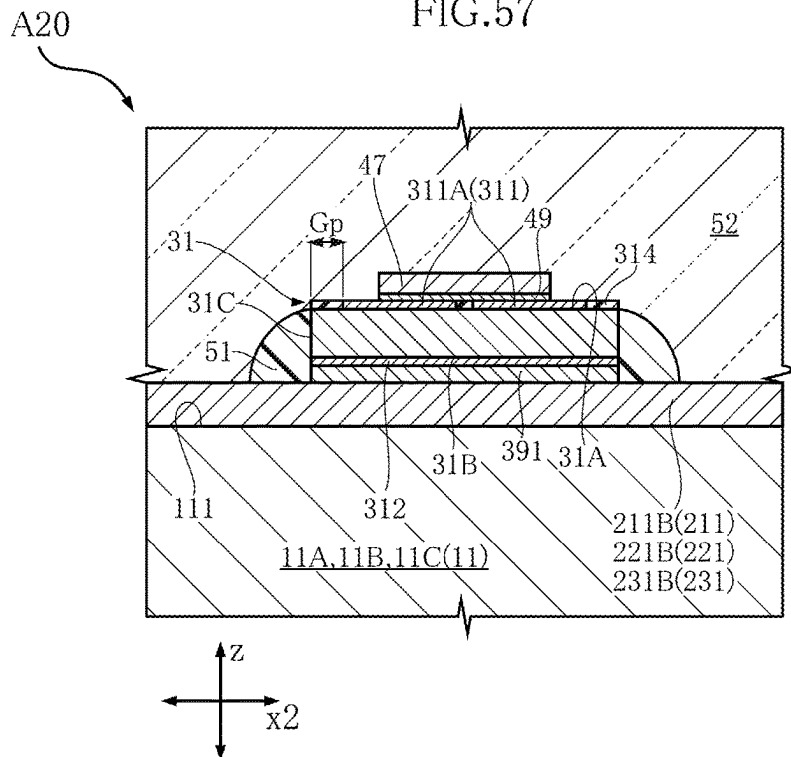


FIG.57



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SEMICONDUCTOR DEVICE**TECHNICAL FIELD**

The present disclosure relates to a semiconductor device 5 provided with a plurality of switching elements.

BACKGROUND ART

A semiconductor device having a plurality of switching 10 elements such as MOSFET which are electrically-coupled is known. Such a semiconductor device may include a case made of a synthetic resin and a wiring board supported by the case. The switching elements are electrically connected to the wiring board. The case and the wiring board surrounds 15 a space, which may be filled with a sealing resin such as silicone gel. The switching elements are covered with the sealing resin.

In recent years, semiconductor devices with relatively 20 high rating voltages are increasingly demanded in areas with tropical climate near/on the equator. In tropical areas, semiconductor devices are placed in a high temperature and high humidity environment. For a semiconductor device to operate stably in such an environment, it is desirable that the semiconductor device passes the H3TRB (High Humidity 25 High Temperature Reverse Bias) test. The H3TRB test estimates the withstand time (unit: hours) of a semiconductor device when it is driven at 80% of its rated DC voltage under high temperature and high humidity conditions (temperature: 85° C., humidity: 85%). In the H3TRB test, 30 semiconductor devices with withstand time of 1000 h or more are considered as acceptable. The semiconductor devices accepted by this test are expected to operate stably under high temperature and high humidity conditions.

By performing the H3TRB test on the above-described 35 semiconductor devices, the inventors have found that the withstand time of the devices is highly likely to be short of 1000 h. If moisture enters the sealing resin of a semiconductor device placed under high temperature and high humidity conditions, the dielectric breakdown voltage of the 40 sealing resin will deteriorate, which may allow leak current in the switching elements. If the leak current reaches the wiring board, at least one of the switching elements may be destroyed, and consequently, the withstand time of the device will be shortened. With a higher rating voltage, a semiconductor device tends to have a shorter withstand 45 time. Thus, in order to operate stably under high temperature and high humidity conditions, a semiconductor device may need to pass, as a criterion, the H3TRB test for the desired rating voltage.

SUMMARY OF THE INVENTION

In light of the above circumstances, the present disclosure 55 aims to provide a semiconductor device capable of operating stably under high temperature and high humidity conditions.

According to the present disclosure, a semiconductor device is provided. The semiconductor device includes a substrate, a mounting layer, switching elements, a moisture-resistant layer and a sealing resin. The substrate has a front surface facing in a thickness direction. The mounting layer is electrically conductive and disposed on the front surface. Each of the switching elements includes an element front surface facing in a same direction in which the front surface faces along the thickness direction, a back surface facing in the opposite direction of the element front surface, and a side surface connected to both of the element front surface and

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the back surface. The switching elements are electrically bonded to the mounting layer with the back surface facing the front surface. The moisture-resistant layer covers at least one of the side surfaces. The sealing resin covers both of the switching elements and the moisture-resistant layer. The moisture-resistant layer is held in contact with both of the mounting layer and the side surface so as to be spanned between the mounting layer and the side surface in the thickness direction.

Other features and advantages of the present disclosure will become apparent from the following detailed description with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a semiconductor device according to a first embodiment of the present disclosure;

FIG. 2 is a plan view of the semiconductor device shown in FIG. 1;

FIG. 3 is a plan view of the semiconductor device shown in FIG. 1 (as seen through a sealing resin, a moisture-resistant layer and a top plate);

FIG. 4 is a front view of the semiconductor device shown in FIG. 1;

FIG. 5 is a right side view of the semiconductor device shown in FIG. 1;

FIG. 6 is a left side view of the semiconductor device shown in FIG. 1;

FIG. 7 is a bottom view of the semiconductor device shown in FIG. 1;

FIG. 8 shows a right portion (near the first substrate) of FIG. 3 as enlarged;

FIG. 9 shows a left portion (near the second substrate) of FIG. 3 as enlarged;

FIG. 10 shows a central portion (near the third substrate) of FIG. 3 as enlarged;

FIG. 11 is a sectional view taken along line XI-XI in FIG. 3;

FIG. 12 is a sectional view taken along line XII-XII in FIG. 3;

FIG. 13 is a sectional view taken along line XIII-XIII in FIG. 3;

FIG. 14 is a sectional view taken along line XIV-XIV in FIG. 3;

FIG. 15 shows a portion of FIG. 3 (a switching element and a protective element bonded to an upper arm mounting layer) as enlarged;

FIG. 16 is a sectional view taken along line XVI-XVI in FIG. 15;

FIG. 17 is a sectional view taken along line XVII-XVII in FIG. 15;

FIG. 18 shows a portion of FIG. 3 (a switching element and a protective element bonded to a lower arm mounting layer) as enlarged;

FIG. 19 is a sectional view taken along line XIX-XIX in FIG. 18;

FIG. 20 is a sectional view taken along line XX-XX in FIG. 18;

FIG. 21 is a circuit diagram of the semiconductor device shown in FIG. 1;

FIG. 22 shows a portion of FIG. 16 as enlarged;

FIG. 23 is a sectional view showing a portion of a semiconductor device of a comparative example (a switching element bonded to an upper arm layer) as enlarged;

FIG. 24 is a sectional view showing a portion of a semiconductor device (a switching element and a protective

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element bonded to the upper arm layer) according to a first variation of the first embodiment of the present disclosure;

FIG. 25 is a sectional view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm layer) according to the first variation of the first embodiment of the present disclosure;

FIG. 26 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a second variation of the first embodiment of the present disclosure;

FIG. 27 is a sectional view taken along line XXVII-XXVII in FIG. 26;

FIG. 28 is a sectional view taken along line XXVIII-XXVIII in FIG. 26;

FIG. 29 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the second variation of the first embodiment of the present disclosure;

FIG. 30 is a sectional view taken along line XXX-XXX in FIG. 29;

FIG. 31 is a sectional view taken along line XXXI-XXXI in FIG. 29;

FIG. 32 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a third variation of the first embodiment of the present disclosure;

FIG. 33 is a sectional view taken along line XXXIII-XXXIII in FIG. 32;

FIG. 34 is a sectional view taken along line XXXIV-XXXIV in FIG. 32;

FIG. 35 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the third variation of the first embodiment of the present disclosure;

FIG. 36 is a sectional view taken along line XXXVI-XXXVI in FIG. 35;

FIG. 37 is a sectional view taken along line XXXVII-XXXVII in FIG. 35;

FIG. 38 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a fourth variation of the first embodiment of the present disclosure;

FIG. 39 is a sectional view taken along line XXXIX-XXXIX in FIG. 38;

FIG. 40 is a sectional view taken along line XL-XL in FIG. 38;

FIG. 41 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the fourth variation of the first embodiment of the present disclosure;

FIG. 42 is a sectional view taken along line XLII-XLII in FIG. 41;

FIG. 43 is a sectional view taken along line XLIII-XLIII in FIG. 41;

FIG. 44 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a fifth variation of the first embodiment of the present disclosure;

FIG. 45 is a sectional view taken along line XLV-XLV in FIG. 44;

FIG. 46 is a sectional view taken along line XLVI-XLVI in FIG. 45;

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FIG. 47 is a plan view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the fifth variation of the first embodiment of the present disclosure;

FIG. 48 is a sectional view taken along line XLVIII-XLVIII in FIG. 47;

FIG. 49 is a sectional view taken along line XLIX-XLIX in FIG. 47;

FIG. 50 shows test results based on variations in thicknesses of the moisture-resistant layer of the semiconductor device according to the fourth variation of the first embodiment of the present disclosure;

FIG. 51 shows results of the H3TRB test on the semiconductor device according to the fourth variation of the first embodiment of the present disclosure and the semiconductor device of the comparative example;

FIG. 52 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a second embodiment of the present disclosure;

FIG. 53 is a sectional view taken along line LIII-LIII in FIG. 52;

FIG. 54 is a sectional view taken along line LIV-LIV in FIG. 52;

FIG. 55 is a plan view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the second embodiment of the present disclosure;

FIG. 56 is a sectional view taken along line LVI-LVI in FIG. 55; and

FIG. 57 is a sectional view taken along line LVII-LVII in FIG. 55.

MODE FOR CARRYING OUT THE INVENTION

Modes for carrying out the disclosure (hereinafter referred to as embodiments) are described below with reference to the accompanying drawings.

First Embodiment

With reference to FIGS. 1 to 23, a semiconductor device A10 according to a first embodiment of the present disclosure is described. The semiconductor device A10 includes a substrate 11, a first mounting layer 211, a second mounting layer 221, a third mounting layer 231, switching elements 31, a moisture-resistant layer 51 and a sealing resin 52. Of these, the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231 are examples of the "mounting layer" as set forth in the appended claims of the present disclosure. In addition to these, the semiconductor device A10 further includes a first electroconductive layer 212, a second electroconductive layer 222, a third electroconductive layer 232, a power supply terminal 24, an output terminal 25, a connecting electroconductive member 261, protective elements 32, wires 41, a heat sink 61 and a case 70. Of these, the first electroconductive layer 212, the second electroconductive layer 222 and the third electroconductive layer 232 are examples of the "electroconductive layer" as set forth in the appended claims of the present disclosure. The power supply terminal 24 includes a first power supply terminal 24A and a second power supply terminal 24B. For easier understanding, FIG. 3 shows a view seen through the moisture-resistant layer 51, the sealing resin 52 and a top plate 79. In FIG. 3, the line XI-XI and the

line XII-XII are indicated by dash-dotted lines. In FIGS. 11 and 12, illustration of the moisture-resistant layer 51 is omitted.

The semiconductor device A10 shown in FIG. 1 is a power module. The semiconductor device A10 may be used for inverter devices of various electric products. As shown in FIGS. 1 and 2, the semiconductor device A10 is rectangular as viewed in the thickness direction z of the substrate 11. For convenience of explanation, a direction that is perpendicular to the thickness direction z of the substrate 11 (hereinafter simply "thickness direction z") is referred to as the "first direction x1". The direction that is perpendicular to both of the thickness direction z and the first direction x1 is referred to as the "second direction x2". The longitudinal direction of the semiconductor device A10 is the second direction x2.

The substrate 11 is an electrically insulating member on which the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the electroconductive layer (the first electroconductive layer 212, the second electroconductive layer 222 and the third electroconductive layer 232) are disposed, as shown in FIG. 3. The substrate 11 has three sections, namely a first substrate 11A, a second substrate 11B and a third substrate 11C. The first substrate 11A, the second substrate 11B and the third substrate 11C are spaced apart from each other in the second direction x2. In the second direction x2, the third substrate 11C is located between the first substrate 11A and the second substrate 11B. Unlike this configuration, the substrate 11 may have two sections, namely the first substrate 11A and the second substrate 11B, or may only have a single section. As shown in FIG. 11, each of the first substrate 11A, the second substrate 11B and the third substrate 11C has a front surface 111 and a back surface 112 facing away from each other in the thickness direction z.

The substrate 11 is made of a ceramic with excellent thermal conductivity. Examples of such a ceramic include aluminum nitride (AlN). A DBC (Direct Bonding Copper) substrate, which has copper (Cu) foils bonded to the front surface 111 and the back surface 112, may be used as the substrate 11. By using a DBC substrate, the mounting layer and the electroconductive layer can be easily formed through patterning of the copper foil bonded to the front surface 111. The copper foil bonded to the back surface 112 can be formed into a heat transfer layer 62 (described later).

As shown in FIGS. 3 and 8, on the front surface 111 of the first substrate 11A are disposed the first mounting layer 211, the first electroconductive layer 212, a first gate layer 213, a first detection layer 214 and a thermistor mounting layer 215. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver (Ag), for example.

As shown in FIG. 8, switching elements 31 and protective elements 32 are electrically bonded to the first mounting layer 211. The first mounting layer 211 includes a first upper arm mounting layer 211A and a first lower arm mounting layer 211B.

As shown in FIG. 8, the first upper arm mounting layer 211A is offset toward one end of the first substrate 11A (upper side in FIG. 8) in the first direction x1. The first upper arm mounting layer 211A is in the form of a strip extending along the second direction x2. Three switching elements 31 and three protective elements 32 are electrically bonded to the first upper arm mounting layer 211A. Note that the number of switching elements 31 and the number of the protective elements 32 to be electrically bonded to the first upper arm mounting layer 211A are not limited to three. On

the first upper arm mounting layer 211A, both of the switching elements 31 and the protective elements 32 are aligned in the second direction x2. The first upper arm mounting layer 211A is formed with a first power supply pad 211C in the form of a strip extending along the first direction x1 at an end close to the case 70 in the second direction x2. The first power supply pad 211C is electrically connected to the first power supply terminal 24A.

As shown in FIG. 8, the first lower arm mounting layer 211B is located between the first upper arm mounting layer 211A and the first electroconductive layer 212 in the first direction x1. The first lower arm mounting layer 211B is in the form of a strip extending along the second direction x2. Three switching elements 31 and three protective elements 32 are electrically bonded to the first lower arm mounting layer 211B. Note that the number of switching elements 31 and the number of the protective elements 32 to be electrically bonded to the first lower arm mounting layer 211B are not limited to three. On the first lower arm mounting layer 211B, both of the switching elements 31 and the protective elements 32 are aligned in the second direction x2. As shown in FIG. 15, the first lower arm mounting layer 211B is electrically connected, via wires 41, to both of the front surface electrodes 311 (described later) of the switching elements 31 and anode electrodes 321 (described later) of the protective elements 32 that are electrically bonded to the first upper arm mounting layer 211A.

As shown in FIGS. 8 and 18, the first electroconductive layer 212 is electrically connected, via wires 41, to the front surface electrodes 311 of the switching elements 31 and the anode electrodes 321 of the protective elements 32 that are electrically bonded to the first lower arm mounting layer 211B. The first electroconductive layer 212 is offset toward the other end of the first substrate 11A (lower side in FIG. 8) in the first direction x1. The first electroconductive layer 212 is in the form of a strip extending along the second direction x2. The first electroconductive layer 212 is formed with a second power supply pad 212A in the form of a strip extending along the first direction x1 at an end close to the case 70 in the second direction x2. The second power supply pad 212A is electrically connected to the second power supply terminal 24B.

As shown in FIGS. 15 and 18, the first gate layer 213 is electrically connected, via first gate wires 421, to gate electrodes 313 (described later) of the switching elements 31 electrically bonded to the first mounting layer 211. The first gate layer 213 is in the form of a strip extending along the second direction x2 and faces the switching elements 31 as viewed in the thickness direction z. The first gate layer 213 includes a first upper arm gate layer 213A and a first lower arm gate layer 213B.

As shown in FIG. 8, the first upper arm gate layer 213A is located between the first upper arm mounting layer 211A and the case 70 in the first direction x1. As viewed in the thickness direction z, the first upper arm gate layer 213A faces the switching elements 31 electrically bonded to the first upper arm mounting layer 211A. As shown in FIG. 15, the first upper arm gate layer 213A is electrically connected, via first gate wires 421, to the gate electrodes 313 of the switching elements 31 electrically bonded to the first upper arm mounting layer 211A.

As shown in FIG. 8, the first lower arm gate layer 213B is located between the first lower arm mounting layer 211B and the first electroconductive layer 212 in the first direction x1. As viewed in the thickness direction z, the first lower arm gate layer 213B faces the switching elements 31 electrically bonded to the first lower arm mounting layer 211B. As

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shown in FIG. 18, the first lower arm gate layer 213B is electrically connected, via first gate wires 421, to the gate electrodes 313 of the switching elements 31 electrically bonded to the first lower arm mounting layer 211B.

As shown in FIGS. 15 and 18, the first detection layer 214 is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the first mounting layer 211. The first detection layer 214 is in the form of a strip extending along the second direction x2 and faces the switching elements 31 as viewed in the thickness direction z. The first detection layer 214 includes a first upper arm detection layer 214A and a first lower arm detection layer 214B.

As shown in FIG. 8, the first upper arm detection layer 214A is located between the first upper arm mounting layer 211A and the first upper arm gate layer 213A in the first direction x1. As viewed in the thickness direction z, the first upper arm detection layer 214A faces the switching elements 31 electrically bonded to the first upper arm mounting layer 211A. As shown in FIG. 15, the first upper arm detection layer 214A is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the first upper arm mounting layer 211A.

As shown in FIG. 8, the first lower arm detection layer 214B is located between the first lower arm mounting layer 211B and the first lower arm gate layer 213B in the first direction x1. The first lower arm detection layer 214B is in the form of an L-shaped strip with a part extending in the first direction x1 and a part extending in the second direction x2. As viewed in the thickness direction z, the part extending in the second direction x2 faces the switching elements 31 electrically bonded to the first lower arm mounting layer 211B. As shown in FIG. 18, the first lower arm detection layer 214B is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the first lower arm mounting layer 211B.

As shown in FIG. 8, a thermistor 33 is electrically bonded to the thermistor mounting layer 215. The thermistor mounting layer 215 is located close to a corner of the first substrate 11A. The thermistor mounting layer 215 is surrounded by the first upper arm mounting layer 211A, the first upper arm gate layer 213A and the first upper arm detection layer 214A. The thermistor mounting layer 215 has a pair of sections spaced apart from each other in the second direction x2. The positive electrode of the thermistor 33 is electrically bonded to one of these sections, whereas the negative electrode of the thermistor 33 is electrically bonded to the other one of these sections.

As shown in FIGS. 3 and 9, on the front surface 111 of the second substrate 11B are disposed the second mounting layer 211, the second electroconductive layer 222, a second gate layer 223 and a second detection layer 224. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver, for example.

As shown in FIG. 9, switching elements 31 and protective elements 32 are electrically bonded to the second mounting layer 221. The second mounting layer 221 includes a second upper arm mounting layer 221A and a second lower arm mounting layer 221B.

As shown in FIG. 9, the second upper arm mounting layer 221A is offset toward one end of the second substrate 11B (upper side in FIG. 9) in the first direction x1. The second upper arm mounting layer 221A is in the form of a strip extending along the second direction x2. Three switching

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elements 31 and three protective elements 32 are electrically bonded to the second upper arm mounting layer 221A. Note that the number of switching elements 31 and the number of the protective elements 32 to be electrically bonded to the second upper arm mounting layer 221A are not limited to three. On the second upper arm mounting layer 221A, both of the switching elements 31 and the protective element 32 are aligned in the second direction x2.

As shown in FIG. 9, the second lower arm mounting layer 221B is located between the second upper arm mounting layer 221A and the second electroconductive layer 222 in the first direction x1. The second lower arm mounting layer 221B is in the form of a strip extending along the second direction x2. Three switching elements 31 and three protective elements 32 are electrically bonded to the second lower arm mounting layer 221B. Note that the number of switching elements 31 and the number of the protective elements 32 to be electrically bonded to the second lower arm mounting layer 221B are not limited to three. On the second lower arm mounting layer 221B, both of the switching elements 31 and the protective element 32 are aligned in the second direction x2. As shown in FIG. 15, the second lower arm mounting layer 221B is electrically connected, via wires 41, to the front surface electrodes 311 of the switching elements 31 and the anode electrodes 321 of the protective elements 32 that are electrically bonded to the second upper arm mounting layer 221A. The second lower arm mounting layer 221B is formed with an output pad 221C in the form of a strip extending along the first direction x1 at an end close to the case 70 in the second direction x2. In the second direction x2, the output pad 221C is close to both of the second upper arm mounting layer 221A and the second electroconductive layer 222. The output pad 221C is electrically connected to the output terminal 25.

As shown in FIGS. 9 and 18, the second electroconductive layer 222 is electrically connected, via wires 41, to the front surface electrodes 311 of the switching elements 31 and the anode electrodes 321 of the protective elements 32 that are electrically bonded to the second lower arm mounting layer 221B. The second electroconductive layer 222 is offset toward the other end of the second substrate 11B (lower side in FIG. 9) in the first direction x1. The second electroconductive layer 222 is in the form of a strip extending along the second direction x2.

As shown in FIGS. 15 and 18, the second gate layer 223 is electrically connected, via first gate wires 421, to the gate electrodes 313 of the switching elements 31 electrically bonded to the second mounting layer 221. The second gate layer 223 is in the form of a strip extending along the second direction x2 and faces the switching elements 31 as viewed in the thickness direction z. The second gate layer 223 includes a second upper arm gate layer 223A and a second lower arm gate layer 223B.

As shown in FIG. 9, the second upper arm gate layer 223A is located between the second upper arm mounting layer 221A and the case 70 in the first direction x1. As viewed in the thickness direction z, the second upper arm gate layer 223A faces the switching elements 31 electrically bonded to the second upper arm mounting layer 221A. As shown in FIG. 15, the second upper arm gate layer 223A is electrically connected, via first gate wires 421, to the gate electrodes 313 of the switching elements 31 electrically bonded to the second upper arm mounting layer 221A.

As shown in FIG. 9, the second lower arm gate layer 223B is located between the second lower arm mounting layer 221B and the second electroconductive layer 222 in the first direction x1. As viewed in the thickness direction z, the

second lower arm gate layer **223B** faces the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**. As shown in FIG. **18**, the second lower arm gate layer **223B** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**.

As shown in FIGS. **15** and **18**, the second detection layer **224** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second mounting layer **221**. The second detection layer **224** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The second detection layer **224** includes a second upper arm detection layer **224A** and a second lower arm detection layer **224B**.

As shown in FIG. **9**, the second upper arm detection layer **224A** is located between the second upper arm mounting layer **221A** and the second upper arm gate layer **223A** in the first direction **x1**. As viewed in the thickness direction **z**, the second upper arm detection layer **224A** faces the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**. As shown in FIG. **15**, the second upper arm detection layer **224A** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**.

As shown in FIG. **9**, the second lower arm detection layer **224B** is located between the second lower arm mounting layer **221B** and the second lower arm gate layer **223B** in the first direction **x1**. As viewed in the thickness direction **z**, the second lower arm detection layer **224B** faces the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**. As shown in FIG. **18**, the second lower arm detection layer **224B** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**.

As shown in FIGS. **3** and **10**, on the front surface **111** of the third substrate **11C** are disposed the third mounting layer **231**, the third electroconductive layer **232**, a third gate layer **233** and a third detection layer **234**. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver, for example.

As shown in FIG. **10**, switching elements **31** and protective elements **32** are electrically bonded to the third mounting layer **231**. The third mounting layer **231** includes a third upper arm mounting layer **231A** and a third lower arm mounting layer **231B**.

As shown in FIG. **10**, the third upper arm mounting layer **231A** is offset toward one end of the third substrate **11C** (upper side in FIG. **10**) in the first direction **x1**. The third upper arm mounting layer **231A** is in the form of a strip extending along the second direction **x2**. Two switching elements **31** and two protective elements **32** are electrically bonded to the third upper arm mounting layer **231A**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the third upper arm mounting layer **231A** are not limited to two. On the third upper arm mounting layer **231A**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**.

As shown in FIG. **10**, the third lower arm mounting layer **231B** is located between the third upper arm mounting layer **231A** and the third electroconductive layer **232** in the first

direction **x1**. The third lower arm mounting layer **231B** is in the form of a strip extending along the second direction **x2**. Two switching elements **31** and two protective elements **32** are electrically bonded to the third lower arm mounting layer **231B**. Note that the number of switching elements **31** and the number of the protective element **32** to be electrically bonded to the third lower arm mounting layer **231B** are not limited to two. On the third lower arm mounting layer **231B**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**. As shown in FIG. **15**, the third lower arm mounting layer **231B** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the third upper arm mounting layer **231A**.

As shown in FIGS. **10** and **18**, the third electroconductive layer **232** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the third lower arm mounting layer **231B**. The third electroconductive layer **232** is offset toward the other end of the third substrate **11C** (lower side in FIG. **10**) in the first direction **x1**. The third electroconductive layer **232** is in the form of a strip extending along the second direction **x2**.

As shown in FIGS. **15** and **18**, the third gate layer **233** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third mounting layer **231**. The third gate layer **233** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The third gate layer **233** includes a third upper arm gate layer **233A** and a third lower arm gate layer **233B**.

As shown in FIG. **10**, the third upper arm gate layer **233A** is located between the third upper arm mounting layer **231A** and the case **70** in the first direction **x1**. As viewed in the thickness direction **z**, the third upper arm gate layer **233A** faces the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**. As shown in FIG. **15**, the third upper arm gate layer **233A** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**.

As shown in FIG. **10**, the third lower arm gate layer **233B** is located between the third lower arm mounting layer **231B** and the third electroconductive layer **232** in the first direction **x1**. The third lower arm gate layer **233B** is in the form of an L-shaped strip with a part extending in the first direction **x1** and a part extending in the second direction **x2**. As viewed in the thickness direction **z**, the part extending in the second direction **x2** faces the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**. As shown in FIG. **18**, the third lower arm gate layer **233B** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**.

As shown in FIGS. **15** and **18**, the third detection layer **234** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the third mounting layer **231**. The third detection layer **234** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The third detection layer **234** includes a third upper arm detection layer **234A** and a third lower arm detection layer **234B**.

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As shown in FIG. 10, the third upper arm detection layer 234A is located between the third upper arm mounting layer 231A and the third upper arm gate layer 233A in the first direction x1. As viewed in the thickness direction z, the third upper arm detection layer 234A faces the switching elements 31 electrically bonded to the third upper arm mounting layer 231A. As shown in FIG. 15, the third upper arm detection layer 234A is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the third upper arm mounting layer 231A.

As shown in FIG. 10, the third lower arm detection layer 234B is located between the third lower arm mounting layer 231B and the third lower arm gate layer 233B in the first direction x1. As viewed in the thickness direction z, the third lower arm detection layer 234B faces the switching elements 31 electrically bonded to the third lower arm mounting layer 231B. As shown in FIG. 18, the third lower arm detection layer 234B is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the third lower arm mounting layer 231B.

The first upper arm mounting layer 211A, the second upper arm mounting layer 221A and the third upper arm mounting layer 231A respectively correspond to sections of the "upper arm mounting layer" as set forth in the appended claims of the present disclosure. The first lower arm mounting layer 211B, the second lower arm mounting layer 221B and the third lower arm mounting layer 231B respectively correspond to sections of the "lower arm mounting layer" as set forth in the appended claims of the present disclosure.

As shown in FIGS. 2 and 3, the power supply terminal 24 is an element of an external connection terminal provided in the semiconductor device A10. As described before, the power supply terminal 24 includes a first power supply terminal 24A and a second power supply terminal 24B. The power supply terminal 24 is supported on the case 70 and connected to a DC power supply arranged outside the semiconductor device A10. The power supply terminal 24 is made of a thin metal plate such as a copper plate. The surface of the thin metal plate may be plated with nickel (Ni). The first power supply terminal 24A is the positive electrode (P-terminal) of the semiconductor device A10. The second power supply terminal 24B is the negative electrode (N-terminal) of the semiconductor device A10. The first power supply terminal 24A and the second power supply terminal 24B are spaced apart from each other in the first direction x1. The first power supply terminal 24A and the second power supply terminal 24B have the same shape.

As shown in FIG. 11, the power supply terminal 24 is bent into a hook shape as viewed in the first direction x1. The power supply terminal 24 is formed with a coupling hole 241 penetrating the terminal in the thickness direction z at a portion exposed outside the semiconductor device A10 and extending perpendicularly to the thickness direction z. Into the coupling hole 241, a fastening member such as a bolt is inserted. As shown in FIG. 8, a connection member 242 with electrical conductivity is connected to a portion of the power supply terminal 24 that is located inside the case 70 and that extends perpendicularly to the thickness direction z. For example, the connection member 242 includes a plurality of wires made of aluminum (Al). The connection member 242 connected to the first power supply terminal 24A is connected at its other end to the power supply pad 211C of the first upper arm mounting layer 211A. Thus, with this connection member 242, the first power supply terminal 24A is electrically connected to the first upper arm mounting layer

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211A. The connection member 242 connected to the second power supply terminal 24B is connected, at its other end, to the second power supply pad 212A of the first electroconductive layer 212. Thus, with this connection member 242, the second power supply terminal 24B is electrically connected to the first electroconductive layer 212.

As shown in FIGS. 2 and 3, the output terminal 25 is an element of an external connection terminal provided in the semiconductor device A10. The output terminal 25 is divided into two, namely a first output terminal 25A and a second output terminal 25B. Note that the output terminal 25 may be configured as a single unit that is not divided into multiple parts. The output terminal 25 is supported on the case 70 and connected to a driving target such as a motor arranged outside the semiconductor device A10. The output terminal 25 is located opposite to the power supply terminal 24 across the substrate 11 in the second direction x2. The output terminal 25 is made of the same thin metal film as the power supply terminal 24. The surface of the thin metal plate may be plated with nickel. The first output terminal 25A and the second output terminal 25B are connected in parallel to the second lower arm mounting layer 221B. The first output terminal 25A and the second output terminal 25B are each connected to a driving target of the semiconductor device A10 that is externally arranged. In the second direction x2, the first output terminal 25A faces the first power supply terminal 24A, whereas the second output terminal 25B faces the second power supply terminal 24B. The first output terminal 25A and the second output terminal 25B are spaced apart from each other in the first direction x1. The first output terminal 25A and the second output terminal 25B have the same shape.

As shown in FIG. 11, the output terminal 25 is bent into a hook shape as viewed in the first direction x1. The output terminal 25 is formed with a coupling hole 251 penetrating the terminal in the thickness direction z at a portion exposed outside the semiconductor device A10 and extending perpendicularly to the thickness direction z. Into the coupling hole 251, a fastening member such as a bolt is inserted. As shown in FIG. 9, a connection member 252 with electrical conductivity is connected to a portion of the output terminal 25 that is located inside the case and that extends perpendicularly to the thickness direction z. For example, the connection member 252 includes a plurality of wires made of aluminum. The connection member 252 connected to the output terminal 25 is connected at its other end to the output pad 221C of the second lower arm mounting layer 221B disposed on the second substrate 11B. Thus, with the connection member 25, the output terminal 25 is electrically connected to the second lower arm mounting layer 221B.

As shown in FIG. 10, the connecting electroconductive member 261 connects the first mounting layer 211 and the third mounting layer 231 to each other, and also connects the second mounting layer 221 and the third mounting layer 231 to each other. Thus, the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231 are electrically connected to each other via the connecting electroconductive member 261. Further, as shown in FIG. 10, the connecting electroconductive member 261 connects the first electroconductive layer 212 and the third electroconductive layer 232 to each other, and also connects the second electroconductive layer 222 and the third electroconductive layer 232 to each other. Thus, the first electroconductive layer 212, the second electroconductive layer 222 and the third electroconductive layer 232 are electrically connected to each other via the connecting electroconduc-

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tive member **261**. For example, the connecting electroconductive member **261** includes a plurality of wires made of aluminum.

As shown in FIG. 10, the connecting electroconductive member **261** includes a first part **261A**, a second part **261B** and a third part **261C**. All of the first part **261A**, the second part **261B** and the third part **261C** extend in the second direction **x2**. The first part **261A** connects the first upper arm mounting layer **211A** and the third upper arm mounting layer **231A** to each other, and also connects the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A** to each other. Thus, the first upper arm mounting layer **211A** and the second upper arm mounting layer **221A** are electrically connected to each other via the first part **261A**. The second part **261B** connects the first lower arm mounting layer **211B** and the third lower arm mounting layer **231B** to each other, and also connects the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B** to each other. Thus, the first lower arm mounting layer **211B** and the second lower arm mounting layer **221B** are electrically connected to each other via the second part **261B**. The third part **261C** connects the first electroconductive layer **212** and the third electroconductive layer **232** to each other, and also connects the second electroconductive layer **222** and the third electroconductive layer **232** to each other. Thus, the first electroconductive layer **212** and the second electroconductive layer **222** are electrically connected to each other via the third part **261C**.

As shown in FIG. 10, first electroconductive members **262** connect the first gate layer **213** and the third gate layer **233** to each other, and also connect the second gate layer **223** and the third gate layer **233** to each other. Thus, the first gate layer **213**, the second gate layer **223** and the third gate layer **233** are electrically connected to each other via the first electroconductive members **262**. For example, the first electroconductive members **262** are wires made of aluminum. All of the first electroconductive members **262** extend in the second direction **x2** and may consist of four first electroconductive members **262**. The first one of the first electroconductive members **262** connects the first upper arm gate layer **213A** and the third upper arm gate layer **233A**. The second one of the first electroconductive members **262** connects the second upper arm gate layer **223A** and the third upper arm gate layer **233A**. The third one of the first electroconductive members **262** connects the first lower arm gate layer **213B** and the third lower arm gate layer **233B**. The fourth one of the first electroconductive members **262** connects the second lower arm gate layer **223B** and the third lower arm gate layer **233B**.

As shown in FIG. 10, second electroconductive members **263** connect the first detection layer **214** and the third detection layer **234** to each other, and also connect the second detection layer **224** and the third detection layer **234** to each other. Thus, the first detection layer **214**, the second detection layer **224** and the third detection layer **234** are electrically connected to each other via the second electroconductive members **263**. For example, the second electroconductive members **263** are wires made of aluminum. All of the second electroconductive members **263** extend in the second direction **x2** and may consist of four second electroconductive members **263**. The first one of the second electroconductive members **263** connects the first upper arm detection layer **214A** and the third upper arm detection layer **234A**. The second one of the second electroconductive members **263** connects the second upper arm detection layer **224A** and the third upper arm detection layer **234A**. The third one of the second electroconductive members **263**

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connects the first lower arm detection layer **214B** and the third lower arm detection layer **234B**. The fourth one of the second electroconductive members **263** connects the second lower arm detection layer **224B** and the third lower arm detection layer **234B**.

As shown in FIGS. 2-4, a gate terminal **27** is an element of an external connection terminal provided in the semiconductor device **A10**. The gate terminal **27** is connected to an externally arranged driving circuit (e.g. gate driver) for the semiconductor device **A10**. The gate terminal **27** is disposed to face the substrate **11** as viewed in the thickness direction **z** and supported on the case **70**. The gate terminal **27** projects in the same direction in which the front surface **111** of the substrate **11** faces (along the thickness direction **z**). For example, the gate terminal **27** is in the form of a metal rod made of copper. The surface of the metal rod is plated with tin (Sn). Nickel plating may be provided between the surface of the metal rod and the tin plating. As shown in FIG. 12, the gate terminal **27** is bent into a hook shape at its end closer to the substrate **11** in the thickness direction **z**, thereby having a portion extending along the first direction **x1**. The gate terminal **27** includes a first gate terminal **27A** and a second gate terminal **27B**. Paired second gate wires **422** are connected to the first gate terminal **27A** and the second gate terminal **27B**. For example, the paired second gate wires **422** are made of aluminum.

As shown in FIG. 10, the first gate terminal **27A** is disposed close to the second upper arm gate layer **223A** to face the second substrate **11B** as viewed in the thickness direction **z**. The second gate wire **422** connected at one end to the first gate terminal **27A** is connected at the other end to the third upper arm gate layer **233A**. Thus, the first gate terminal **27A** is electrically connected to the gate electrodes **313** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**.

As shown in FIG. 10, the second gate terminal **27B** is disposed close to the third lower arm gate layer **233B** to face the third substrate **11C** as viewed in the thickness direction **z**. The second gate wire **422** connected at one end to the second gate terminal **27B** is connected at the other end to the third lower arm gate layer **233B**. Thus, the second gate terminal **27B** is electrically connected to the gate electrodes **313** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B**.

As shown in FIGS. 2-4, a device current detection terminal **281** is an element of an external connection terminal provided in the semiconductor device **A10**. The device current detection terminal **281** is connected to an externally arranged control circuit for the semiconductor device **A10**. The device current detection terminal **281** is disposed to face the substrate **11** and supported on the case **70**. The device current detection terminal **281** projects in the same direction in which the gate terminal **27** projects along the thickness direction **z**. The device current detection terminal **281** is made of a metal rod of the same material as the gate terminal **27**. The device current detection terminal **281** has the same shape as the gate terminal **27**. Thus, the device current detection terminal **281** is bent into a hook shape at its end closer to the substrate **11** in the thickness direction **z**, thereby having a portion extending along the first direction **x1**. The device current detection terminal **281** includes a first detection terminal **281A** and a second detection terminal **281B**. Paired second detection wires **432** are connected to the first

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detection terminal **281A** and the second detection terminal **281B**. For example, the paired second detection wires **432** are made of aluminum.

As shown in FIG. 10, the first detection terminal **281A** is disposed close to the second upper arm detection layer **224A** to face the second substrate **11B** as viewed in the thickness direction *z*, and is also close to the first gate terminal **27A**. The second detection wire **432** connected at one end to the first detection terminal **281A** is connected at the other end to the second upper arm detection layer **224A**. Thus, the first detection terminal **281A** is electrically connected to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**.

As shown in FIGS. 2-4 and 9, the second detection terminal **281B** is disposed close to the first lower arm detection layer **214B** to face the first substrate **11A** as viewed in the thickness direction *z*, and is also close to the second gate terminal **27B**. The second detection wire **432** connected at one end to the second detection terminal **281B** is connected at the other end to the first lower arm detection layer **214B**. Thus, the second detection terminal **281B** is electrically connected to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B**.

As shown in FIGS. 2-4 and 9, a power supply current detection terminal **281** is an element of an external connection terminal provided in the semiconductor device **A10**. The power supply current detection terminal **282** is connected to an externally arranged control circuit for the semiconductor device **A10** and supported on the case **70**. The power supply current detection terminal **282** projects in the same direction in which the gate terminal **27** projects along the thickness direction *z*. The power supply current detection terminal **28** is made of a metal rod of the same material as the gate terminal **27**. The power supply current detection terminal **282** is located at the same position as the first gate terminal **27A** and the first detection terminal **281A** in the first direction *x1* and spaced apart from the first detection terminal **281A** toward the first output terminal **25A** in the second direction *x2*. The power supply current detection terminal **282** is disposed close to the second upper arm mounting layer **221A** to face the second substrate **11B** in the first direction *x1*. The power supply current detection terminal **282** has the same shape as the gate terminal **27**. Thus, the power supply current detection terminal **282** is bent into a hook shape at its end closer to the second substrate **11B** in the thickness direction *z*, thereby having a portion extending along the first direction *x1*. To this end of the power supply current detection terminal, one end of a power supply current detection wire **44** is connected. The other end of the power supply current detection wire **44** is connected to the second upper arm mounting layer **221A**. Thus, the power supply current detection terminal **282** is electrically connected to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**. For example, the power supply current detection wire **44** is made of aluminum.

As shown in FIGS. 2-4 and 8, a pair of thermistor terminals **29** are an element of an external connection terminal provided in the semiconductor device **A10**. The paired thermistor terminals **29** are connected to an externally arranged control circuit for the semiconductor device **A10** and supported on the case **70**. The paired thermistor terminal **29** project in the same direction in which the gate terminal

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27 projects along the thickness direction *z*. The paired thermistor terminals **29** are made of a metal rod of the same material as the gate terminal **27**. The paired thermistor terminals **29** are located at the same position as the first gate terminal **27A** and the first detection terminal **281A** in the first direction *x1* and spaced apart from the first gate terminal **27A** toward the first power supply terminal **24A** in the second direction *x2*. The paired thermistor terminal **29** are disposed close to the thermistor mounting layer **215** to face the first substrate **11A** in the first direction *x1*. The paired thermistor terminals **29** have the same shape as the gate terminal **27**. Thus, each of the paired thermistor terminals is bent into a hook shape at its end closer to the first substrate **11A** in the thickness direction *z*, thereby having a portion extending along the first direction *x1*. To this end of each of the paired thermistor terminals **29**, one end of a corresponding one of a paired thermistor wires **45** is connected. The other ends of the paired thermistor wires **45** are connected to paired sections of the thermistor mounting layer **215**. Thus, the thermistor terminals **29** are electrically connected to the thermistor **33**. For example, the paired thermistor wires **45** are made of aluminum.

As shown in FIG. 3, the switching elements **31** are semiconductor elements electrically bonded to and aligned, in the second direction *x2*, on each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**. The switching elements **31** are rectangular (square in the semiconductor device **A10**) as viewed in the thickness direction *z*. The switching elements **31** are MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) made of a semiconductor material mainly composed of silicon carbide (SiC). Note that the switching elements **31** are not limited to MOSFETs and may be IGBTs (Insulated Gate Bipolar Transistors). In the semiconductor device **A10**, it is assumed that the switching elements **31** are n-channel MOSFETs made of a semiconductor material mainly composed of silicon carbide. In the semiconductor device **A10**, the switching elements **31** are 400 μm or less and preferably 150 μm or less in thickness. The breakdown voltage of the switching elements **31** is 1,200 V or more.

As shown in FIGS. 15-20, each of the switching elements **31** has a front surface **31A**, a back surface **31B**, a side surface **31C**, a front surface electrode **311**, a back surface electrode **312**, a gate electrode **313** and an insulating film **314**. The front surface **31A**, the back surface **31B** and the side surface **31C** correspond to the “first element front surface”, the “first element back surface” and the “first element side surface”, respectively, as set forth in the appended claims of the present disclosure. The front surface **31A** faces in the same direction in which the front surface **111** of the substrate **11** faces along the thickness direction *z*. The back surface **31B** faces in the opposite direction of the front surface **31A**. The switching elements **31** are electrically bonded to the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**, with the back surfaces **31B** facing the front surface **111**. The side surface **31C** is connected to both of the front surface **31A** and the back surface **31B**. The side surface **31C** includes a plurality of sections (four sections in the semiconductor device **A10**) each facing in the first direction *x1* or the second direction *x2*.

As shown in FIGS. 15-20, the front surface electrode **311** is provided on the front surface **31A**. A source current flows through the front surface electrode **311**. The front surface electrode **311** has a pair of first pads **311A** and a pair of second pads **311B**. The paired first pads **311A** are sections of a front surface electrode **311** that are spaced apart from each

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other in the second direction x , so are the paired second pads 311B. In each of the switching elements 31 electrically bonded to the first mounting layer 211, the second pads 311B are located opposite to the first lower arm mounting layer 211B or the first electroconductive layer 212 across the paired first pads 311A in the first direction $x1$. The above-described positional relationship between the paired first pads 311A and the paired second pads 311B also applies to the switching elements 31 electrically bonded to the second mounting layer 221 or the third mounting layer 231.

As shown in FIG. 15, in each of the switching elements 31 electrically bonded to the first upper arm mounting layer 211A, the second upper arm mounting layer 221A or the third upper arm mounting layer 231A, one end of the first detection wire 431 is connected to one of the paired second pads 311B. The other end of the first detection wire 431 is connected to the first upper arm detection layer 214A, the second upper arm detection layer 224A or the third upper arm detection layer 234A. As shown in FIG. 18, in each of the switching elements 31 electrically bonded to the first lower arm mounting layer 211B, the second lower arm mounting layer 221B or the third lower arm mounting layer 231B, one end of the first detection wire 431 is connected to one of the paired first pads 311A. The other end of the first detection wire 431 is connected to the first lower arm detection layer 214B, the second lower arm detection layer 224B or the third lower arm detection layer 234B. In this way, the front surface electrodes 311 are electrically connected to the first detection layer 214, the second detection layer 224 or the third detection layer 234 via the first detection wires 431. For example, the first detection wires 431 are made of gold (Au).

As shown in FIGS. 16-20 (excluding FIG. 18), the back surface electrode 312 is provided on the entirety of the back surface 31B. A drain current flows through the back surface electrode 312. The back surface electrode 312 is electrically bonded to one of the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231 via a first bonding layer 391. The first bonding layer 391 is electrically conductive. The first bonding layer 391 is sandwiched between the back surface electrode 312 and the first mounting layer 211, the second mounting layer 221 or the third mounting layer 231. For example, the first bonding layer 391 is made of lead-free solder mainly composed of tin. The first bonding layer 391 electrically connects each of the back surface electrodes 312 to one of the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231.

As shown in FIGS. 15 and 18, the gate electrode 313 is provided on the front surface 31A. A gate voltage for driving each of the switching elements 31 is applied to the gate electrode 313. As shown in FIG. 15, in each of the switching elements 31 electrically bonded to the first upper arm mounting layer 211A, the second upper arm mounting layer 221A or the third upper arm mounting layer 231A, the gate electrode 313 is close to the paired second pads 311B of the front surface electrode 311. One end of a first gate wire 421, which is connected at its other end to the first upper arm gate layer 213A, the second upper arm gate layer 223A or the third upper arm gate layer 233A, is connected to the gate electrode 313. As shown in FIG. 18, in each of the switching elements 31 electrically bonded to the first lower arm mounting layer 211B, the second lower arm mounting layer 221B or the third lower arm mounting layer 231B, the gate electrode 313 is close to the paired first pads 311A of the front surface electrode 311. One end of a first gate wire 421, which is connected at its other end to the first lower arm gate

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layer 213B, the second lower arm gate layer 223B or the third lower arm gate layer 233B, is connected to the gate electrode 313. In this way, the gate electrodes 313 are electrically connected to the first gate layer 213, the second gate layer 223 or the third gate layer 233 via the first gate wires 421. For example, the first gate wires 421 are made of gold.

As shown in FIGS. 15-20, the insulating film 314 is provided on the front surface 31A. The insulating film 314 is electrically insulating. As viewed in the thickness direction z , the insulating film 314 surrounds the front surface electrode 311. The insulating film 314 may be formed by laminating a silicon dioxide (SiO_2) layer, a silicon nitride (Si_3N_4) layer, and a polybenzoxazole (PBO) layer on the front surface 31A in the mentioned order. For the insulating film 314, a polyimide layer may be used instead of the polybenzoxazole layer. In FIGS. 15-20, the length from the edge 314A of the insulating film 314 to the front surface electrode 311 as viewed in the thickness direction z is indicated as a gap G_p in each of the switching elements 31. The gap G_p is a length along the first direction $x1$ or the second direction $x2$. The edge 314A is rectangular (square in the semiconductor device A10) as viewed in the thickness direction z . As viewed in the thickness direction z , the ratio of the length of the gap G_p to the length of one side of the edge 314 (shorter side when the edge 314 is rectangular) is set to 5% to 25%. The longer the gap G_p is, the higher the dielectric breakdown voltage of the switching element 31 is.

As shown in FIG. 3, the protective elements 32 are semiconductor elements electrically bonded to and aligned in the second direction $x2$ on each of the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231. The protective elements 32 are rectangular as viewed in the thickness direction z . The protective elements 32 are arranged to be electrically connected to the switching elements 31, respectively. The protective elements 32 are electrically connected to both of the front surface electrodes 311 and the back surface electrodes 312 of the switching elements 31. Thus, each of the switching elements 31 and a corresponding one of the protective elements 32 form a parallel circuit. For example, the protective elements 32 are schottky-barrier diodes made by using a semiconductor material mainly composed of silicon carbide. In the semiconductor device A10, the protective elements 32 are 400 μm or less and preferably 150 μm or less in thickness. The breakdown voltage of the protective elements 32 is 1,200 V or more.

As shown in FIGS. 15-20, each of the protective elements 32 has a front surface 32A, a back surface 32B, a side surface 32C, an anode electrode 321, a cathode electrode 322 and an insulating film 323. The front surface 32A corresponds to the "second element front surface" as set forth in the appended claims of the present disclosure. The front surface 32A faces in the same direction in which the front surface 111 of the substrate 11 faces along the thickness direction z . The back surface 32B faces in the opposite direction of the front surface 32A. The protective elements 32 are electrically bonded to the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231, with the back surfaces 32B facing the front surface 111. The side surface 32C is connected to both of the front surface 32A and the back surface 32B. The side surface 32C includes a plurality of sections (four sections in the semiconductor device A10) each facing in the first direction $x1$ or the second direction $x2$.

As shown in FIGS. 15-20, the anode electrode 321 is provided on the front surface 32A. The anode electrode 321

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is electrically connected to the front surface electrode **311** of the switching element **31** with which that protective element **32** is associated.

As shown in FIGS. **16** and **19**, the cathode electrode **322** is provided on the entirety of the back surface **32B**. The cathode electrode **322** is electrically bonded to one of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** via a second bonding layer **392**. The second bonding layer **392** is electrically conductive. The second bonding layer **392** is sandwiched between the cathode electrode **322** and the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**. The second bonding layer **392** is made of the same material as that for the first bonding layer **391**. With the second bonding layer **392**, the cathode electrode **322** is electrically connected, via the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**, to the back surface electrode **312** of the switching element **31** with which the protective element **32** of that cathode electrode **322** is associated.

As shown in FIGS. **16** and **19**, the insulating film **323** is provided on the front surface **32A**. The insulating film **323** is electrically insulating. As shown in FIGS. **15** and **18**, the insulating film **323** surrounds the anode electrode **321** as viewed in the thickness direction **z**. The insulating film **323** may be formed by laminating a silicon dioxide layer, a silicon nitride layer, and a polybenzoxazole layer on the front surface **31A** in the mentioned order. For the insulating film **323**, a polyimide layer may be used instead of the polybenzoxazole layer.

As shown in FIGS. **3** and **8**, the thermistor **33** is an element electrically bonded to the thermistor mounting layer **215**. For example, the thermistor **33** is an NTC (Negative Temperature Coefficient) thermistor. An NTC thermistor has the property of decreasing the resistance with increasing temperature. The thermistor **33** is used as a temperature detection sensor of the semiconductor device **A10**.

As shown in FIGS. **15-17**, wires **41** are connected to the front surface electrodes **311** of switching elements **31** and the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIGS. **18-20**, wires **41** are connected to the front surface electrodes **311** of switching elements **31** and the first electroconductive layer **212**, the second electroconductive layer **222** or the third electroconductive layer **232**. For example, the wires **41** are made of aluminum. The wires **41** are larger in diameter than the first gate wires **421** and the first detection wires **431**.

As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, wires **41** are connected to the front surface electrodes **311** and the first lower arm mounting layer **211B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, wires **41** are connected to the front surface electrodes **311** and the first electroconductive layer **212**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the first mounting layer **211** are electrically connected to the first lower arm mounting layer **211B** or the first electroconductive layer **212**.

As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**, wires **41** are connected to the front surface electrodes **311** and the second lower arm mounting layer **221B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**, wires **41** are connected to the front surface electrodes

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311 and the second electroconductive layer **222**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the second mounting layer **221** are electrically connected to the second lower arm mounting layer **221B** or the second electroconductive layer **222**.

As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**, wires are connected to the front surface electrodes **311** and the third lower arm mounting layer **231B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**, wires **41** are connected to the front surface electrodes **311** and the third electroconductive layer **232**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the third mounting layer **231** are electrically connected to the third lower arm mounting layer **231B** or the third electroconductive layer **232**.

As shown in FIGS. **15-20**, the wires **41** extend in the first direction **x1**. Each of the wires **41** has a first bonding portion **411**. The first bonding portions **411** are held in contact with the front surface electrodes **311** of the switching elements **31**. The wires **41** for each of the switching elements **31** include a pair of inner wires **41A** and a pair of outer wires **41B**. The paired inner wires **41A** are flanked by the paired outer wires **41B** in the second direction **x2**.

With reference to FIGS. **15-17**, description is given below of the configuration of the wires **41** for each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**. As shown in FIG. **17**, the first bonding portions **411** of the paired inner wires **41A** are held in contact with the paired first pads **311A** of the front surface electrode **311**. As shown in FIG. **16**, the first bonding portions **411** of the paired outer wires **41B** are held in contact with both of the paired first pads **311A** and the paired second pads **311B** of the front surface electrode **311**. As shown in FIGS. **15** and **16**, each of the first bonding portions **411** of the paired outer wires **41B** has a first connect portion **411A**, a second connect portion **411B** and a joint portion **411C**. The first connect portion **411A** is held in contact with a first pad **311A**. The second connect portion **411B** is held in contact with a second pad **311B**. The joint portion **411C** is sandwiched between the first connect portion **411A** and the second connect portion **411B** in the first direction **x1**. The joint portion **411C** projects in the same direction in which the front surface **31A** of the switching element **31** faces along the thickness direction **z**.

As shown in FIGS. **15** and **16**, each of the wires **41** for each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A** has a second bonding portion **412**. The second bonding portion **412** is held in contact with the anode electrode **321** of the protective element **32**. Thus, the anode electrodes **321** of the protective elements **32** electrically bonded to the first upper arm mounting layer **211A** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the first lower arm mounting layer **211B**. The anode electrodes **321** of the protective elements **32** electrically bonded to the second upper arm mounting layer **221A** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the second lower arm mounting layer **221B**. The anode electrodes **321** of the protective elements **32** electrically bonded to the third upper arm mounting layer **231A** are electrically connected to both of

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the front surface electrodes **311** of the corresponding switching elements **31** and the third lower arm mounting layer **231B**.

With reference to FIGS. **18-20**, description is given below of the configuration of the wires **41** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIG. **20**, the first bonding portions **411** of the paired inner wires **41A** are held in contact with the paired first pads **311A** of the front surface electrode **311**. As shown in FIG. **19**, the first bonding portions **411** of the paired outer wires **41B** are held in contact with both of the paired first pads **311A** and the paired second pads **311B** of the front surface electrode **311**. As shown in FIGS. **18** and **19**, each of the first bonding portions **411** of the paired outer wires **41B** has a first connect portion **411A**, a second connect portion **411B** and a joint portion **411C**. The first connect portion **411A** is held in contact with a first pad **311A**. The second connect portion **411B** is held in contact with a second pad **311B**. The joint portion **411C** is sandwiched between the first connect portion **411A** and the second connect portion **411B** in the first direction **x1**. The joint portion **411C** projects in the same direction in which the front surface **31A** of the switching elements **31** faces along the thickness direction **z**.

As shown in FIGS. **18** and **19**, each of the paired outer wires **41B** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B** has a second bonding portion **412**. The second bonding portion **412** is held in contact with the anode electrode **321** of the protective element **32**. Thus, the anode electrodes **321** of the protective elements **32** electrically bonded to the first lower arm mounting layer **211B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the first electroconductive layer **212**. The anode electrodes **321** of the protective elements **32** electrically bonded to the second lower arm mounting layer **221B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the second electroconductive layer **222**. The anode electrodes **321** of the protective elements **32** electrically bonded to the third lower arm mounting layer **231B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the third electroconductive layer **232**.

As shown in FIG. **18**, a pair of auxiliary wires **46** are connected to the anode electrode **321** of each of the protective elements **32** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. The auxiliary wires **46** are connected at the other ends thereof to paired second pads **311B** of the front surface electrode **311** of the switching element **31** with which the protective element **32** is associated. The auxiliary wires **46** are located between paired outer wires **41B** in the second direction **x2**. The auxiliary wires **46** are made of the same material as that for the wires **41**. The diameter of the auxiliary wires **46** are equal to that of the wires **41**.

As shown in FIGS. **15-20**, the moisture-resistant layer **51** covers the side surface **31C** of switching elements **31**. As the material for the moisture-resistant layer **51**, an electrically insulating material having a high resistance to temperature cycling and a lower moisture permeability than the sealing resin **52** (silicone gel in the semiconductor device **A10**) is selected. As such an electrically insulating material for the moisture-resistant layer **51**, polyimide and silicone gel are

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selected. The ratio of content by weight of polyimide to silicone gel in the moisture-resistant layer **51** is 1.5:1 to 7.0:1. That is, in the moisture-resistant layer **51**, the weight of polyimide is larger than that of silicone gel. In the moisture-resistant layer **51**, polyimide molecules and silicone gel molecules are mixed. Preferably, polyimide molecules and silicone gel molecules are uniformly dispersed throughout the moisture-resistant layer **51**. This effectively prevents the moisture-resistant layer **51** from cracking due to temperature cycling, so that the moisture-resistant layer **51** maintains the function of preventing moisture intrusion. Although the moisture-resistant layer **51** consisting of polyimide and silicone gel alone is explained for the semiconductor device **A10**, other materials may be added to these materials to form the moisture-resistant layer **51**. Further, although a mixture of polyimide and silicone gel is selected as the material for the moisture-resistant layer **51** in the semiconductor device **A10**, other materials having low moisture permeability may be selected. For example, the moisture-resistant layer **51** may be made of a mixture of polybenzoxazole and silicone gel.

An example of a method for forming the moisture-resistant layer **51** of the semiconductor device **A10** is described below. A liquefied synthetic resin material containing polyimide, silicone gel and solvent is prepared. Note that the solvent is volatile. Then, the synthetic resin material is dropped with a dispenser onto the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**). As a result, the synthetic resin material spreads on the side surfaces **31C** of the switching elements **31**, so that the side surfaces **31C** are covered with the synthetic resin material. Finally, the synthetic resin material is heat-cured to obtain the moisture-resistant layer **51**. In this process, the solvent volatilizes. With this method, the moisture-resistant layer **51** covering the side surfaces **31C** of the switching elements **31** is easily formed.

As shown in FIGS. **15-17**, the moisture-resistant layer **51** is held in contact with one of the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A** and with the side surface **31C** of at least one of the switching elements **31**. In the thickness direction **z**, the moisture-resistant layer **51** extends to be spanned between the side surface **31C** and the first upper arm mounting layer **211A** or the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**, thereby crossing over the bonding layer **39** and the back surface electrode **312**.

As shown in FIGS. **18-20**, the moisture-resistant layer **51** is held in contact with the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B** and with the side surface **31C** of switching elements **31**. In the thickness direction **z**, the moisture-resistant layer **51** extends to be spanned between the side surface **31C** and the first lower arm mounting layer **211B** or the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**, thereby crossing over the bonding layer **39** and the back surface electrode **312**.

Thus, the moisture-resistant layer **51** is held in contact with the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231** and with at least one of the side surfaces **31C**. In the thickness direction, the moisture-resistant layer **51** extends to be spanned between the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231** and the side surface **31C**.

As shown in FIGS. **15**, **16**, **18** and **19**, the moisture-resistant layer **51** integrally covers the side surface **31C** of a

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switching element 31 and the side surface 32C of the protective element 32 paired with that switching element 31 (the protective element 32 connected anti-parallel to the switching element 31). In the example shown in these figures, the moisture-resistant layer 51 is provided correspondingly to the pairs of switching elements 31 and protective element 32. That is, the moisture-resistant layer 51 is divided into a plurality of sections such that each section covers a pair of the side surface 31C of a switching element 31 and the side surface 32C of a protective element 32. Unlike this configuration, the moisture-resistant layer 51 may be configured to integrally cover the side surfaces 31 of a plurality of switching elements 31 on each of the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231.

As shown in FIGS. 11 and 12, the sealing resin 52 is housed in a region surrounded by the case 70 and the heat sink 61. As shown in FIGS. 16, 17, 19 and 20, the sealing resin 52 covers both of the switching elements 31 and the moisture-resistant layer 51. The sealing resin 52 also covers the protective elements 32. It is preferable that the sealing resin 52 is an electrically insulating synthetic resin with excellent heat resistance and adhesion. For example, the sealing resin 52 is silicone gel mainly composed of thermosetting organopolysiloxane. The sealing resin 52 is exposed to the atmosphere.

As shown in FIGS. 11 and 12, the heat sink 61 is bonded to the back surface 112 of the substrate 11. In the semiconductor device A10, the heat sink 61 is bonded to each of the back surface 112 of the first substrate 11A, the back surface 112 of the second substrate 11B and the back surface 112 of the third substrate 11C via a heat transfer layer 62 and a substrate bonding layer 69 (both described later). For example, the heat sink 61 is made of a metal plate such as a copper plate. The surface of the metal plate may be plated with nickel. As shown in FIGS. 7-9, the heat sink 61 is provided with a plurality of support holes 611 at its four corners as viewed in the thickness direction z. Each of the support holes 611 penetrates the heat sink 61 in the thickness direction z. The support holes 611 are used to support the heat sink 61, which is bonded to the substrate 11, on the case 70.

As shown in FIGS. 11 and 12, the heat transfer layer 62 is disposed on the back surface 112 of the substrate 11. The heat transfer layer 62 is made of a metallic material such as a copper foil. The heat transfer layer 62 transfers the heat generated by driving the switching elements 31 to the heat sink 61.

The substrate bonding layer 69 is a bonding material interposed between the heat sink 61 and the heat transfer layer 62, as shown in FIGS. 11 and 12. In the semiconductor device A10, the substrate bonding layer 69 is made of lead-free solder mainly composed of tin. The substrate bonding layer 69 bonds the heat sink 61 to the substrate 11.

The case 70 is an electrically insulating member surrounding the substrate 11 as viewed in the thickness direction z, as shown in FIG. 3. The case 70 is in the form of a frame. The case 70 is made of an electrically insulating synthetic resin with excellent heat resistance such as PPS (polyphenylene sulfide). The case 70 has a pair of side walls 71, a pair of terminal seats 72, mount parts 73, a power supply terminal base 74 and an output terminal base 75.

As shown in FIGS. 2, 3, 5 and 6, the paired side walls 71 are spaced apart from each other in the first direction x1 and in the form of a groove. Each of the side walls 71 is arranged along both of the second direction x2 and the thickness direction z, and one end of each side wall in the thickness

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direction z is held in contact with the heat sink 61. Opposite ends of each side wall 71 in the second direction x2 are connected to the paired terminal seats 72. In one of the side walls 71, the first gate terminal 27A, the first detection terminal 281A, the power supply current detection terminal 282 and the paired thermistor terminals 29 are disposed. In the other side wall 71, the second gate terminal 27B and the second detection terminal 281B are disposed. As shown in FIGS. 8-10, the ends of these terminals that are close to the substrate 11 in the thickness direction z are supported on the side walls 71.

As shown in FIGS. 3, 8 and 9, the paired terminal seats 72 are spaced apart from each other in the second direction x2. Each of the terminal seats 72 is disposed along the second direction x2. To one of the terminal seats 72 is connected the power supply terminal base 74 that projects outward in the second direction x2, and a part of the power supply terminal 24 is supported on the terminal seat 72. To the other one of the terminal seats 72 is connected the output terminal base 75 that projects outward in the second direction x2, and a part of the output terminal 25 is supported on the terminal seat 72.

As shown in FIGS. 2, 8 and 9, the mount parts 73 are provided at four corners of the case 70 as viewed in the thickness direction z. Each of the mount parts 73 is provided with a mount hole 731 penetrating the mount part 73 in the thickness direction z. The positions of the mount holes 731 correspond to the support holes 611 provided in the heat sink 61. The heat sink 61 is supported on the case 70 by inserting fastening members such as pins into the mount holes 731 and the support holes 611.

As shown in FIGS. 2, 5 and 8, the power supply terminal base 74, along with the terminal seat 72 connected thereto, supports the power supply terminal 24. The power supply terminal base 74 includes a first terminal base 741 and a second terminal base 742. The first terminal base 741 and the second terminal base 742 are spaced apart from each other in the first direction x1. On the first terminal base 741 is supported a part of the first power supply terminal 24A, and the supported part is exposed outside the semiconductor device A10. On the second terminal base 742 is supported a part of the second power supply terminal 24B, and the supported part is exposed outside the semiconductor device A10. As shown in FIGS. 8 and 13, a nut 743 is disposed in each of the first terminal base 741 and the second terminal base 742. Each nut 743 corresponds in the thickness direction z to the coupling hole 241 provided in the first power supply terminal 24A or the second power supply terminal 24B. The fastening member such as a bolt inserted in the coupling hole 241 is in threaded engagement with a nut 743.

As shown in FIGS. 2, 6 and 9, the output terminal base 75, along with the terminal seat 72 connected thereto, supports the output terminal 25. The output terminal base 75 includes a first terminal base 751 and a second terminal base 752. The first terminal base 751 and the second terminal base 752 are spaced apart from each other in the first direction x1. On the first terminal base 751 is supported a part of the first output terminal 25A, and the supported part is exposed outside the semiconductor device A10. On the second terminal base 752 is supported a part of the second output terminal 25B, and the supported part is exposed outside the semiconductor device A10. As shown in FIGS. 9 and 14, a nut 753 is disposed in each of the first terminal base 751 and the second terminal base 752. Each nut 753 corresponds in the thickness direction z to the coupling hole 251 provided in the first output terminal 25A or the second output terminal 25B. The

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fastening member such as a bolt inserted in the coupling hole 251 is in threaded engagement with a nut 753.

As shown in FIGS. 2, 11 and 12, the top plate 79 closes the interior of the semiconductor device A10 defined by the heat sink 61 and the case 70. The top plate 79 is supported on the paired side walls 71 of the case 70, facing the front surface 111 of the substrate 11 and being spaced apart from the front surface 111 in the thickness direction z. The top plate 79 is made of an electrically insulating synthetic resin.

Next, the circuit configuration in the semiconductor device A10 is described with reference to FIG. 21.

As shown in FIG. 21, two switching circuits, namely an upper arm circuit 81 and a lower arm circuit 82 are formed in the semiconductor device A10. The upper arm circuit 81 may be made up of the first upper arm mounting layer 211A, the second upper arm mounting layer 221A, the third upper arm mounting layer 231A, and the switching elements 31 and the protective elements 32 electrically bonded to these mounting layers. The switching elements 31 and the protective elements 32 electrically bonded to these mounting layers are connected in parallel between the first power supply terminal 24A and the output terminal 25. The gate electrodes 313 of the switching elements 31 in the upper arm circuit 81 are connected in parallel to the first gate terminal 27A. The switching elements 31 in the upper arm circuit 81 are driven simultaneously by the application of a gate voltage to the first gate terminal 27A using a driving circuit such as a gate driver disposed outside the semiconductor device A10.

The front surface electrodes 311 of the switching elements 31 in the upper arm circuit 81 are connected in parallel to the first detection terminal 281A. The source current flowing through the switching elements 31 in the upper arm circuit 81 is inputted to a control circuit disposed outside the semiconductor device A10 via the first detection terminal 281A.

In the upper arm circuit 81, the voltage applied by the first power supply terminal 24A and the second power supply terminal 24B to the first upper arm mounting layer 211A, the second upper arm mounting layer 221A and the third upper arm mounting layer 231A is inputted to the control circuit disposed outside the semiconductor device A10 via the power supply current detection terminal 282.

The lower arm circuit 82 may be made up of the first lower arm mounting layer 211B, the second lower arm mounting layer 221B, the third lower arm mounting layer 231B, and the switching elements 31 electrically bonded to these mounting layers, and the protective elements 32 are connected in parallel between the output terminal 25 and the second power supply terminal 24B. The gate electrodes 313 of the switching elements 31 in the lower arm circuit 82 are connected in parallel to the second gate terminal 27B. The switching elements 31 in the lower arm circuit 82 are driven simultaneously by the application of a gate voltage to the second gate terminal 27B using a driving circuit such as a gate driver disposed outside the semiconductor device A10.

The front surface electrodes 311 of the switching elements 31 in the lower arm circuit 82 are connected in parallel to the second detection terminal 281B. The source current flowing through the switching elements 31 in the lower arm circuit 82 is inputted to a control circuit disposed outside the semiconductor device A10 via the second detection terminal 281B.

Alternating voltages of various frequencies are output from the output terminal 25 by connecting a DC power supply to the first power supply terminal 24A and the second power supply terminal 24B and driving the switching ele-

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ments 31 in the upper arm circuit 81 and the lower arm circuit 82. The alternating voltage output from the output terminal 25 is supplied to a power supply target such as a motor.

The advantages of the semiconductor device A10 are described below.

With the configuration of the semiconductor device A10, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. When moisture enters the sealing resin 52 due to the effect of high temperature and high humidity, a leakage current L_C is likely to be generated from the front surface electrode 311 of the switching element 31, as shown in FIG. 22. On the switching element 31, the leakage current L_C tries to flow along the front surface of the insulating film 314 and the side surface 31C. The provision of the moisture-resistant layer 51 makes the path of the leakage current L_C longer, making it difficult for the leakage current L_C to flow. Since the leakage current L_C is thus prevented from reaching the mounting layer, breakdown of the switching element 31 due to the flow of the leakage current L_C is prevented. Thus, the semiconductor device A10 operates stably under high temperature and high humidity conditions. When the switching element 31 or the protective element 32 has a relatively small thickness of 150 μm or less, the path of the leakage current L_C becomes relatively short. When a voltage of 1,200 V or more is applied to such a semiconductor device, the leakage current L_C flows relatively easily. Provision of the moisture-resistant layer 51 is particularly effective for such a relatively thin switching element 31.

In the comparative example B10 shown in FIG. 23, which does not include a moisture-resistant layer 51, the leakage current L_C is conducted to the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) along the side surface 31C of the switching element 31. This causes a short circuit between the front surface electrode 311 and the back surface electrode 312 of the switching element 31, resulting in breakdown of the switching element 31.

It is preferable that the moisture-resistant layer 51 contains polyimide. Polyimide is an electrically insulating material that is resistant to temperature cycling and not easily affected by moisture. Thus, by containing polyimide, the moisture-resistant layer 51 reliably prevents the leakage current L_C from flowing along the side surface 31C as shown in FIG. 22 even under high temperature and high humidity conditions.

It is preferable that the moisture-resistant layer 51 contains silicone gel in addition to polyimide. Such a moisture-resistant layer 51 has a lower Young's modulus as compared with a moisture-resistant layer 51 made of polyimide alone. Thus, the moisture-resistant layer 51 easily follows the thermal strain of the switching element 31 during the use of the semiconductor device A10. This reduces the shearing stress acting on the switching element 31.

Moreover, containing polyimide and silicone gel in the moisture-resistant layer 51 enhances the resistance of the moisture-resistant layer 51 to temperature cycling. A semiconductor device A10 with a moisture-resistant layer 51 made of polyimide alone was subjected to a temperature cycling test in a range of -40 to 125°C . As a result, the semiconductor device A10 was broken after about 20 cycles. Presumably, a crack was formed in the moisture-resistant

layer 51, and moisture entering through the crack caused the breakdown. A semiconductor device A10 with a moisture-resistant layer 51 made of polyimide and silicone gel was subjected to the same temperature cycling test, and the semiconductor device A10 was not broken even after 1000 cycles. This is because the Young's modulus of this moisture-resistant layer 51 is lower than that of the moisture-resistant layer 51 made of polyimide alone, and hence the shearing stress acting on the moisture-resistant layer 51 due to thermal expansion or contraction is reduced. Thus, it is preferable that the moisture-resistant layer 51 contains polyimide and silicone gel.

The semiconductor device A10 includes wires 41 connected to the front surface electrodes 311 of the switching elements 31, and the wires 41 extend in the first direction x1. Thus, the moisture-resistant layer 51 that covers the side surface 31C of the switching element 31 can be formed without being hindered by the wires 41.

The first bonding portion 411 of each of the paired outer wires 41B has the first connect portion 411A held in contact with the first pad 311A, the second connect portion 411B held in contact with the second pad 311B, and the joint portion 411C located between the first connect portion 411A and the second connect portion 411B. The joint portion 411C projects in the same direction in which the front surface 31A of the switching element 31 faces along the thickness direction z. As shown in FIGS. 16 and 19, it is preferable that the height H of the joint portion 411C in the thickness direction z from the front surface of the front surface electrode 311 of the switching element 31 to the top C of the joint portion 411C is not less than three times the diameter of the wires 41. For example, when the diameter of the wires 41 is 300 μm , it is preferable that the height H of the joint portion 411C is 900 μm or more. With such a configuration, the joint portion 411C functions as an elastic member capable of elastically deforming in the first direction x1 to reduce the shearing stress acting on the first connect portion 411A and the second connect portion 411B. Thus, detachment of the first bonding portion 411 from the front surface electrode 311 due to shearing stress is prevented. In the semiconductor device A10, the diameter of the wires 41 is 400 μm , and the height H of the joint portion 411C is 1,600 μm . Note that when the height H of the joint portion 411C in the semiconductor device A10 is 800 μm , at least one of the first connect portion 411A or the second connect portion 411B may be detached in the ΔT_j power cycling test described later.

The semiconductor device A10 has the heat sink 61 bonded to the back surface 112 of the substrate 11. Thus, the heat generated at the switching elements 31 is efficiently dissipated outside the semiconductor device A10. In this case, it is preferable that the substrate 11 is made of a ceramic with excellent thermal conductivity (e.g. aluminum nitride).

FIGS. 24-50 show semiconductor devices A11 to A15 that are variations of the semiconductor device A10. [First Variation] A semiconductor device A11 according to a first variation of the semiconductor device A10 is described with reference to FIGS. 24 and 25. The semiconductor device A11 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is smaller than that in the foregoing semiconductor device A10. Note that FIG. 24 is a sectional view taken along the same plane as FIG. 16. FIG. 25 is a sectional view taken along the same plane as FIG. 19.

As shown in FIGS. 24 and 25, the moisture-resistant layer 51 covers a part of the side surface 31C of the switching element 31.

The advantages of the semiconductor device A11 are described below.

With the configuration of the semiconductor device A11, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A11 also operates stably under high temperature and high humidity conditions.

[Second Variation]

A semiconductor device A12 according to a second variation of the semiconductor device A10 is described with reference to FIGS. 26-31. The semiconductor device A12 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is larger than that in the foregoing semiconductor device A10.

As shown in FIGS. 26-31, at the switching element 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314.

The advantages of the semiconductor device A12 are described below.

With the configuration of the semiconductor device A12, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A12 also operates stably under high temperature and high humidity conditions.

At the switching element 31 of the semiconductor device A12, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314. Such a configuration makes the path of the leakage current L_C shown in FIG. 22 longer than that in the semiconductor device A10, making it more difficult for the leak current L_C to flow than in the semiconductor device A10. Moreover, since the moisture-resistant layer 51 covers the insulating film 314, the insulating film 314 is protected from external factors.

[Third Variation]

A semiconductor device A13 according to a third variation of the semiconductor device A10 is described with reference to FIGS. 32-37. The semiconductor device A13 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is larger than that in the foregoing semiconductor device A12.

As shown in FIGS. 32-37, at the switching element 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314. Moreover, the moisture-resistant layer 51 is held in contact with at least a part of the front surface electrode 311.

The advantages of the semiconductor device A13 are described below.

With the configuration of the semiconductor device A13, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A13 also operates stably under high temperature and high humidity conditions.

At the switching element 31 of the semiconductor device A13, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314, and also in contact with at least a part of the front surface electrode 311. As shown in FIGS. 32 and 35, as viewed in the thickness direction z, the moisture-resistant layer 51 surrounds the front surface electrode 311 while overlapping with a part of the front surface electrode 311. This configuration improves the dielectric breakdown voltage of the side surface 31C as compared with that in the semiconductor device A12, making it more difficult for the leak current L_C to flow than in the semiconductor device A12. Moreover, since the moisture-resistant layer 51 covers the insulating film 314, the insulating film 314 is protected from external factors.

[Fourth Variation]

A semiconductor device A14 according to a fourth variation of the semiconductor device A10 is described with reference to FIGS. 38-43. The semiconductor device A14 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is larger than that in the foregoing semiconductor device A13.

As shown in FIGS. 38-43, at the switching element 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314. Moreover, the moisture-resistant layer 51 is held in contact with the front surface electrode 311 and at least a part of the first bonding portions 411 of the wires 41. Thus, the switching elements 31 are entirely covered with the moisture-resistant layer 51. However, the first bonding portions 411 are not completely covered with the moisture-resistant layer 51, and the upper ends of the first bonding portions 411 are exposed from the moisture-resistant layer 51. That is, the thickness of the moisture-resistant layer 51 covering the front surface 31A of the switching element 31 is smaller than the diameter of the wires 41.

As shown in FIGS. 38, 39, 41 and 42, the moisture-resistant layer 51 covers the entirety of the front surface of the protective element 32 associated with the switching element 31. However, the second bonding portions 412 of the wires 41 are not completely covered with the moisture-resistant layer 51, and the upper ends of the second bonding portions 411 are exposed from the moisture-resistant layer 51. That is, the thickness of the moisture-resistant layer 51 covering the front surface 32A of the protective element 32 is smaller than the diameter of the wires 41.

An example of a method for forming the moisture-resistant layer 51 of the semiconductor device A14 is described below. A liquefied synthetic resin material containing polyimide, silicone gel and solvent is prepared. Note that the solvent is volatile. Then, after the switching elements 31 and the protective elements 32 are electrically connected to the mounting layer on which these elements are mounted (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231), the synthetic

resin material is dropped onto respective upper surfaces of a switching element 31 and a protective element 32 with a dispenser. Since the synthetic resin material has fluidity, it spreads over the entirety of the upper surface of the switching element 31 including the front surface electrode 311, the gate electrode 313 and the insulating film 314, and further spreads from the side surface 31C of the switching element 31 onto the mounting layer. Similarly, at the protective element 32, the entirety of the front surface of the protective element 32 is covered with the synthetic resin material. Thus, the entirety of the front surface of the switching element 31 is covered with the synthetic resin material. Due to the surface tension of the synthetic resin material, the thickness of the synthetic resin material on the upper surface of the switching element 31 becomes generally uniform. Finally, the synthetic resin material is heat-cured to obtain the moisture-resistant layer 51. In this process, the solvent volatilizes. With this method, the moisture-resistant layer 51 covering the switching element 31 and the protective element 32 is easily formed.

The advantages of the semiconductor device A14 are described below.

With the configuration of the semiconductor device A14, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A14 also operates stably under high temperature and high humidity conditions.

At each of the switching elements 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 is also held in contact with the front surface electrode 311 and at least a part of the first bonding portions 411 of the wires 41. Since the entirety of the switching element 31 is covered with the moisture-resistant layer 51 in this way, moisture entering the sealing resin 52 is prevented from reaching the front surface of the switching element 31. Thus, dielectric breakdown of the switching elements 31 caused by the leakage current L_C shown in FIG. 22, which is due to the influence of moisture, is effectively prevented. Moreover, since the moisture-resistant layer 51 covers the insulating film 314, the insulating film 314 is protected from external factors.

At each of the switching elements 31, the paired outer wires 41B, each of which has the joint portion 411C of the first bonding portion 411 that projects in the thickness direction z, are arranged on the opposite sides of the paired inner wires 41A in the second direction x2. Thus, the synthetic resin material for forming the moisture-resistant layer 51 can be dropped from above the front surface 31A of the switching element 31 without interfering with the joint portion 411C. As shown in FIG. 38, in the semiconductor device A14, the joint portions 411C are provided only in the first bonding portions 411 of the paired outer wires 41B. It may be considered, as another example, to provide the joint portions 411C not only in the first bonding portions 411 of the paired outer wires 41B but also in the first bonding portions 411 of the paired inner wires 41A. With such a configuration, however, two adjacent joint portions 411C are located close to each other, which makes it difficult to drop the synthetic resin material for forming the moisture-resistant layer 51 onto the switching element 31. Moreover, when the synthetic resin material is dropped on the switching

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element 31, the synthetic resin material may rise to the upper end of the joint portions 411C. In such a case, since the Young's modulus of the moisture-resistant layer 51 is relatively high, heat generated from the switching elements 31 exerts a large shearing stress on the joint portions 411C. This may result in detachment of the first connect portions 411A and the second connect portions 411B of the first bonding portions 411 from the front surface electrode 311 of the switching element 31. Thus, in terms of the reliability of the semiconductor device A14, it is preferable that a larger distance is secured between two adjacent joint portions 411C in the second direction x2.

The moisture-resistant layer 51 also covers the entirety of the protective element 32 associated with the switching element 31 which the moisture-resistant layer covers. Thus, protective element 32 is effectively protected from external factors. On the other hand, the first bonding portions 411 are not completely covered with the moisture-resistant layer 51, and the upper ends of the first bonding portions 411 are exposed from the moisture-resistant layer 51. That is, the thickness of the moisture-resistant layer 51 covering the front surface 31A of the switching element 31 is smaller than the diameter of the wires 41. With this configuration, an excessively large shearing stress is less likely to act on the first bonding portions 411, as compared with the configuration in which the first bonding portions 411 are completely covered with the moisture-resistant layer 51. Thus, detachment of the first bonding portions 411 from the front surface electrode 311 of the switching element 31 is prevented, which enhances the reliability of the semiconductor device A14.

Next, with reference to FIG. 50, description is given of a favorable thickness of the moisture-resistant layer 51 in the semiconductor device A14. FIG. 50 shows the results of a H3TRB test and a ΔT_j power cycling test with varying thicknesses of the moisture-resistant layer 51 of the semiconductor device A14. The thickness of the moisture-resistant layer 51 shown in FIG. 50 is the thickness at the corner of the insulating film 314 (i.e., the portion connected to both of the edge 314A and the side surface 31C) of a switching element 31. Before the H3TRB test is conducted, the semiconductor device A14 has been subjected to temperature cycles from -40 to 125°C . The number of the temperature cycles was 300. In the H3TRB test, the semiconductor device A14 was driven at a DC voltage of 1,360 V as explained later. In the ΔT_j power cycling test, the temperature ΔT_j of the first bonding layer 391 for electrically bonding the switching elements 31 to the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) was set to 100°C . In light of this, the range of the temperature cycles in the ΔT_j power cycling test was from 50 to 150°C .

The left vertical axis in FIG. 50 is for indicating the withstand time of the semiconductor device A14 in the H3TRB test. The "withstand time" means the time from the start of the test to the time when dielectric breakdown is observed in at least one of the switching elements 31 of the semiconductor device A14. The right vertical axis in FIG. 50 is for indicating the number of temperature cycles done in the ΔT_j power cycling test before the first bonding portion 411 of a wire 41, connected to the front surface electrode 311 of a switching element 31, was detached from the front surface electrode 311 (hereinafter referred to as " ΔT_j power cycle"). The desirable number of temperature cycles (or the standard value of the ΔT_j power cycle, indicated in FIG. 50) is 15,000 times. The horizontal axis in FIG. 50 represents the thickness of the moisture-resistant layer 51.

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As shown in FIG. 50, the withstand time of the semiconductor device A14 increases sharply when the thickness of the moisture-resistant layer 51 exceeds $10\text{ }\mu\text{m}$. This indicates that the resistance to breakdown of the switching element 31 due to moisture intrusion (or reliability related to moisture absorption) improves with increasing thickness of the moisture-resistant layer 51. On the other hand, the ΔT_j power cycle gradually reduces with increasing thickness of the moisture-resistant layer 51. This indicates that increasing the thickness of the moisture-resistant layer 51 leads to an increased risk of detachment of the first bonding portion 411 of the wire 41 from the front surface electrode 311 of the switching element 31 or detachment of the second bonding portion 412 of the wire 41 from the anode electrode 321 of the protective element 32. These test results reveal that preferable thickness of the moisture-resistant layer 51 is in the range from 40 to $200\text{ }\mu\text{m}$. More preferable range of the thickness of the moisture-resistant layer 51 may be from 50 to $100\text{ }\mu\text{m}$. Experiments have confirmed that the thickness of the moisture-resistant layer 51 on the upper surface of the switching element 31 is 1.2 times the thickness of the moisture-resistant layer 51 at the corners. Accordingly, the preferable thickness of the moisture-resistant layer 51 on the upper surface of the switching element 31 is from 48 to $240\text{ }\mu\text{m}$, and more preferably, from 60 to $120\text{ }\mu\text{m}$.

FIG. 51 shows the results (unit: h) of the H3TRB test performed on the semiconductor device A14 and a comparative example B10 that does not include the moisture-resistant layer 51 shown in FIG. 23. As described before, a semiconductor device determined to be acceptable in the H3TRB test (the device withstand time is 1000 h or more) is expected to operate stably under high temperature and high humidity conditions. In the H3TRB test, when the rating voltage is $1,700\text{ V}$, the DC voltage for driving the semiconductor device A14 and the comparative example B10 is set to $1,360\text{ V}$ (80% of the rating voltage). As a result of the H3TRB test performed based on this DC voltage, the withstand time of the semiconductor device A14 was found to 1000 h or more, which is acceptable. Thus, the semiconductor device A14 is expected to operate stably under high temperature and high humidity conditions. On the other hand, the withstand time of the comparative example B10 was 10 to 500 h , which is not acceptable. The comparative example B10 is considered to be inferior to the semiconductor device A14 in terms of the capability of stable operation under high temperature and high humidity conditions.

As shown in FIG. 51, the insulation resistance reduction rate (unit: %) of the sealing resin 52 during the H3TRB test was 20% in the semiconductor device A14 and 84% in the comparative example B10. Presumably, the insulation resistance reduction rate of the sealing resin 52 shown in FIG. 51 were obtained because even when moisture enters the sealing resin 52 due to a high temperature and high humidity environment, the moisture-resistant layer 51 hinders the leakage current L_C shown in FIG. 22 from flowing along the side surfaces 31C of the switching element.

[Fifth Variation]

A semiconductor device A15 according to a fifth variation of the semiconductor device A10 is described with reference to FIGS. 44-49. The semiconductor device A15 is an example in which the thickness of the moisture-resistant layer 51 on the upper surface of a switching element 31 is larger than that in the foregoing semiconductor device A14.

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As shown in FIGS. 44-49, at the switching elements 31, the moisture-resistant layer 51 covers both of the switching elements 31 and the first bonding portions 411 of the wires 41.

As shown in FIGS. 44, 45, 47 and 48, the moisture-resistant layer 51 covers the entirety of the front surface of the protective element 32 associated with the switching element 31 and the second bonding portions 412 of the wires 41 connected to the anode electrode 321 of the protective element 32.

The advantages of the semiconductor device A15 are described below.

With the configuration of the semiconductor device A15, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A15 also operates stably under high temperature and high humidity conditions.

At the switching element 31, the moisture-resistant layer covers both of the switching element 31 and the first bonding portions 411 of the wires 41. Since the entirety of the switching element 31 is covered with the moisture-resistant layer 51 in this way, moisture entering the sealing resin 52 is prevented from reaching the front surface 31A of the switching element 31. Thus, dielectric breakdown of the switching elements 31 caused by the leakage current I_C shown in FIG. 22, which is due to the influence of moisture, is effectively prevented. Moreover, since the moisture-resistant layer 51 covers the insulating film 314, the insulating film 314 is protected from external factors. In the semiconductor device A15 again, it is preferable that the thickness of the moisture-resistant layer 51 on the upper surface of the switching element 31 is from 48 to 240 μm .

Second Embodiment

A semiconductor device A20 according to a second embodiment of the present disclosure is described below with reference to FIGS. 52-57. In these figures, the elements that are identical or similar to those of the foregoing semiconductor device A10 are designated by the same reference signs as those used for the foregoing embodiment and, descriptions thereof are omitted.

The semiconductor device A20 differs from the foregoing semiconductor device A10 in that it includes clips 47 instead of the wires 41.

As shown in FIGS. 52-54, the clips 47 are electrically bonded to the front surface electrodes 311 of the switching elements 31 and the first lower arm mounting layer 211B, the second lower arm mounting layer 221B or the third lower arm mounting layer 231B. As shown in FIGS. 55-57, the clips 47 are electrically bonded to the front surface electrodes 311 of the switching elements 31 and the first electroconductive layer 212, the second electroconductive layer 222 or the third electroconductive layer. The clips 47 are formed by bending a thin metal plate such as a copper plate. As shown in FIGS. 52 and 55, the clips 47 are each in the form of a strip extending in the first direction x1 as viewed in the thickness direction z. As shown in FIGS. 53 and 56, the clips 47 have a hook-like shape as viewed in the second direction x2. As shown in FIGS. 53 and 56, the clips 47 are electrically bonded to an object such as the front surface electrode 311 by using a clip bonding layer 49. The

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clip bonding layer 49 is electrically conductive. For example, the clip bonding layer 49 is made of lead-free solder mainly composed of tin. To use the clip bonding layer 49, a plating layer of nickel or gold, for example, is applied to the front surface of the front surface electrode 311. In the semiconductor device A20, when the front surface electrode 311 is covered with the moisture-resistant layer 51, the clip bonding layer 49 and the plating layer are also covered with the moisture-resistant layer 51.

As shown in FIGS. 52-54, in the switching elements 31 electrically bonded to the first upper arm mounting layer 211A, the clips 47 are electrically bonded to the front surface electrodes 311 and the first lower arm mounting layer 211B. As shown in FIGS. 55-57, in the switching elements 31 electrically bonded to the first lower arm mounting layer 211B, the clips 47 are electrically bonded to the front surface electrodes 311 and the first electroconductive layer 212. Thus, the front surface electrodes 311 of the switching elements 31 electrically bonded to the first mounting layer 211 are electrically connected to the first lower arm mounting layer 211B or the first electroconductive layer 212.

As shown in FIGS. 52-54, in the switching elements 31 electrically bonded to the second upper arm mounting layer 221A, the clips 47 are electrically bonded to the front surface electrodes 311 and the second lower arm mounting layer 221B. As shown in FIGS. 55-57, in the switching elements 31 electrically bonded to the second lower arm mounting layer 221B, the clips 47 are electrically bonded to the front surface electrodes 311 and the second electroconductive layer 222. Thus, the front surface electrodes 311 of the switching elements 31 electrically bonded to the second mounting layer 221 are electrically connected to the second lower arm mounting layer 221B or the second electroconductive layer 222.

As shown in FIGS. 52-54, in the switching elements 31 electrically bonded to the third upper arm mounting layer 231A, the clips 47 are electrically bonded to the front surface electrodes 311 and the third lower arm mounting layer 231B. As shown in FIGS. 55-57, in the switching elements 31 electrically bonded to the third lower arm mounting layer 231B, the clips 47 are electrically bonded to the front surface electrodes 311 and the third electroconductive layer 232. Thus, the front surface electrodes 311 of the switching elements 31 electrically bonded to the third mounting layer 231 are electrically connected to the third lower arm mounting layer 231B or the third electroconductive layer 232.

With reference to FIGS. 52 and 53, description is given below of the configuration of the clips 47 for each of the switching elements 31 electrically bonded to the first upper arm mounting layer 211A, the second upper arm mounting layer 221A or the third upper arm mounting layer 231A. As shown in FIG. 53, each clip 47 is electrically bonded also to the anode electrode 321 of the protective element 32 associated with the switching element 31 by using the clip bonding layer 49. Thus, the anode electrode 321 of the protective element 32 electrically bonded to the first upper arm mounting layer 211A is electrically connected to both of the front surface electrode 311 of the corresponding switching element 31 and the first lower arm mounting layer 211B. The anode electrode 321 of the protective element 32 electrically bonded to the second upper arm mounting layer 221A is electrically connected to both of the front surface electrode 311 of the corresponding switching element 31 and the second lower arm mounting layer 221B. Also, the anode electrode 321 of the protective element 32 electrically bonded to the third upper arm mounting layer 231A is

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electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the third lower arm mounting layer **231B**.

As shown in FIGS. **52** and **53**, each of the clips **47** has an opening **471** penetrating in the thickness direction **z**. The opening **471** is located between the front surface electrode **311** of the switching element **31** and the anode electrode **321** of the protective element **32** in the first direction **x**. As viewed in the thickness direction **z**, the edge **314A** of the insulating film **314** of the switching element **31** is visible through the opening **471**. When clips **47** are electrically bonded to the front surface electrodes **311** of the switching elements **31**, most of the switching elements **31** are covered with the clips **47**. By forming an opening **471** in each clip **47** at a location overlapping with the switching element **31** as viewed in the thickness direction, the synthetic resin material for forming the moisture-resistant layer **51** can be dropped under the clip **47**. Thus, the synthetic resin material can be dropped uniformly over the entirety of the switching element **31**. Although the semiconductor device **A20** is described as having an opening **471** in each clip **47**, a cutout penetrating in the thickness direction **z** may be formed, instead of the opening **471**, in each clip **46** at a location overlapping with the switching element **31** as viewed in the thickness direction **z**.

With reference to FIGS. **55** and **56**, description is given below of the configuration of the clips **47** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIG. **56**, each clip **47** is electrically bonded also to the anode electrode **321** of the protective element **32** associated with the switching element **31** by using the clip bonding layer **49**. Thus, the anode electrode **321** of the protective element **32** electrically bonded to the first lower arm mounting layer **211B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the first electroconductive layer **212**. The anode electrode **321** of the protective element **32** electrically bonded to the second lower arm mounting layer **221B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the second electroconductive layer **222**. Also, the anode electrode **321** of the protective element **32** electrically bonded to the third lower arm mounting layer **231B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the third electroconductive layer **232**. Since the clip **47** is electrically bonded to the anode electrode **321**, the paired auxiliary wires **46** shown in FIG. **18** are not connected to the anode electrode **321** in the semiconductor device **A20**.

As shown in FIGS. **55** and **56**, each of the clips **47** has a pair of openings **471** penetrating in the thickness direction **z**. The paired openings **471** are located on the opposite sides of the front surface electrode **311** of the switching element **31** in the first direction **x1**. As viewed in the thickness direction **z**, the edge **314A** of the insulating film **314** of the switching element **31** is visible through the paired openings **471**.

The advantages of the semiconductor device **A20** are described below.

With the configuration of the semiconductor device **A20**, the moisture-resistant layer **51** is held in contact with both of the side surfaces **31C** of the switching elements **31** and the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**. In the thickness direction, the moisture-resistant layer **51** extends to be spanned between the first mounting layer **211**, the second mounting layer **221**

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or the third mounting layer **231** and the side surfaces **31C**. Thus, the semiconductor device **A20** also operates stably under high temperature and high humidity conditions.

The configuration of the moisture-resistant layer **51** of the semiconductor device **A20** is the same as that of the semiconductor device **A10**. Note however that the configuration of the moisture-resistant layer **51** in the semiconductor devices **A11** to **A15** may be employed in the semiconductor device **A20**.

The semiconductor device **A20** has clips **47** instead of the wires **41**. The cross sectional area (the area in cross section along the second direction **x2**) of the clip **47** is larger than that of the wires **41**. Thus, the electric resistance of the clip **47** is lower than that of the wires **41**. Thus, the parasitic resistance of the semiconductor device **A20** is lower than that of the semiconductor device **A10**, so that the power loss of the semiconductor device **A20** is reduced as compared with the semiconductor device **A10**.

Since the cross sectional area of the clip **47** is larger than that of the wires **41**, the clip **47** conducts more heat in the first direction **x1** than the wires **41**. Thus, heat generated from the switching elements **31** is dissipated more efficiently. For example, on the first substrate **11A**, the heat generated from the switching elements **31** is likely to be accumulated in the first upper arm mounting layer **211A** that constitutes the upper arm circuit **81** shown in FIG. **21**. The clips **47** efficiently dissipate the heat accumulated in the first upper arm mounting layer **211A** to the first lower arm mounting layer **211B** and the first electroconductive layer **212**.

The number of the switching elements electrically bonded to each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** can be set appropriately in accordance with the required power conversion. The first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** are examples of a "mounting layer" as set forth in the appended claims of the present disclosure. The number of sections constituting the "mounting layer" is not limited to six as is in the present disclosure, and may be set appropriately.

The foregoing embodiments show the example in which the moisture-resistant layer **51** covers the switching elements **31** and the protective elements **32** connected antiparallel to the switching elements **31**. However, in a semiconductor device that does not use the protective elements **32** but uses the switching elements **31** alone (the configuration that does not use an external free wheeling diode), the moisture-resistant layer **51** may cover the switching elements **31**. Further, the present disclosure is applicable not only to switching elements but also to rectifier elements. For example, the present disclosure is applicable to a semiconductor device that includes a plurality of schottky-barrier diodes. In this case, the configuration such as the material, thickness or formation area of the moisture-resistant layer **51** is the same as that of the foregoing embodiments.

In the foregoing embodiments, as one example, the semiconductor device includes a substrate **11** on which electrically conductive members (the mounting layer and the electroconductive layer) made of a thin metal film are disposed, and the switching elements **31** electrically bonded to the electrically conductive members. The present disclosure is not limited to such an example and also applicable to a resin package-type semiconductor device that includes a lead frame on which elements such as switching elements or rectifier elements are electrically bonded and resin-molded. Since such a semiconductor device also has a risk of moisture intrusion through the sealing resin, covering the

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entire surfaces or side surfaces of the switching elements or rectifier elements with the moisture-resistant layer according to the present disclosure provides the same advantages. Note that the connection structure by wire bonding using the wires 41 in the semiconductor device A10 or the connection structure using a thin metal plate or clips 47 in the semiconductor device A20 is also applicable to a resin package-type semiconductor device.

The present disclosure is not limited to foregoing embodiments. The specific configuration of each part of the present disclosure may be varied in many ways.

The present disclosure at least includes the configurations related to the following clauses in addition to the configurations as set forth in the appended claims.

Clause 1.

A semiconductor device comprising:

- a first electroconductive layer;
 - a second electroconductive layer spaced apart from the first electroconductive layer;
 - a semiconductor element including a semiconductor layer, a front surface electrode provided on an upper surface of the semiconductor layer, and a back surface electrode provided on a lower surface of the semiconductor layer, the semiconductor element being mounted on the first electroconductive layer with the back surface electrode electrically connected to the first electroconductive layer;
 - a connection structure electrically connected to the front surface electrode and the second electroconductive layer;
 - a first insulating layer covering at least a side surface of the semiconductor element; and
 - a second insulating layer covering the first insulating layer,
- wherein the first insulating layer is made of a material having a lower moisture permeability than the second insulating layer. The first insulating layer functions as a barrier film for preventing moisture intrusion.

Clause 2.

The semiconductor device according to clause 1, wherein the first insulating layer covers an entirety of the semiconductor element.

Clause 3.

The semiconductor device according to clause 1, wherein the connection structure includes a connecting portion held in contact with the front surface electrode, and

the first insulating layer covers an entirety of the semiconductor element except the connecting portion.

Clause 4.

The semiconductor device according to clause 1, wherein the first insulating layer is from 40 to 200 μm in thickness at a corner located between an upper surface and the side surface of the semiconductor element, and

the first insulating layer is from 48 to 240 μm in thickness on the upper surface of the semiconductor element.

Clause 5.

The semiconductor device according to clause 4, wherein the first insulating layer is from 50 to 100 μm in thickness at the corner located between the upper surface and the side surface of the semiconductor element, and

the first insulating layer is from 60 to 120 μm in thickness on the upper surface of the semiconductor element.

Clause 6.

The semiconductor device according to clause 1, wherein the connection structure includes a connection structure using a wire, and

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a thickness of the first insulating layer on an upper surface of the semiconductor element is smaller than a diameter of the wire.

Clause 7.

The semiconductor device according to clause 6, wherein the wire includes a connecting portion held in contact with the front surface electrode, and

the thickness of the first insulating layer on the upper surface of the semiconductor element is smaller than a height of the connecting portion (a distance from a front surface of the front surface electrode to a top of the connecting portion). That is, the top of the connecting portion is exposed from the first insulating layer. The connecting portion is crushed with a wedge tool during a bonding process, and the height of the connecting portion is smaller than the diameter of the wire.

Clause 8.

The semiconductor device according to clause 1, wherein the semiconductor layer is 400 μm or less in thickness.

Clause 9.

The semiconductor device according to clause 8, wherein the semiconductor layer is 150 μm or less in thickness.

Clause 10.

The semiconductor device according to clause 1, wherein the semiconductor element further comprises a voltage withstanding structure including an insulating layer covering an upper surface of the semiconductor layer and surrounding an edge of the front surface electrode, and the first insulating layer covers the voltage withstanding structure. In the voltage withstanding structure, an oxide film or a nitride film is formed on the semiconductor layer, and a layer such as a polyimide layer or a polybenzoxazole layer is formed thereon as the insulating layer.

Clause 11.

The semiconductor device according to clause 1, wherein the first insulating layer contains a synthetic resin that is polyimide or polybenzoxazole.

Clause 12.

The semiconductor device according to clause 11, wherein the first insulating layer contains silicone gel.

Clause 13.

The semiconductor device according to clause 12, wherein the synthetic resin and the silicone gel are uniformly dispersed in the first insulating layer.

Clause 14.

The semiconductor device according to clause 12 or 13, wherein a content by weight of the synthetic resin in the first insulating layer is larger than that of the silicone gel.

Clause 15.

The semiconductor device according to clause 14, wherein the ratio of content by weight of the synthetic resin to the silicone gel in the first insulating layer is from 1.5:1 to 7.0:1.

Clause 16.

The semiconductor device according to clause 1, wherein the connection structure includes a connection structure using a wire and a connection structure using a thin metal plate.

Clause 17.

The semiconductor device according to clause 1, wherein the semiconductor layer is made of a semiconductor material mainly composed of silicon carbide.

Clause 18.

The semiconductor device according to clause 17, wherein the semiconductor element includes a MOSFET or a schottky-barrier diode.

Clause 19.

The semiconductor device according to clause 1, wherein breakdown voltage of the protective element is 1,200 V or more.

Clause 20.

The semiconductor device according to clause 1, wherein the first electroconductive layer and the second electroconductive layer are made of a lead frame, and

the second insulating layer comprises a resin package sealing the first electroconductive layer, the second electroconductive layer, the semiconductor element and the connection structure.

Clause 21.

The semiconductor device according to clause 1, wherein the first electroconductive layer and the second electroconductive layer comprise a metal layer disposed on an insulating substrate, and

the second insulating layer includes a resin package sealing the insulating substrate, the first electroconductive layer, the second electroconductive layer, the semiconductor element and the connection structure. The sealing resin contains silicone gel.

Clause 22.

The semiconductor device according to clause 1, wherein the second insulating layer has a front surface exposed to outside air (atmosphere).

Clause 23.

A method for manufacturing a semiconductor device as set forth in any one of clauses 1 to 22, the method comprising the steps of:

preparing a synthetic resin material containing: a material having a lower moisture permeability than a material forming the second insulating layer, or a precursor thereof; and a volatile solvent,

electrically connecting the back surface electrode to the first electroconductive layer,

dropping the synthetic resin material onto an upper surface of the semiconductor element to cover the semiconductor element with the synthetic resin material, and

heat-curing the synthetic resin material with the semiconductor element covered with the synthetic resin material, to thereby form the first insulating layer. Before the step of heat-curing the synthetic resin material, the synthetic resin material does not need to have the function of the first insulating layer. It is only necessary that the synthetic resin material has the function of the first insulating layer after it is heat-cured. For example, polyimide is dissolved in the solvent in the state of a precursor, and it becomes polyimide through "imidization" after heat-curing, to thereby have the function as the first insulating layer.

Clause 24.

The method for manufacturing the semiconductor device according to clause 23, further comprising the step of connecting the connection structure to the front surface electrode and the second electroconductive layer before the step of covering the semiconductor element with the synthetic resin material.

The invention claimed is:

1. A semiconductor device comprising:

a first mounting layer including an upper surface facing a vertical direction upwardly;

a bonding layer that is electrically conductive and disposed on the upper surface of the first mounting layer;

a switching element which includes a first element upper surface facing the vertical direction upwardly, a first

element lower surface facing an opposite direction of the first element upper surface, and a first element side surface connected to both of the first element upper surface and the first element lower surface, the switching element being electrically bonded to the first mounting layer with the first element lower surface facing the upper surface of the first mounting layer;

a conductive member connected to an upper surface electrode of the first element upper surface;

a resin layer that covers at least the first element side surface; and

a sealing resin that covers the switching element, the resin layer, and at least a part of the conductive member,

wherein the resin layer is held in contact with both of a side surface of the bonding layer and the first element side surface in a manner such that the resin layer straddles the side surface of the bonding layer and the first element side surface in a cross section along the vertical direction.

2. The semiconductor device according to claim 1, wherein the side surface of the bonding layer is flush with a side surface of the switching element at their joint.

3. The semiconductor device according to claim 1, further comprising a second mounting layer that is electrically connected to the conductive member, wherein the resin layer does not cover the second mounting layer.

4. The semiconductor device according to claim 1, wherein the resin layer does not cover the first element upper surface.

5. The semiconductor device according to claim 1, wherein the resin layer covers at least a part of the first element upper surface.

6. The semiconductor device according to claim 1, wherein the switching element is a MOSFET formed on a SiC substrate.

7. The semiconductor device according to claim 1, wherein the first mounting layer comprises a lead frame.

8. The semiconductor device according to claim 1, further comprising a substrate on a surface of a conductive layer opposite to a surface of the conductive layer on which the switching element is mounted.

9. The semiconductor device according to claim 8, wherein the substrate is made of a ceramic with excellent thermal conductivity.

10. The semiconductor device according to claim 8, wherein the resin layer is disposed on the first mounting layer, and is not in contact with the substrate.

11. The semiconductor device according to claim 8, wherein the sealing resin does not cover a side of the substrate.

12. The semiconductor device according to claim 8, wherein a side surface and a lower surface of the substrate are exposed from the sealing resin.

13. The semiconductor device according to claim 1, wherein the conductive member comprises wires.

14. The semiconductor device according to claim 1, wherein the conductive member comprises a metal plate.

15. The semiconductor device according to claim 1, further comprising protective elements parallelly connected to the switching element, wherein the resin layer is arranged between the protective elements.

16. The semiconductor device according to claim 1, wherein the resin layer comprises polyimide.

17. The semiconductor device according to claim 15, wherein the resin layer comprises silicone gel.

18. The semiconductor device according to claim 15, wherein the resin layer covers the protective elements.

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19. A semiconductor device comprising:
- a first lead frame including an upper surface facing a vertical direction upwardly;
 - a bonding layer that is electrically conductive and disposed on the upper surface of the first lead frame;
 - a switching element which includes a first element upper surface facing the vertical direction upwardly, a first element lower surface facing an opposite direction of the first element upper surface, and a first element side surface connected to both of the first element upper surface and the first element lower surface, the switching element being electrically bonded to the first lead frame with the first element lower surface facing the upper surface of the first lead frame;
 - a conductive member connected to an upper surface electrode of the first element upper surface;
 - a resin layer that covers at least the first element side surface; and
 - a sealing resin that covers the switching element, the resin layer, and at least a part of the conductive member, wherein the resin layer is held in contact with both of a side surface of the bonding layer and the first element side surface in a manner such that the resin layer straddles the side surface of the bonding layer and the first element side surface in a cross section along the vertical direction.
20. The semiconductor device according to claim 19, wherein the side surface of the bonding layer is flush with a side surface of the switching element at their joint.

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21. The semiconductor device according to claim 19, further comprising a second lead frame that is electrically connected to the conductive member, wherein the resin layer does not cover the second lead frame.
22. The semiconductor device according to claim 19, wherein the resin layer does not cover the first element upper surface.
23. The semiconductor device according to claim 19, wherein the resin layer covers at least a part of the first element upper surface.
24. The semiconductor device according to claim 19, wherein the switching element is a MOSFET formed on a SiC substrate.
25. The semiconductor device according to claim 19, wherein the conductive member comprises wires.
26. The semiconductor device according to claim 19, wherein the conductive member comprises a thin metal plate.
27. The semiconductor device according to claim 19, further comprising protective elements parallelly connected to the switching element, wherein the resin layer is arranged between the protective elements.
28. The semiconductor device according to claim 19, wherein the resin layer comprises polyimide.
29. The semiconductor device according to claim 19, wherein the resin layer comprises silicone gel.
30. The semiconductor device according to claim 27, wherein the resin layer covers the protective elements.

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